



PROJECT REPORT

DPS Quantum Key Distribution over FSO Channels

Simulation and Comparative Analysis

Theory, worked calculations, turbulence, machine learning and trusted satellite extensions

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Institution	IIT Delhi, Bharti School of Telecommunication Technology and Management
Prepared by	Limnesh Augustine
Supervisor	Prof. Neel Kanth Kundu
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Abstract

This project develops a reproducible numerical framework for differential phase shift quantum key distribution (DPS-QKD) over free-space optical (FSO) channels. The report begins with the long-term confidentiality problem and establishes the roles of quantum communication, authenticated classical discussion and conventional encrypted data transport. BB84, BBM92 and DPS teaching examples introduce the main protocol approaches; QBER estimation, syndrome-based information reconciliation and Toeplitz privacy amplification then establish the common key post-processing chain. The study then returns to DPS for the interferometer, detector-click and photon-statistics models before evaluating a 20 km terrestrial optical link, atmospheric fading, machine-learning-assisted design search, LEO satellite, trusted-relay, multi-satellite and banking application scenarios.

At the 20 km baseline, the total efficiency is 0.003464884, physical raw detections are approximately 5,236/s, and physical click-conditioned QBER is 1.374%. The separate compact model gives 3.180% modeled bit QBER and 843.5 modeled bit/s at a 10 MHz clock. The turbulence-pooled result is 843.7 modeled bit/s. An RBF kernel-ridge surrogate reaches held-out R^2 values of 0.9921 for rate and 0.9755 for modeled QBER; its proposed operating point yields 2,940.0 modeled bit/s under the original physics recheck. The sampled single-LEO pass provides about 95,195 modeled proxy bits. Trusted relay and network results are interpreted as minimum integrated hop budgets.

Introduction and Problem Statement

This section defines the long-term confidentiality problem that motivates the project. It first explains the quantum threat to widely used public-key mechanisms, then positions post-quantum cryptography and QKD as complementary quantum-safe approaches before stating the project objectives and scope.

Long-Term Confidentiality and the Quantum Threat

Modern digital communication relies heavily on cryptographic algorithms to protect sensitive information transmitted across public and private networks. Although current encryption systems provide strong protection against conventional computing attacks, many widely deployed security protocols depend on public-key algorithms such as RSA and Elliptic Curve Cryptography (ECC) for key establishment and authentication. The rapid advancement of quantum computing introduces a significant long-term security concern because a sufficiently powerful, fault-tolerant quantum computer could break these algorithms using Shor's algorithm. This creates the Harvest-Now, Decrypt-Later (HNDL) threat, in which adversaries intercept and store encrypted communications today with the intention of decrypting them when future quantum computing capabilities become available. Information requiring confidentiality for many years, including financial records, healthcare data, government communications, intellectual property and critical infrastructure information, is particularly exposed to this risk. Although strong symmetric encryption such as AES-256 is not immediately considered broken by the same quantum attack, the compromise of vulnerable key-establishment mechanisms could expose the keys used to protect recorded communications. Consequently, relying exclusively on conventional, quantum-vulnerable public-key cryptography may not provide sufficient long-term confidentiality.

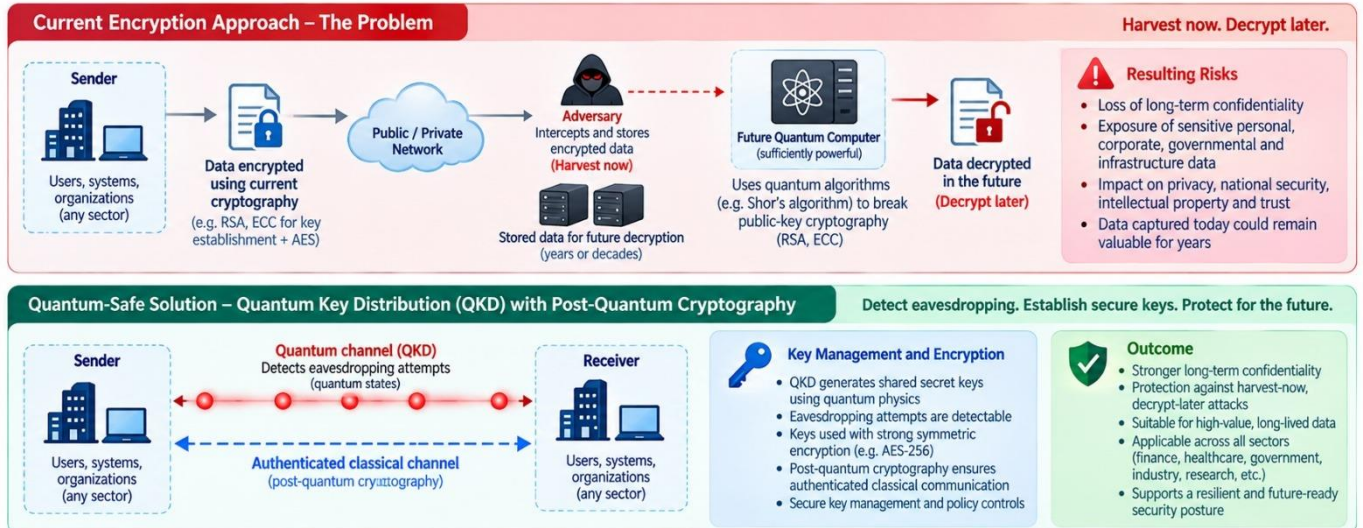


Figure 1. Current public-key confidentiality risk and the QKD-assisted quantum-safe key-distribution concept.

Shor's Algorithm

RSA relies on a problem that is easy to perform in one direction and computationally difficult to reverse: multiplying two very large prime numbers is straightforward, while recovering those factors from their product is impractical for current conventional computers at cryptographic sizes. Shor's algorithm changes this security assumption because a sufficiently capable fault-tolerant quantum computer could factor large integers efficiently. NIST notes that factoring sufficiently large integers has been estimated to require extremely long times on conventional computers, whereas a cryptographically relevant quantum computer could reduce the task dramatically; exact RSA-breaking times depend on the assumed quantum architecture, error correction and available resources. Today's quantum computers cannot break RSA-2048

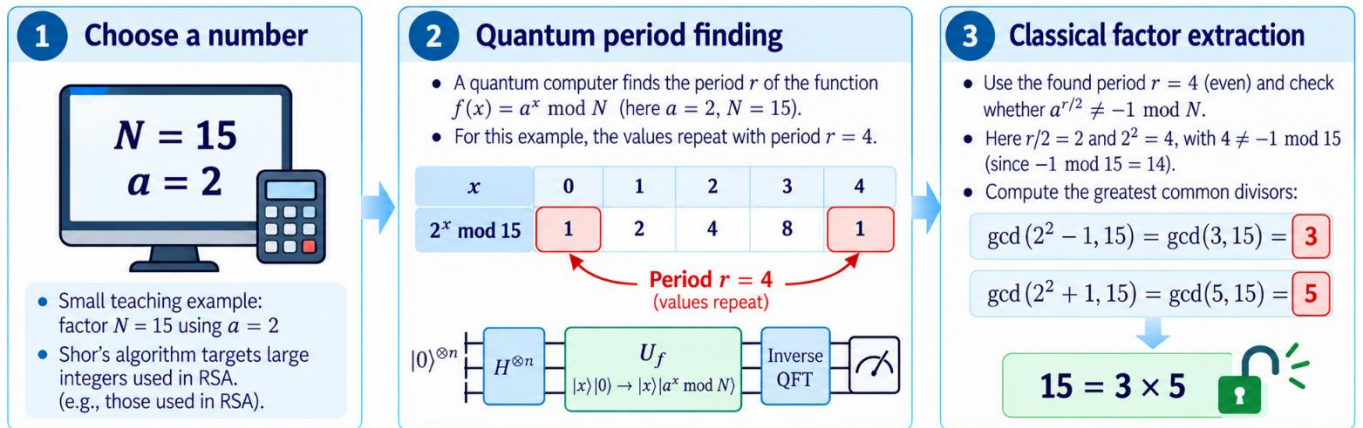


Figure 2. Illustrative overview of Shor's factoring workflow using a small teaching example.

Quantum-Safe Approaches: PQC and QKD

This problem motivates the investigation of quantum-resistant approaches, including post-quantum cryptography and Quantum Key Distribution (QKD), which uses quantum-mechanical principles to establish shared secret keys. When appropriately implemented alongside authenticated communication, secure key management and strong symmetric encryption, QKD offers an alternative key-distribution mechanism for protecting sensitive information against future cryptographic threats.

Example: Suppose an organization has data that must remain confidential for **10 years**. An attacker records encrypted traffic today. If the session keys were established using a public-key method that is vulnerable to a future quantum computer, and a sufficiently capable quantum-computing service becomes available before



the 10-year confidentiality period expires, the attacker could potentially recover the old session secrets and decrypt the stored traffic while the information is still sensitive.

Possible organizational impact includes:

- exposure of confidential customer, financial, technical, or business data;
- breach of contractual or regulatory confidentiality obligations;
- loss of intellectual property or competitive information;
- incident-response, legal, notification, and recovery costs;
- reputational damage even though the original traffic was captured years earlier.

The purpose of quantum-safe key establishment is therefore not only to protect communications against attacks available **today**, but also to protect information whose required secrecy lifetime extends into the period when large fault-tolerant quantum computers may exist.

Project Objectives

The key objective of this project is to **develop a simulation framework to evaluate DPS-QKD QBER and a modeled secret-key-rate proxy over free-space optical channels**. The core model is extended to study **log-normal atmospheric turbulence, ML-assisted parameter optimization** and **satellite/trusted-relay scenarios**.

Scope of the Study

The study moves from the long-term confidentiality problem and QKD fundamentals through BB84, BBM92 and DPS, then covers QBER estimation, information reconciliation and privacy amplification before returning to the detailed DPS physical model. The model is applied to a 20 km terrestrial FSO link, atmospheric turbulence, ML-assisted parameter optimization, satellite and trusted-relay scenarios, and a banking application case study.

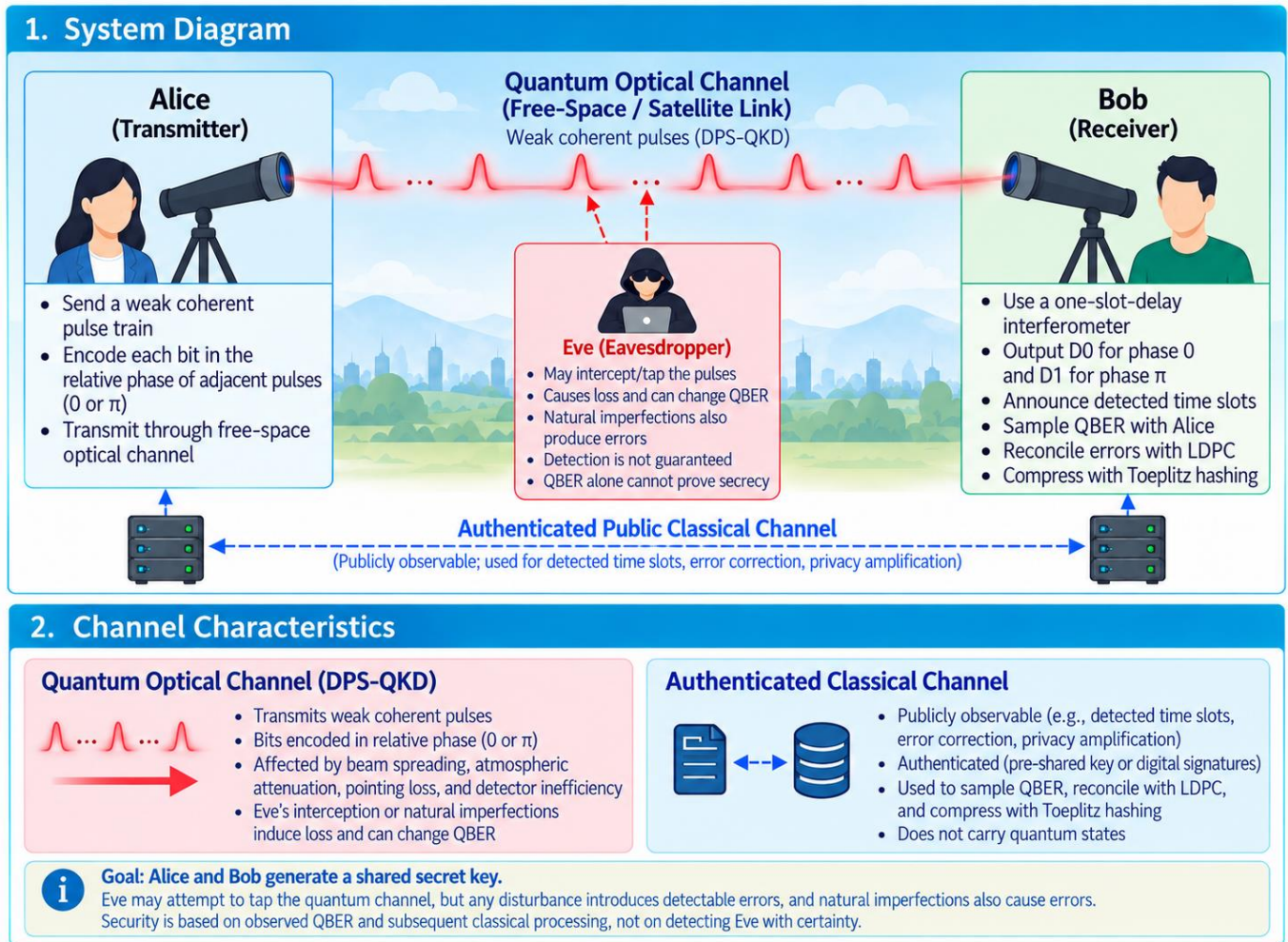


Figure 3. DPS-QKD system view showing the quantum optical channel and authenticated classical channel.

Research Questions

The primary research questions addressed in this project are:

- How do distance, aperture, mean photon number, visibility, detector background and atmospheric fading change DPS QBER and modeled SKR?
- Does averaging a fluctuating channel before applying the nonlinear detector, QBER and modeled-rate equations bias the result compared with evaluating each fading state individually and then pooling the statistics?
- Can a validated ML surrogate identify promising operating points while respecting a QBER constraint, and do those points survive rechecking with the original physics model?

Fundamentals of Quantum Key Distribution

This section introduces the communication model common to the QKD protocols used in this study. It distinguishes the quantum channel, the authenticated classical channel and the classical post-processing stages that transform raw detection events into shared key material.

Quantum and Classical Channels, Threats and Authentication

Depending on the protocol, Alice prepares quantum signals for Bob (for example BB84/DPS) or Alice and Bob measure signals distributed from an entangled source (for example BBM92). They use an **authenticated** classical channel for the required public discussion—such as basis information, detection positions,



parameter estimation, reconciliation data and verification—without publishing the undisclosed raw key. They estimate errors, reconcile, verify equality, and compress the key to limit Eve's information. Eve may observe or manipulate the optical channel and public discussion; authentication blocks an otherwise fatal man-in-the-middle impersonation. Photon loss or errors alone cannot identify Eve: hardware imperfections also cause them. Real devices require security analysis against side channels.

Example of why authentication matters: Without it, Eve can separately negotiate a key with Alice and another with Bob, then read and forward messages. A QBER check in each forged session need not expose her, because Eve is acting as an endpoint of two separate sessions rather than merely measuring and resending individual signals.

Why QKD Still Requires Authentication

QKD can reveal disturbances under its protocol and device assumptions, but it does **not automatically prove the identity of the communicating parties**.

The classical channel used for:

- Basis announcements,
- Parameter estimation,
- Syndrome exchange,
- Error correction, and
- Privacy-amplification coordination

must therefore be **authenticated**.

If the classical channel is not authenticated, Eve can perform a **man-in-the-middle (MITM) attack**.

Instead of allowing Alice and Bob to establish one common key,

$$\text{Alice} \leftrightarrow \text{Bob}$$

Eve can create two independent connections:

$$\text{Alice} \leftrightarrow \text{Eve}$$

and

$$\text{Eve} \leftrightarrow \text{Bob}$$

Alice may believe that her key is shared with Bob, while it is actually shared with Eve.

Bob may similarly believe that his key is shared with Alice, while it is actually shared with Eve.

Therefore, **matching keys within a QKD session do not by themselves prove the identity of the remote party**.

Worked authentication example

Eve negotiates two separate keys. Each forged session passes a local equality check, yet Alice and Bob do not share the same key. This example separates identity authentication from error estimation on an optical link.

Example calculation

=== Unauthenticated QKD Man-in-the-Middle Example ===

Session 1: Alice <-> Eve

Alice's key : 10110010

Eve's key : 10110010

Keys match? : True

Session 2: Eve <-> Bob

Eve's key : 01011001

Bob's key : 01011001

Keys match? : True

=== Alice versus Bob ===

Alice thinks her key belongs to Bob:

Alice: 10110010

Bob thinks his key belongs to Alice:

Bob : 01011001

Do Alice and Bob actually share the same key? False



Conclusion:

Alice successfully shares a key with Eve.

Bob successfully shares another key with Eve.

But Alice and Bob have NOT authenticated each other.

QKD Key-Processing Flow

At a high level, QKD proceeds from quantum-state preparation or entangled-state measurement to valid-event selection, parameter estimation and QBER sampling, information reconciliation, key verification and privacy amplification. These steps are coordinated over an authenticated classical channel without publishing the undisclosed raw key; the resulting shared key can then be supplied to conventional cryptographic protection.

Major QKD Protocols: BB84, BBM92 and DPS

Table 1. Comparison of BB84, BBM92 and DPS protocol concepts.

Protocol	Optical idea	Example bit extraction	Main difference
BB84	Alice prepares one of four states in two bases, e.g. H/V and D/A	Keep a detected position if Alice's and Bob's announced bases match	Basis sifting; weak coherent implementations usually need decoy-state analysis
BBM92	A source sends entangled photon pairs to Alice and Bob	Keep compatible basis/coincident detections and correct correlation convention	Both arms must arrive; source need not hold a copy of their outcomes
DPS	Alice phase-modulates a coherent pulse train	Interfere pulse k with $k - 1$; one detector represents 0, the other 1	Relative phase, no BB84 random-basis sifting; security proof is protocol-specific

The following sections work through BB84 and BBM92 before developing the DPS model.

BB84 Protocol Demonstration

BB84 encodes each classical bit using one of two randomly selected quantum bases.

Two BB84 bases

The **rectilinear basis (Z basis)** uses:

- Bit 0 -> Horizontal polarization, H
- Bit 1 -> Vertical polarization, V

The **diagonal basis (X basis)** uses:

- Bit 0 -> +45 degree polarization, D
- Bit 1 -> -45 degree polarization, A

In this simulation we represent the bases numerically as:

- 0 -> Z basis (H/V)
- 1 -> X basis (D/A)

Alice randomly generates:

1. A raw binary key.
2. A random basis for every bit.
3. The corresponding quantum state.

Bob does not know Alice's basis while measuring the incoming photons. He therefore independently chooses a random measurement basis for every photon.

If Bob chooses the **same basis as Alice**, he obtains Alice's bit correctly in this idealized simulation.

If Bob chooses the **wrong basis**, quantum mechanics gives a random result, so Bob has a 50% probability of obtaining either 0 or 1.

After the quantum transmission is complete, Alice and Bob publicly announce **only their basis choices** over an authenticated classical channel.

They do **not** reveal their secret bit values.

They keep only positions where their bases matched. This process is called **basis sifting**.

For random independent basis choices,

$$P(\text{basis match}) = \frac{1}{2}$$

so approximately half of the transmitted bits survive BB84 sifting.

This first example represents an ideal BB84 channel with no Eve, no detector noise and no optical errors. Therefore, whenever Alice and Bob use the same basis, their retained bits are identical.

With seed 42, Alice prepares 16 random bits and independently chooses their bases. Bob independently chooses measurement bases. Matching bases recover Alice's bit in this ideal example; mismatched bases return a random outcome. The keep mask determines the retained positions.

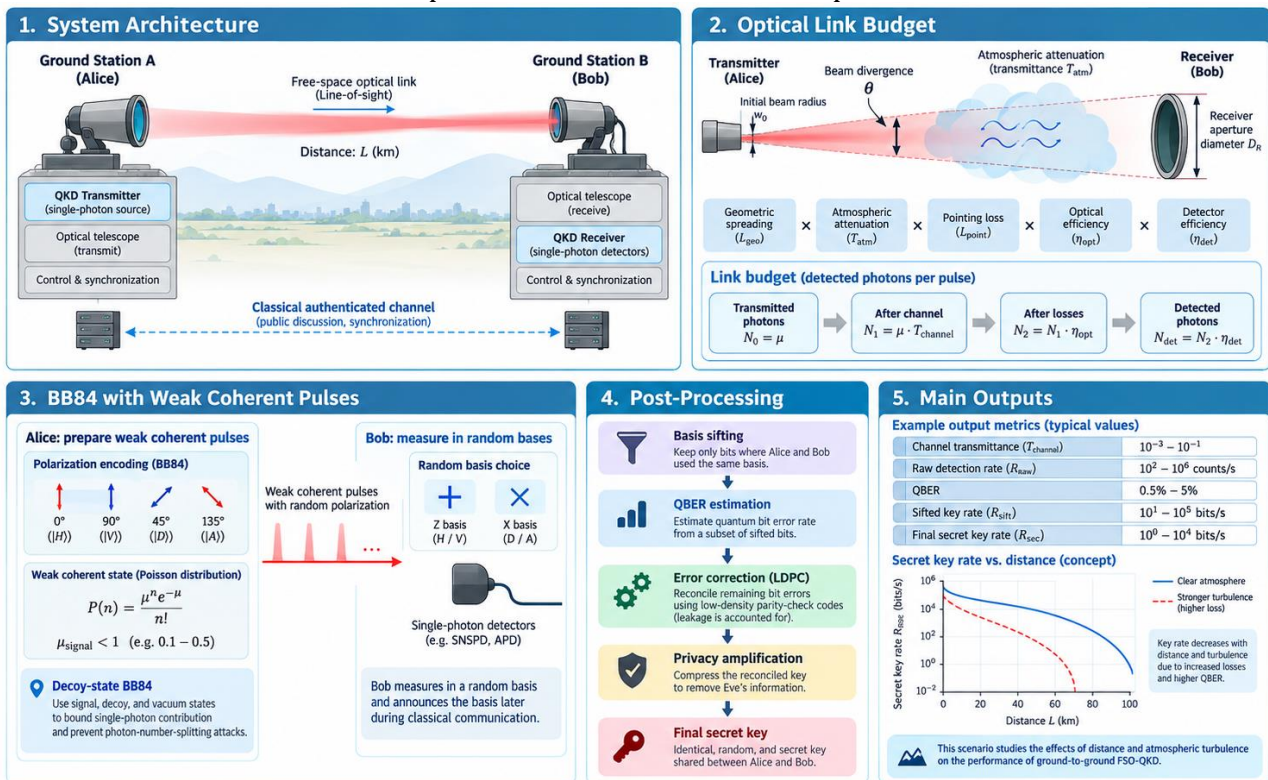


Figure 4. BB84: preparation, measurement and sifting
Example calculation

STEP 1 - Alice generates a random raw key

Alice bits:

1010110111110000

STEP 2 - Alice randomly selects encoding bases

Alice bases:

0010101111000111

STEP 3 - Alice prepares BB84 quantum states

States:

V H A H A V D A A A V V H D D D



STEP 4 - Bob randomly selects measurement bases

Bob bases:

1110100010111011

STEP 5 - Bob's measurement results

Bob bits:

1110111111010100

STEP 6 - Public basis announcement

Alice bases: 0010101111000111

Bob bases: 1110100010111011

STEP 7 - Matching basis positions

Keep mask:

0011110010000011

Matching positions (0-based):

[2, 3, 4, 5, 8, 14, 15]

STEP 8 - Sifted keys

Alice sifted key: 1011100

Bob sifted key: 1011100

STEP 9 - Compare sifted keys

Sifted keys identical? True

STEP 10 - Sifting statistics

Photons transmitted : 16

Bits retained : 7

Sifting ratio : 43.75%

Theoretical average : ~50%

Each row identifies Alice's bit and encoding basis, the prepared polarization, Bob's basis and outcome, and the sifting decision. Discarded mismatched-basis observations do not enter the final comparison.

Table 2. Complete BB84 transmission record

Index	Alice bit	Alice basis	State	Bob basis	Bob bit	Match	Keep
0	1	Z (H/V)	V	X (D/A)	1	NO	DROP
1	0	Z (H/V)	H	X (D/A)	1	NO	DROP
2	1	X (D/A)	A	X (D/A)	1	YES	YES
3	0	Z (H/V)	H	Z (H/V)	0	YES	YES
4	1	X (D/A)	A	X (D/A)	1	YES	YES
5	1	Z (H/V)	V	Z (H/V)	1	YES	YES
6	0	X (D/A)	D	Z (H/V)	1	NO	DROP
7	1	X (D/A)	A	Z (H/V)	1	NO	DROP
8	1	X (D/A)	A	X (D/A)	1	YES	YES
9	1	X (D/A)	A	Z (H/V)	1	NO	DROP
10	1	Z (H/V)	V	X (D/A)	0	NO	DROP
11	1	Z (H/V)	V	X (D/A)	1	NO	DROP
12	0	Z (H/V)	H	X (D/A)	0	NO	DROP
13	0	X (D/A)	D	Z (H/V)	1	NO	DROP
14	0	X (D/A)	D	X (D/A)	0	YES	YES
15	0	X (D/A)	D	X (D/A)	0	YES	YES



BB84 sifted-key check

QBER is calculated only on the retained positions. Seven of sixteen positions survive in this finite example, so the observed 43.75% sifting ratio differs from its 50% expectation.

Calculation

Sifted key length : 7

Errors : 0

QBER : 0.00%

BBM92 Protocol Demonstration

BBM92 is an **entanglement-based quantum key distribution protocol**.

Unlike BB84, Alice does not prepare a classical bit and encode it into a photon.

Instead, an entangled-photon source generates pairs of photons. One photon is sent to Alice and the other photon is sent to Bob.

In this educational example, the source produces the Bell state:

$$\Phi^+ = \frac{HH + VV}{\sqrt{2}}$$

The same state can also be written in the diagonal basis as:

$$\Phi^+ = \frac{DD + AA}{\sqrt{2}}$$

Therefore, if Alice and Bob measure their photons using the **same basis**, their measurement results are perfectly correlated in an ideal system.

Measurement bases

Alice and Bob independently select one of two measurement bases:

- 0 -> Z basis (H/V)
- 1 -> X basis (D/A)

The bit mapping used in this simulation is:

Z basis

- H -> bit 0
- V -> bit 1

X basis

- D (+45 degrees) -> bit 0
- A (-45 degrees) -> bit 1

Alice and Bob independently and randomly select their measurement bases.

If both select the same basis:

$$Basis_A = Basis_B$$

their measurement results are correlated:

$$Bit_A = Bit_B$$

for the ideal Bell state used here.

If their bases are different, the outcomes do not provide deterministic shared-key information and those positions are discarded during sifting.

After all measurements are complete, Alice and Bob publicly announce **only their basis choices** over an authenticated classical channel.

They keep only positions where their bases matched.

For independently random basis choices:

$$P(\text{basis match}) = \frac{1}{2}$$

so approximately half of the detected entangled photon pairs survive basis sifting.

This first BBM92 simulation assumes an ideal entangled source, perfect detectors, no channel noise and no Eve. Therefore the QBER of the sifted key should be 0%.

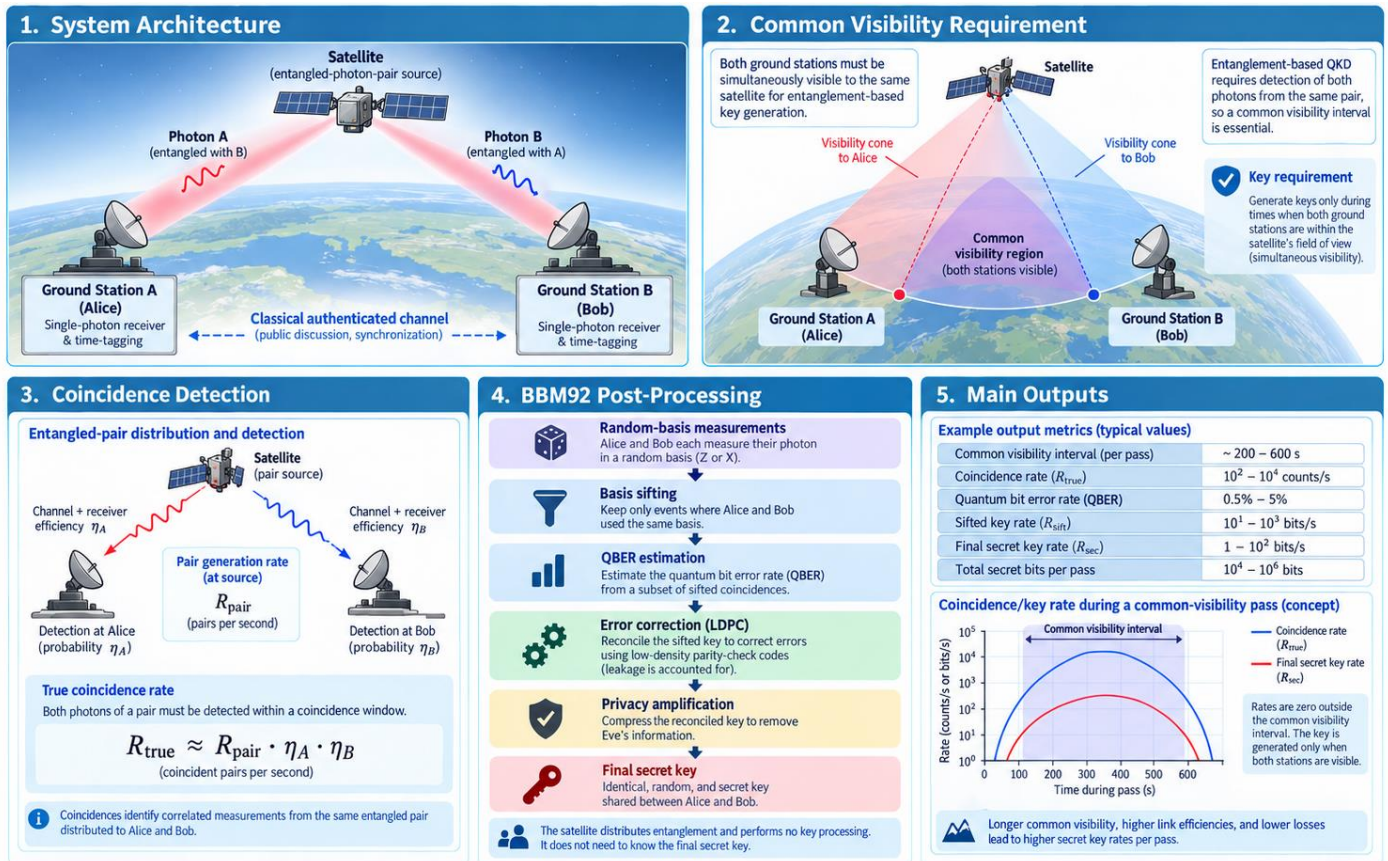


Figure 5. BBM92: create the ideal pair example

The example defines sixteen entangled-pair positions and the correlated outcome convention for the Bell state Phi-plus. It samples the ideal measurement outcomes; it does not propagate a laboratory source or density matrix.

Example calculation

STEP 1 - Entangled photon source

Bell state:

$$|\Phi^+\rangle = (|HH\rangle + |VV\rangle) / \sqrt{2}$$

Number of entangled pairs produced: 16

BBM92: Alice selects measurement bases

Alice independently chooses Z or X for every pair. Z denotes H/V and X denotes D/A.

STEP 2 - Alice selects random measurement bases

Alice basis:

1010110111110000

BBM92: Bob selects measurement bases

Bob's basis sequence is independent of Alice's. The public comparison occurs after measurement.

STEP 3 - Bob selects random measurement bases

Bob basis:

0010101111000111

BBM92: sample correlated outcomes

For the chosen Bell state, same-basis outcomes agree ideally. Incompatible bases give random relative outcomes.



STEP 4 - Measurement outcomes

Alice measurement bits:

0111111101011010

Bob measurement bits:

0111110101011100

BBM92: map bits to detector outcomes

The sampled binary results are translated back to physical polarization labels using each station's selected basis.

STEP 5 - Physical detector outcomes

Alice:

D V A V A A V A D A D A V H V H

Bob:

H V A V A V D A D A H V V A D D

BBM92: public basis announcement

Only basis choices are announced for sifting; the undisclosed measurement bits remain private within the ideal teaching example.

STEP 6 - Public basis announcement

Alice bases: 1010110111110000

Bob bases: 0010101111000111

BBM92: identify compatible positions

The Boolean keep mask selects matching-basis pairs. Positions are indexed from zero throughout these worked examples.

STEP 7 - Matching basis positions

Keep mask:

0111100111001000

Matching positions (0-based):

[1, 2, 3, 4, 7, 8, 9, 12]

BBM92: extract the sifted strings

Alice and Bob retain the same subset of pair positions and form their respective sifted strings.

STEP 8 - BBM92 sifted keys

Alice sifted key: 11111011

Bob sifted key: 11111011

BBM92: verify ideal correlations

The comparison checks the correlation convention rather than testing a real entangled source.

STEP 9 - Compare sifted keys

Alice: 11111011

Bob: 11111011

Sifted keys identical? True

BBM92: count retained pairs

The retained fraction is an outcome of this finite random example, with a theoretical matching-basis probability of one half.

STEP 10 - Sifting statistics

Entangled pairs generated : 16

Pairs retained : 8

Sifting ratio : 50.00%

Theoretical average : ~50%

The complete row-by-row record shows basis compatibility and the resulting retained pair outcomes.



Table 3. Complete BBM92 measurement record

Pair	Alice basis	Alice state	Alice bit	Bob basis	Bob state	Bob bit	Keep
0	X (D/A)	D	0	Z (H/V)	H	0	DROP
1	Z (H/V)	V	1	Z (H/V)	V	1	YES
2	X (D/A)	A	1	X (D/A)	A	1	YES
3	Z (H/V)	V	1	Z (H/V)	V	1	YES
4	X (D/A)	A	1	X (D/A)	A	1	YES
5	X (D/A)	A	1	Z (H/V)	V	1	DROP
6	Z (H/V)	V	1	X (D/A)	D	0	DROP
7	X (D/A)	A	1	X (D/A)	A	1	YES
8	X (D/A)	D	0	X (D/A)	D	0	YES
9	X (D/A)	A	1	X (D/A)	A	1	YES
10	X (D/A)	D	0	Z (H/V)	H	0	DROP
11	X (D/A)	A	1	Z (H/V)	V	1	DROP
12	Z (H/V)	V	1	Z (H/V)	V	1	YES
13	Z (H/V)	H	0	X (D/A)	A	1	DROP
14	Z (H/V)	V	1	X (D/A)	D	0	DROP
15	Z (H/V)	H	0	X (D/A)	D	0	DROP

BBM92 sifted-key error rate

The ideal same-basis strings agree, giving zero mismatches. Atmospheric two-arm coincidence rates are outside this introductory example.

Calculation

BBM92 sifted key length : 8
BBM92 errors : 0
BBM92 QBER : 0.00%

DPS-QKD Differential Phase Encoding

Differential Phase Shift Quantum Key Distribution (DPS QKD) uses a sequence of weak coherent optical pulses.

Unlike BB84, information is not encoded in the absolute polarization of each individual photon. Instead, the secret bit is encoded in the **phase difference between two consecutive optical pulses**. For pulse k ,

$$a_k = \sqrt{\mu} e^{i\phi_k}$$

where:

- μ is the mean photon number per pulse
- ϕ_k is the optical phase of pulse k

The information bit is determined by the phase difference between consecutive pulses:

$$b_{k-1} = \frac{\phi_k - \phi_{k-1}}{\pi} \pmod{2}$$

The encoding rule is:

- Bit 0 -> no phase change

$$\Delta\phi = 0$$

- Bit 1 -> phase change of π

$$\Delta\phi = \pi$$

Therefore, Alice does not encode a bit into one isolated pulse.

Instead:

One DPS bit is represented by the relative phase between two consecutive pulses

For example, consider the DPS data bits:

1 0 1 1 0

We start with an arbitrary reference phase:

$$\phi_0 = 0$$

and generate every following pulse phase according to:

$$\phi_k = \phi_{k-1} + \pi b_{k-1} \pmod{2\pi}$$

This produces six optical pulses for five DPS bits.

Bob uses a delay-line Mach-Zehnder interferometer with a delay equal to one pulse period.

This causes pulse k to interfere with pulse $k - 1$.

In an ideal system:

- Equal phases $\rightarrow$ Detector 0 clicks
- Opposite phases $\rightarrow$ Detector 1 clicks

Therefore:

$$D_0 \rightarrow 0$$

and

$$D_1 \rightarrow 1$$

This first example assumes an ideal optical channel with no loss, detector noise, turbulence or eavesdropper.

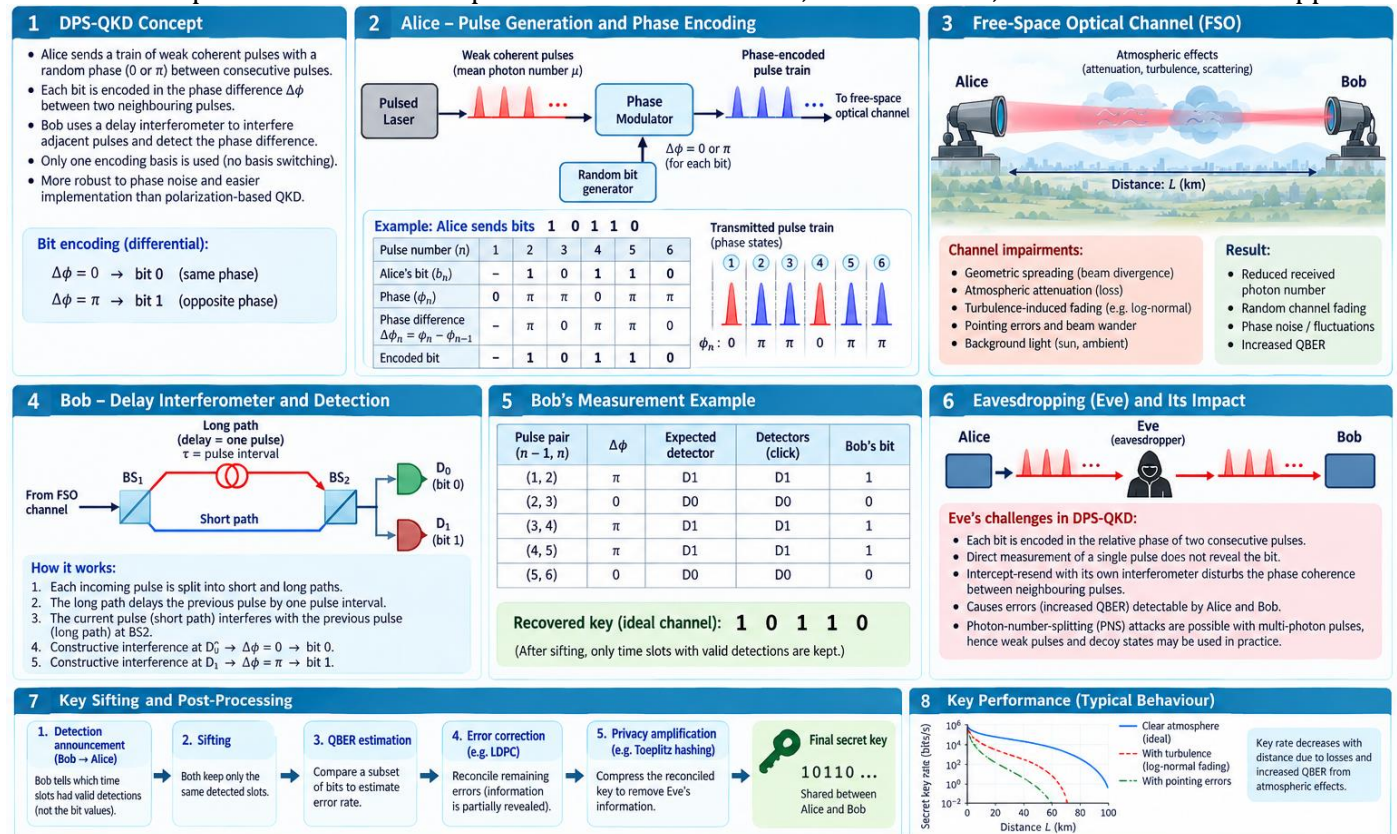


Figure 6. DPS: select the relative bits

The five-bit sequence 10110 specifies the relative phase differences of a six-pulse train. The initial pulse is a phase reference.

Example calculation

STEP 1 - Alice's DPS information bits

Relative bits: 10110

DPS: accumulate the absolute phases

Starting from zero, each bit adds either zero or π , modulo two π . This produces the absolute phase sequence from which adjacent differences carry the key.

STEP 2 - Absolute pulse phases

Pulse phases / π : [0, 1, 1, 0, 1, 1]



DPS: recover adjacent phase differences

Consecutive pulse phases are subtracted modulo two pi. The resulting differences recover the originally encoded relative pattern.

STEP 3 - Consecutive phase differences

Raw Delta phi / pi: [1, 0, -1, 1, 0]

DPS: decode the phase pattern

A relative phase of zero maps to bit zero and a relative phase of pi maps to bit one.

STEP 4 - Decode relative phases

Original DPS bits: 10110

Decoded DPS bits : 10110

Correct decoding? : True

DPS: calculate the interferometer outputs

Using a coherent amplitude $\sqrt{\mu} \exp(i \phi)$ with $\mu = 0.5$ for this ideal demonstration, the delayed and direct fields are added and subtracted at the two output ports.

STEP 5 - Ideal interferometer output

Detector D0 intensity: [0. 0.5 0. 0. 0.5]

Detector D1 intensity: [0.5 0. 0.5 0.5 0.]

DPS: make the ideal detector decision

The stronger ideal output port identifies the relative bit. This deterministic teaching decision precedes the probabilistic noisy detector model.

STEP 6 - Detector decisions

Detector sequence: ['D1', 'D0', 'D1', 'D1', 'D0']

Bob decoded bits: 10110

Alice DPS bits : 10110

Alice == Bob ? : True

The table connects each adjacent pair to its input bit, relative phase, output intensities, selected detector and decoded bit.

Table 4. Complete DPS pulse-pair calculation

Bit	Pulse pair	Previous ϕ/π	Current ϕ/π	$\Delta\phi/\pi$	D0 intensity	D1 intensity	Click	Decoded
0	P0-P1	0	1	1	0.00	0.50	D1	1
1	P1-P2	1	1	0	0.50	0.00	D0	0
2	P2-P3	1	0	1	0.00	0.50	D1	1
3	P3-P4	0	1	1	0.00	0.50	D1	1
4	P4-P5	1	1	0	0.50	0.00	D0	0

Why DPS-QKD Is the Focus of This Project

BB84 and BBM92 are retained as reference teaching protocols. The remainder of this project concentrates on DPS-QKD because the stated project objective is to evaluate DPS QBER and modeled/asymptotic secret-key-rate performance over free-space optical channels, including atmospheric turbulence, machine-learning-assisted parameter optimization and satellite extensions.

QKD Errors, Information Reconciliation and Privacy Amplification

The preceding protocol examples produce raw or sifted bit strings, but those strings are not yet ready for cryptographic use. Practical QKD therefore requires parameter estimation, information reconciliation, key verification and privacy amplification before the final shared key is released.

Error Sources and QBER Estimation from Disclosed Samples

$QBER = a_i \neq b_i / n_{compare}$, for a **randomly disclosed sample of detected positions**. The sample is discarded from the final key; finite sampling uncertainty must be bounded. Higher QBER can mean alignment, phase drift, background, or attack. It is a warning statistic, not a proof of Eve.

Disclosed-sample calculation

A 1,200-bit simulated pair is generated with an input mismatch probability of 0.035. Two hundred positions are disclosed without replacement and then removed; the simulator also inspects the remaining positions to demonstrate sampling variation.

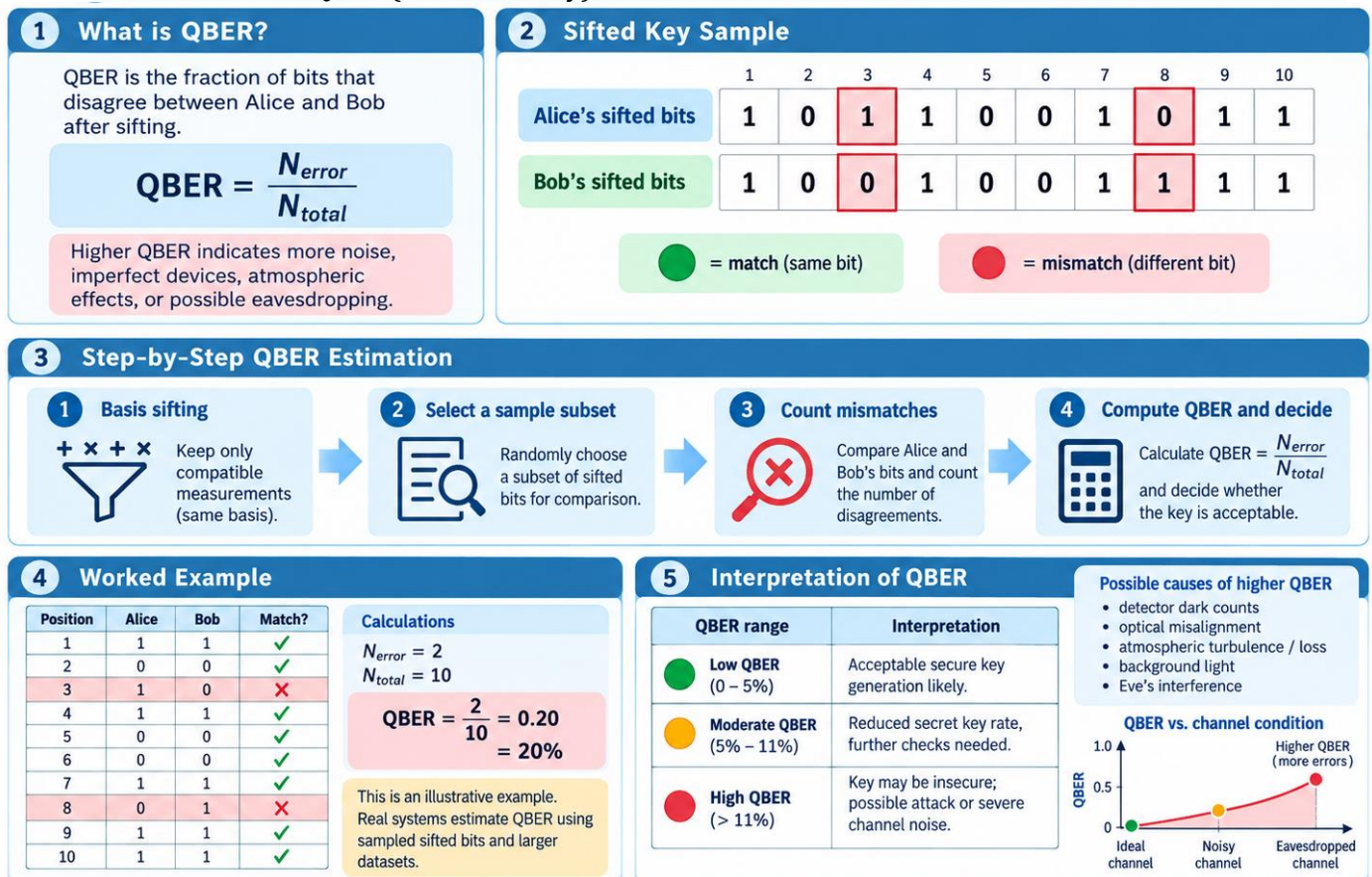
Example calculation

Disclosed sample errors: 5 / 200

Estimated QBER: 2.50%

Undisclosed positions left for reconciliation: 1000

True undisclosed QBER (simulator only): 4.60%



QBER is a key security metric in BB84, DPS-QKD and BBM92 systems.

Figure 7. QBER estimation from a disclosed sample and retained-key sampling variation



Information Reconciliation and Syndrome/LDPC Decoding

After parameter estimation, Alice and Bob retain noisy versions of the same intended key. A binary parity-check matrix H maps a key x to a public syndrome

$$s = Hx(\text{mod}2).$$

In the convention used in the implemented calculation, Bob sends his syndrome

$$s_B = Hx_B(\text{mod}2)$$

over the authenticated classical channel. Alice computes her own syndrome s_A , then forms

$$\Delta s = s_A \oplus s_B = H(x_A \oplus x_B)(\text{mod}2).$$

The vector $e = x_A \oplus x_B$ is the error pattern. For the deliberately tiny 8-bit example below, we know there are exactly two errors, so the code exhaustively tests all

$$\binom{8}{2} = 28$$

two-bit patterns and finds the unique pair whose syndrome equals Δs . Alice flips those two positions and obtains Bob's reconciled key.

Important: this exhaustive two-error search is a **teaching demonstration of syndrome decoding**, not practical LDPC belief-propagation/min-sum decoding. Real QKD reconciliation uses much longer, carefully designed sparse LDPC codes, an estimated channel error probability, iterative decoding, verification, a failure policy, and explicit leakage accounting.

The disclosed syndrome leaks information. For a binary parity-check matrix, the number of independent syndrome bits is at most

$$\text{rank}_{GF(2)}(H),$$

not simply the floating-point numerical rank. That reconciliation leakage must be included when sizing privacy amplification.

Eight-bit syndrome decoding: complete arithmetic

The sparse 4-by-8 matrix has rows 11000001, 01111000, 00010011 and 10100110. Bob supplies his syndrome. Under the explicit exactly-two-errors assumption, all 28 candidate column pairs are searched. Alice flips the unique matching pair; binary row reduction gives the GF(2) rank.

$$H = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Example calculation

Alice's sifted key : 10110010
Bob's sifted key : 10010110
Alice syndrome s_A : 1001
Bob syndrome s_B : 1101
Syndrome difference: 0100
Candidate error pairs: [(2, 5)]
Detected error positions (0-based): (2, 5)
Alice before correction : 10110010
Alice after correction : 10010110
Bob's key : 10010110
Keys match? True
Syndrome bits transmitted: 4
Independent syndrome bits $\leq \text{rank}_{GF(2)}(H) = 4$

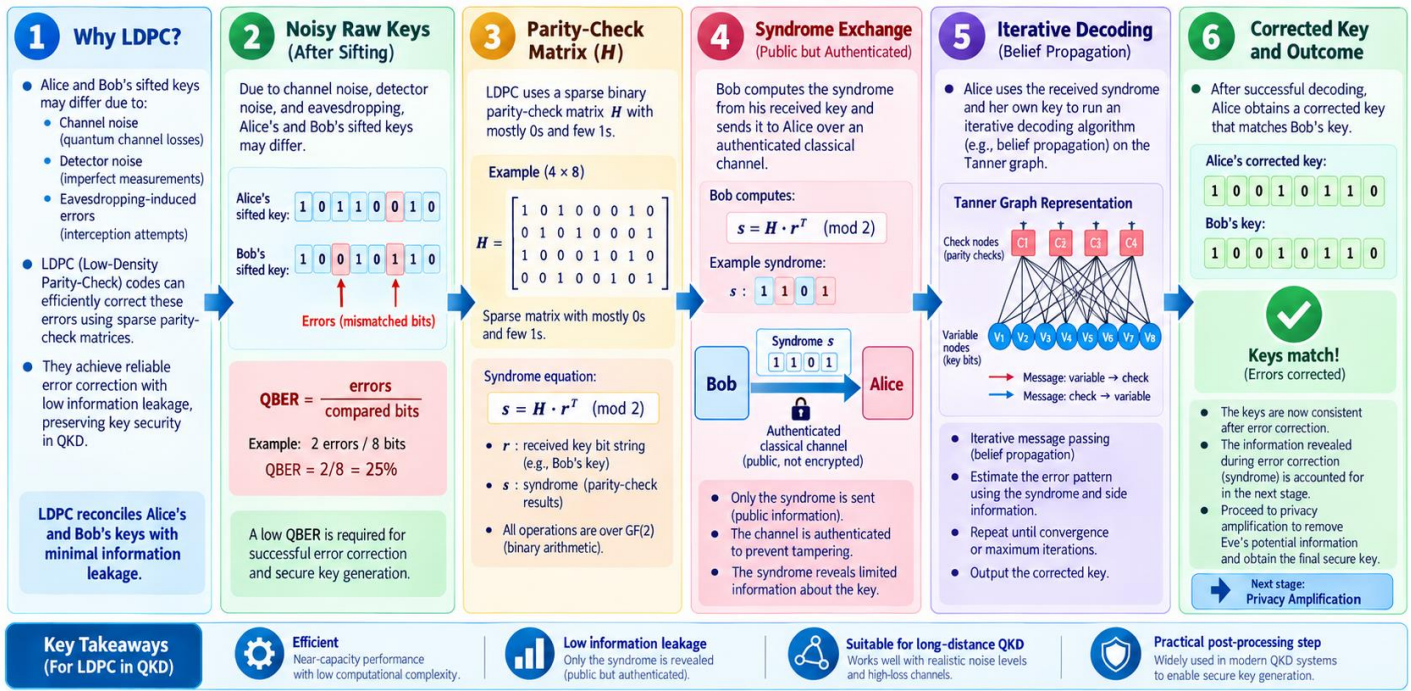


Figure 8. LDPC/syndrome-based error-correction workflow for information reconciliation
Key Verification

After reconciliation, Alice and Bob must verify that their corrected keys agree before using them as input to privacy amplification. The worked syndrome example above demonstrates equality with the reported “Keys match? True” result. In a practical system, verification is normally performed with a short authenticated hash or tag; its information leakage, authentication cost and residual failure probability must be included in the protocol accounting.

Privacy Amplification with Toeplitz Hashing

Error correction makes Alice's and Bob's reconciled keys identical, but it does **not** remove Eve's possible partial knowledge of the key.

Privacy amplification compresses the reconciled key using a **random, publicly known universal hash function**. A commonly used construction in QKD is a binary **Toeplitz matrix**, whose values are constant along each diagonal.

For a reconciled key:

$$x \in \{0,1\}^n$$

Alice and Bob calculate the final key as:

$$K = Tx \pmod{2}$$

where T is an $l \times n$ Toeplitz matrix.

For this example:

$$n = 16, l = 8$$

Therefore:

$$T \in GF(2)^{8 \times 16}$$

An 8×16 Toeplitz matrix requires:

$$n + l - 1 = 16 + 8 - 1 = 23$$

random seed bits.

The Toeplitz matrix and its seed may be **public**. Security does not depend on keeping T secret. In the implemented calculation the seed is fixed only to make the demonstration reproducible; a real protocol chooses the universal hash according to its security procedure.

Example

For a **separate 16-bit privacy-amplification example** (not a continuation of the preceding 8-bit syndrome toy example), Alice and Bob have the same reconciled key:

$$x_A = x_B = 1011001010011010$$



Using the Toeplitz matrix:

$$K_A = Tx_A(mod2)$$

and:

$$K_B = Tx_B(mod2)$$

Both obtain the same 8-bit output:

$$K_A = K_B = 00000011$$

Eve also knows the public Toeplitz matrix. However, suppose Eve's estimate of the reconciled key is:

$$x_E = 1001011010010010$$

Eve applies the same Toeplitz matrix:

$$K_E = Tx_E(mod2)$$

and obtains:

$$K_E = 01010101$$

Therefore, in this example:

$$K_A = K_B \neq K_E$$

What Happens to Eve?

Alice and Bob:

1011001010011010

becomes:

00000011

Eve's incorrect estimate:

1001011010010010

becomes:

01010101

Therefore:

Alice = Bob

but:

Eve != Alice/Bob

Important: Privacy amplification does not depend on Eve being computationally unable to calculate the hash. Eve knows the Toeplitz matrix and can perform the same calculation. Security comes from Eve's uncertainty about the reconciled key and from compressing that key sufficiently so that Eve's remaining information about the final key becomes negligibly small.

Hash-collision caveat: Eve having an incorrect 16-bit estimate does **not** mathematically guarantee a different 8-bit hash value; different inputs can collide under any compression hash. The displayed Eve value is one concrete example. QKD privacy amplification is justified statistically by universal hashing and a lower bound on Eve's conditional smooth min-entropy, not by requiring every wrong Eve input to produce a different hash.

Critical Sizing Rule

In a real QKD implementation, the final key length l must be determined from a proven security bound.

The calculation must account for:

- Eve's estimated information
- Error-correction leakage
- QBER and parameter-estimation results
- Finite-key effects
- Required security margin

Therefore, the **16-bit to 8-bit compression shown here is an educational example only.**

The result:

00000011

is a reproducible privacy-amplification output, but it should not be interpreted as a certified 8-bit secret key unless the required QKD security bound has first been established.

Sixteen-to-eight-bit Toeplitz calculation

This is a separate example with a new 16-bit shared input. The first row is 1010011010110010 and the first column is 10100000. The fixed matrix makes the arithmetic reproducible; its public seed occupies 23 independent positions.

Example calculation

Toeplitz matrix:

```
[[1 0 1 0 0 1 1 0 1 0 1 1 0 0 1 0]
[0 1 0 1 0 0 1 1 0 1 0 1 1 0 0 1]
[1 0 1 0 1 0 0 1 1 0 1 0 1 1 0 0]
[0 1 0 1 0 1 0 0 1 1 0 1 0 1 1 0]
[0 0 1 0 1 0 1 0 0 1 1 0 1 0 1 1]
[0 0 0 1 0 1 0 1 0 0 1 1 0 1 0 1]
[0 0 0 0 1 0 1 0 1 0 0 1 1 0 1 0]
[0 0 0 0 0 1 0 1 0 1 0 0 1 1 0 1]]
```

Toeplitz matrix shape: (8, 16)

Before privacy amplification:

Alice : 1011001010011010

Bob : 1011001010011010

Eve : 1001011010010010

Eve's wrong bit positions: [2 5 12]

Number of Eve errors: 3

After Toeplitz privacy amplification:

Alice final key : 00000011

Bob final key : 00000011

Eve final guess : 01010101

Alice == Bob ? True

Alice == Eve ? False

Public Toeplitz seed bits: 23

Toeplitz Matrix for Privacy Amplification in QKD

Using a random Toeplitz matrix to compress the reconciled key and reduce Eve's information

1 What is a Toeplitz Matrix?

- A Toeplitz matrix has constant values along each diagonal (top-left to bottom-right).
- A fully defined $m \times n$ Toeplitz matrix is determined by $(m + n - 1)$ random bits.
- In privacy amplification, we use a random binary Toeplitz matrix (publicly shared).

Example: 3×5 Toeplitz matrix

```
1 0 1 1 0
0 1 0 1 1
1 0 1 0 1
```

Each diagonal (top-left to bottom-right) has the same value (highlighted).

2 Generating a Random Toeplitz Matrix

- Choose output length m and input length n . Example: $m = 8, n = 16$.
- Generate a random binary seed of length $m + n - 1 = 8 + 16 - 1 = 23$ bits.
- Fill the matrix such that each diagonal (top-left to bottom-right) has the same value.

Seed format for 8×16 matrix (23 bits)

```
t_{-7} t_{-6} ... t_0 ... t_{15}
```

↑ First column (excluding top) 7 bits ($t_{-7} \dots t_{-1}$) Main diagonal 1 bit (t_0) First row (excluding first) 15 bits ($t_1 \dots t_{15}$)

These 23 bits uniquely define the 8×16 Toeplitz matrix.

3 Example: 8×16 Toeplitz Matrix

$T (8 \times 16)$

```
1 0 1 0 0 1 1 0 1 0 1 1 0 0 1 0
0 1 0 1 0 0 1 1 0 1 0 1 1 0 0 1
1 0 1 0 1 0 0 1 1 0 1 0 1 1 0 0
0 1 0 1 0 1 0 0 1 1 0 1 0 1 1 0
0 0 1 0 1 0 1 0 0 1 1 0 1 0 1 1
0 0 0 1 0 1 0 1 0 0 1 1 0 1 0 1
0 0 0 0 1 0 1 0 1 0 0 1 1 0 1 0
0 0 0 0 0 1 0 1 0 1 0 0 1 1 0 1
```

- Each diagonal (top-left to bottom-right) contains the same value.
- Only 23 random bits are needed to generate this entire matrix.
- The matrix is public and does not need to be secret.

4 Parameters Summary

Input key length (n)	16 bits
Output key length (m)	8 bits
Matrix size	8×16
Random seed length	$m + n - 1 = 23$ bits
Compression ratio	$m / n = 1/2$
Matrix elements	Binary (0 or 1)
Matrix type	Toeplitz (constant diagonals)
Shared	Public (not secret)
Security	Reduces Eve's information to a negligible level (information-theoretic security, not computational).

5 Applying the Matrix (Example Calculation)

Reconciled key (input, $n = 16$)

$x = 1 0 1 1 0 0 1 0 1 0 0 1 1 0 1 0$

Compute the final key
 $K = T \times x \pmod{2}$

Each output bit is the XOR of the corresponding subset of input bits defined by a row of T .

First output bit (row 1)

$k_1 = 1 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 \oplus \dots \oplus 0 \oplus 0$
 $= 1 \oplus 0 \oplus 1 \oplus 0 \oplus \dots \oplus 0$
 $= 0 \pmod{2}$

(All operations are modulo 2, i.e., XOR)

Final key (output, $m = 8$)

$K = 0 0 0 0 0 0 1 1$

6 Effect on Eve's Wrong Key

- Eve may have partial information about x due to quantum channel leakage and the information revealed during error correction.
- She also knows the public Toeplitz matrix T and can apply the same hash function.
- If Eve's estimated key is incorrect, her final key will generally be different from Alice and Bob's after privacy amplification.
- This significantly reduces Eve's information about the final key. The security is information-theoretic: by choosing the output length m appropriately, Eve's information can be made negligibly small (according to the security bound).

7 Complete Example with Alice, Bob and Eve

Alice (reconciled key x)	1 0 1 1 0 0 1 0 1 0 0 1 1 0 1 0
Bob (same key)	1 0 1 1 0 0 1 0 1 0 0 1 1 0 1 0
Eve (estimated key)	1 0 0 1 0 1 1 0 1 0 0 1 0 0 1 0
Eve's wrong bit positions: 2, 5, 12 (0-based index)	
After applying the same Toeplitz matrix T (final 8-bit keys)	
Alice final key $K_A =$	0 0 0 0 0 0 1 1
Bob final key $K_B =$	0 0 0 0 0 0 1 1
Eve final key $K_E =$	0 1 0 1 0 1 0 1
Alice = Bob (keys match)	
Eve ≠ Alice/Bob (different key)	

Figure 9. Toeplitz-matrix privacy amplification from reconciled key to compressed final key

DPS-QKD Detailed Physical and Numerical Model

Earlier section defined the weak coherent pulse amplitude and the relative DPS bit. Here we use those definitions to build the optical and detector model used later for QBER and rate calculations. $a_k =$

$$\sqrt{\mu}, e^{i\phi_k} b_{k-1} = (\phi_k - \phi_{k-1})/\pi \pmod{2}$$

Delay-Line Mach-Zehnder Interferometer

Bob uses an unbalanced Mach-Zehnder interferometer whose path-length difference corresponds to exactly one pulse period. Consequently, pulse k interferes with the delayed copy of pulse $k - 1$.

The ideal output fields are proportional to:

$$E_0 = \frac{a_k + a_{k-1}}{2}$$

and:

$$E_1 = \frac{a_k - a_{k-1}}{2}.$$

The corresponding output intensities are:

$$I_0 = \frac{\mu}{2} [1 + \cos(\Delta\phi)]$$

and:

$$I_1 = \frac{\mu}{2} [1 - \cos(\Delta\phi)],$$

where:

$$\Delta\phi = \phi_k - \phi_{k-1}.$$

Ideal Detector Behavior

For equal consecutive phases, $\Delta\phi = 0$ and $\cos(0) = 1$, so:

$$I_0 = \mu, I_1 = 0.$$

Detector 0 is therefore favored and represents bit 0:

$$D_0 \rightarrow 0.$$

For opposite consecutive phases, $\Delta\phi = \pi$ and $\cos(\pi) = -1$, so:

$$I_0 = 0, I_1 = \mu.$$

Detector 1 is therefore favored and represents bit 1:

$$D_1 \rightarrow 1.$$

Number of Usable DPS Bits

If Alice transmits N pulses, information is obtained from adjacent pulse pairs, giving an ideal maximum of:

$$N - 1$$

relative-phase positions. For example, six pulses P0 P1 P2 P3 P4 P5 contain five adjacent pairs: P0-P1, P1-P2, P2-P3, P3-P4, and P4-P5.

Thus:

$$6 \text{ pulses} \rightarrow 5 \text{ DPS bits.}$$

In a practical system, guard intervals, detector dead time and synchronization requirements may reduce the usable number of detection gates. For rate plots later in this report, the pulse clock is set to **10 MHz**. For long pulse trains, the number of interior DPS gates per second is approximated by the pulse clock; train-boundary and dead-time reductions must be applied for a specific hardware implementation.

Channel and Detection Efficiency

Let the total transmission and detection efficiency be η . It should include each physical efficiency or loss exactly once, such as atmospheric transmission, geometric collection efficiency, pointing efficiency, receiver optical efficiency and detector efficiency.

For mean photon number μ , the mean detected signal photon number is:

$$\lambda = \mu\eta.$$

Optical Error Probability

Real interferometers are not perfectly stable. Let V be interferometer visibility and δ the residual phase offset. We use the illustrative optical-error model:

$$e_{opt} = \frac{1 - V\cos(\delta)}{2}.$$

For an ideal interferometer, $V = 1$ and $\delta = 0$, giving $e_{opt} = 0$.



Correct and Wrong Detector Click Probabilities

Let d be the background or dark-count probability **per detector per open detection gate**. The probability that the correct detector clicks is modeled as:

$$p_c = 1 - (1 - d)e^{-\lambda(1-e_{opt})}.$$

The probability that the wrong detector clicks is:

$$p_w = 1 - (1 - d)e^{-\lambda e_{opt}}.$$

Total Detection Probability

Assuming independent detector clicks, the probability that at least one detector clicks is:

$$Q = p_c + p_w - p_c p_w.$$

Here, Q is the detection probability (gain) per usable gate in this illustrative detector model.

QBER

A click only in the wrong detector contributes an error. If both detectors click simultaneously, this model assigns a random bit, which contributes an error probability of $1/2$.

Therefore:

$$E = \frac{p_w(1 - p_c) + \frac{1}{2}p_c p_w}{Q}.$$

Here, E is the Quantum Bit Error Rate (QBER).

Important: This detector model is an illustrative physical model for studying DPS-QKD behavior. It is not by itself a complete DPS security proof or a certified secret-key-rate bound.

Worked physical detector calculation

The introductory operating point uses $\mu = 0.5$, total efficiency 0.10 , visibility 0.98 , zero constant phase offset, and dark probability $2e-6$ per detector per gate. These inputs are distinct from the later terrestrial baseline.

Example calculation

DPS detector model
Mean photon number μ : 0.5000
Total efficiency η : 0.1000
Detected mean photons λ : 0.050000
Visibility V : 0.9800
Phase offset δ : 0.0000 rad
Optical error e_{opt} : 0.010000
Correct click probability : 0.04829675
Wrong click probability : 0.00050187
Total detection probability : 0.04877438
QBER : 1.0041%
Example raw detections/s : 487,744 at 10 MHz

Expected Values Versus Sampled Data

The Poisson click calculation gives an *expectation*. Generate finite detector clicks to show sampling error. A “double click” is randomly assigned to a bit and contributes about half an error on average.

Finite-sample detector experiment

The experiment samples the independent correct and wrong click probabilities for 100,000 gates. Wrong-only events count as errors, and double clicks receive a random error assignment.

Example calculation

Observed detections: 4839 / 100000
Observed QBER: 1.013%
Expected QBER: 1.004%
Sampling difference: +0.008%

Photon Statistics, Modeled QBER and Secret-Key-Rate Proxy

The earlier DPS section showed **how relative phase produces detector clicks**. This section now builds a compact numerical model for QBER and a modeled secret-key-rate proxy.

The calculation starts from two central quantities:

- μ : mean photons emitted per optical pulse,
- η : overall probability that a transmitted photon survives the channel and is detected.

We then add detector dark counts, interferometer visibility, phase noise and error-correction overhead. What we will calculate

Starting from η and μ , we calculate:

1. source photon-number probabilities,
2. detected single-photon and multi-photon probabilities,
3. dark-count probability,
4. device-error probability,
5. DPS bit QBER,
6. photon-number gains G_0 , G_1 and $G_{\geq 2}$,
7. a phase-error proxy Q_{ph} ,
8. binary entropy penalties,
9. a modeled SKR proxy.

We use one numerical operating point repeatedly so that every intermediate value can be followed:

Table 5. Compact-model parameters and definitions.

Parameter	Example value	Meaning
η	0.00346	Overall transmission and detection efficiency
μ	0.15	Mean photons transmitted per pulse
V	0.98	Interferometer visibility
d	2×10^{-6}	Dark/background click probability per detector per gate
σ_{phase}	0.04 rad	Phase-noise standard deviation
f_{EC}	1.16	Error-correction inefficiency
pulse clock	10 MHz	Converts per-gate values into modeled bit/s

Important: the rate calculated here is a **modeled asymptotic proxy** used for engineering comparison inside this study. It is not a certified finite-key secret-key rate.

This derivation intentionally uses rounded efficiency $\eta = 0.00346$, $\mu = 0.15$, visibility 0.98, phase standard deviation 0.04 rad and $f_{EC} = 1.16$. Its approximately 842.1 modeled bit/s therefore differs slightly from the exact 20 km result of 843.5.

Example calculation

Example parameters used

eta = 0.00346

mu = 0.150

visibility V = 0.980

dark/detector/gate = 2.0e-06

phase sigma = 0.040 rad

f_EC = 1.16

pulse clock = 10 MHz

Weak Coherent Source and Poisson Distribution

A practical DPS transmitter normally uses a heavily attenuated laser rather than an ideal source that emits exactly one photon on demand. Practical single-photon sources do exist; this choice describes the weak-coherent-pulse transmitter modeled here, not a claim that single-photon sources are merely theoretical. A laser pulse is well described by a **coherent state**. If its mean photon number is μ , then a photon-number measurement does not always return μ photons. Instead, the number of photons varies randomly from pulse to pulse according to a **Poisson distribution**:



$$P(n) = e^{-\mu} \frac{\mu^n}{n!}.$$

Here:

- $n = 0, 1, 2, \dots$ is the actual photon count in one pulse,
- μ is the **average** number of photons per pulse over many pulses,
- $P(n)$ is the probability that a pulse contains exactly n photons.

So $\mu = 0.5$ does **not** mean every pulse contains half a photon. It means that after sending many pulses, the average photon count approaches 0.5 photons per pulse. Every individual pulse still contains an integer number of photons: 0, 1, 2, 3, and so on.

For QKD it is useful to group the source into three classes:

$$P_0 = e^{-\mu}$$

for a vacuum pulse,

$$P_1 = \mu e^{-\mu}$$

for exactly one photon, and

$$P_{\geq 2} = 1 - P_0 - P_1$$

for two or more photons.

Example: $\mu = 0.1$

$$\begin{aligned} P_0 &= e^{-0.1} \approx 0.90484, \\ P_1 &= 0.1e^{-0.1} \approx 0.09048, \\ P_{\geq 2} &\approx 0.00468. \end{aligned}$$

Out of approximately 10,000 emitted pulses, this corresponds to about:

- 9,048 vacuum pulses,
- 905 one-photon pulses,
- 47 multi-photon pulses.

This produces very few multi-photon pulses, but most pulses are empty.

Example: $\mu = 0.5$

$$\begin{aligned} P_0 &= e^{-0.5} \approx 0.60653, \\ P_1 &= 0.5e^{-0.5} \approx 0.30327, \\ P_{\geq 2} &\approx 0.09020. \end{aligned}$$

Out of approximately 10,000 emitted pulses, this corresponds to about:

- 6,065 vacuum pulses,
- 3,033 one-photon pulses,
- 902 multi-photon pulses.

Compared with $\mu = 0.1$, more one-photon events are available, but the multi-photon fraction is also much larger.

Notebook operating point: $\mu = 0.15$

$$P_0 \approx 0.86071, P_1 \approx 0.12911, P_{\geq 2} \approx 0.01019.$$

So approximately **86.1%** of the pulses are vacuum, **12.9%** contain one photon, and **1.02%** contain two or more photons.

This trade-off is why μ is an important optimization parameter: making μ too small reduces useful detections, while making it too large increases the multi-photon fraction.

These are source probabilities before channel loss. Then channel efficiency is applied to determine which events can actually be detected by Bob. η

Vacuum, one-photon and multiphoton probabilities are evaluated at three source intensities. Their probability sum is checked at the working intensity.

Example calculation

Poisson probabilities evaluated numerically

Table 6. Poisson probabilities evaluated numerically

μ	P(0)	P(1)	P(2)	P(3)	P($n \geq 2$)
0.10	0.904837	0.090484	0.004524	0.000151	0.004679
0.15	0.860708	0.129106	0.009683	0.000484	0.010186
0.50	0.606531	0.303265	0.075816	0.012636	0.090204

Operating point used below: $\mu = 0.15$
 P_0 vacuum = 0.860707976 (86.071%)
 P_1 one photon = 0.129106196 (12.911%)
 $P_{\geq 2}$ photons = 0.010185827 (1.019%)
 Probability sum = 1.000000000

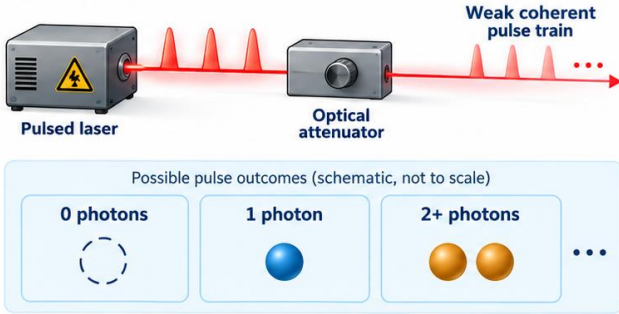
Weak Coherent Source and Poisson Photon Statistics

A heavily attenuated laser emits pulses with integer photon counts following a Poisson distribution.

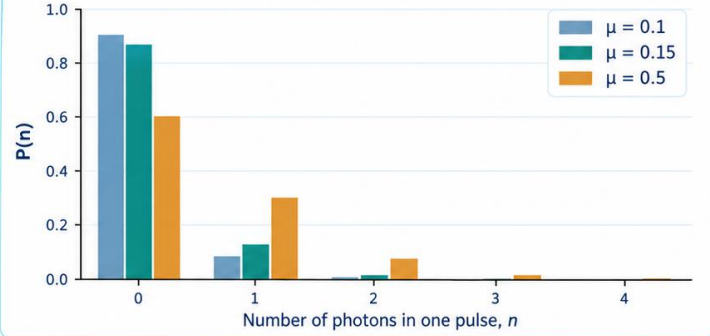
$$P(n) = e^{-\mu} \frac{\mu^n}{n!}$$

μ is the average photon number per pulse.

1 From laser to weak pulses



2 Photon-number distribution



3 Three photon-number classes

μ	$P_0 = e^{-\mu}$ (vacuum)	$P_1 = \mu e^{-\mu}$ (one photon)	$P_{\geq 2} = 1 - P_0 - P_1$ (two or more photons)
0.1	90.48%	9.05%	0.47%
0.15	86.07%	12.91%	1.02%
0.5	60.65%	30.33%	9.02%

Legend: Vacuum (0 photons) (light blue), One photon (1 photon) (dark blue), Two or more photons (2+ photons) (orange)

Source probabilities before channel loss.

Figure 10. Weak coherent source and Poisson photon-number statistics

From Source Probabilities to Detected Events

The source probabilities tell us what Alice emits. Bob sees only a fraction of those photons because the optical path, receiver and detector are imperfect.

Let η be the overall transmission-and-detection efficiency. For the one-photon source class, the detected signal probability is modeled as:

$$P_{\text{signal}} = \eta \mu e^{-\mu} = \eta P_1.$$

Using $\eta = 0.00346$ and $\mu = 0.15$:

$$P_{\text{signal}} \approx 0.00346 \times 0.129106 \approx 4.47 \times 10^{-4}.$$

So only about 0.0447% of gates contain a detected event from the one-photon source class at this operating point.

For the multi-photon source class, the simple approximation used here is:

$$P_{\text{multi}} = \eta(1 - e^{-\mu} - \mu e^{-\mu}) = \eta P_{\geq 2}.$$

For the same values:

$$P_{\text{multi}} \approx 3.52 \times 10^{-5}.$$

Now include dark/background events. The input d is the probability that **one detector** generates a dark click in one gate. With two independent detectors, the probability that at least one dark click occurs is:

$$P_d = 1 - (1 - d)^2.$$

For a daylight study, separate the detector's intrinsic dark-event probability p_{dark} from the light-background event probability p_{bg} **per detector per gate**. If the two are independent, enter their combined probability as

$$d = 1 - (1 - p_{\text{dark}})(1 - p_{\text{bg}}) \approx p_{\text{dark}} + p_{\text{bg}}$$

The approximation on the right is valid when both the intrinsic dark-event and light-background probabilities are small.

The baseline below uses $p_{bg} = 0$, so its d is intrinsic dark noise only. A measured daytime p_{bg} would depend on optical bandwidth, field of view, gate width, ambient radiance and filtering; this study does not derive it from solar irradiance.

With $d = 2 \times 10^{-6}$:

$$P_d \approx 4.0 \times 10^{-6}.$$

We now have three event classes for the QBER calculation: one-photon signal detections, approximate multiphoton detections, and dark/background events.

Detected single-photon, approximate multiphoton and dark terms

The two-detector dark union is calculated once. Single-photon and approximate multiphoton contributions are then multiplied by the total link efficiency.

Example calculation

Detected-event probabilities

$$P_{\text{dark}} = 3.999996000e-06$$

$$P_{\text{signal}} = 4.467074398e-04$$

$$P_{\text{multi approx}} = 3.524296180e-05$$

$$\text{Total event probability} = 4.859503976e-04$$

Device-Error Probability

Even when a real signal photon reaches Bob, the receiver can assign the wrong bit because of imperfect interference, phase drift, timing uncertainty, polarization effects or detector afterpulsing. Afterpulsing can matter for avalanche photodiodes, but its contribution depends on detector technology, count history and hold-off time; it is not universally the dominant QBER source.

We combine these contributions as:

$$e_{\text{device}} = e_{\text{vis}} + e_{\text{phase}} + e_{\text{jitter}} + e_{\text{pol}} + e_{\text{after}}.$$

Here e_{after} is only a placeholder error allowance, **not** an afterpulse probability or a time-correlated detector model. The pulse clock converts per-gate values into rates without a dead-time correction. A hardware prediction needs detector-specific afterpulse and recovery measurements; simply adding a measured P_{after} to e_{device} would generally be incorrect.

The visibility contribution is:

$$e_{\text{vis}} = \frac{1 - V}{2}.$$

For $V = 0.98$:

$$e_{\text{vis}} = \frac{1 - 0.98}{2} = 0.01.$$

So a visibility of 98% contributes a 1% error term in this model.

For Gaussian phase fluctuations with standard deviation σ_{phase} :

$$e_{\text{phase}} = \frac{1}{2} \left(1 - e^{-\sigma_{\text{phase}}^2/2} \right).$$

With $\sigma_{\text{phase}} = 0.04$ rad:

$$e_{\text{phase}} \approx 3.9984 \times 10^{-4}.$$

In the baseline example, jitter, polarization and afterpulse contributions are initially set to zero so that the arithmetic remains easy to follow. Therefore:

$$e_{\text{device}} = 0.01 + 0.00039984 \approx 0.01039984.$$

Physical interferometer visibility must satisfy:

$$0 \leq V \leq 1.$$

Device-error budget

Visibility and phase-noise contributions are evaluated separately. Additional jitter, polarization and afterpulse QBER allowances are zero in this calculation.

Example calculation

Device-error components

$$e_{\text{visibility}} = 0.010000000$$

$$e_{\text{phase}} = 0.000399840$$

$$e_{\text{jitter}} = 0.000000000$$

$$e_{\text{polarization}} = 0.000000000$$

$e_{\text{afterpulse}} = 0.000000000$
 $e_{\text{device total}} = 0.010399840$ (1.040%)

DPS Bit QBER

The modeled bit-error rate is the probability of error events divided by the total probability of relevant detection events:

$$Q_{\text{bit}} = \frac{\frac{1}{2}P_d + e_{\text{device}}P_{\text{signal}} + \frac{1}{4}P_{\text{multi}}}{P_{\text{signal}} + P_d + P_{\text{multi}}}.$$

The numerator contains three error contributions:

- $\frac{1}{2}P_d$: dark/background events are random, so half are expected to have the wrong bit value;
- $e_{\text{device}}P_{\text{signal}}$: a fraction of legitimate signal detections are wrong because of receiver imperfections;
- $\frac{1}{4}P_{\text{multi}}$: this simplified model assigns a 25% error contribution to the multi-photon class.

For the current operating point:

$$\begin{aligned}\frac{1}{2}P_d &\approx 2.00 \times 10^{-6}, \\ e_{\text{device}}P_{\text{signal}} &\approx 4.65 \times 10^{-6}, \\ \frac{1}{4}P_{\text{multi}} &\approx 8.81 \times 10^{-6}.\end{aligned}$$

Dividing the total error probability by the total event probability gives approximately:

$$Q_{\text{bit}} \approx 0.03181 = 3.18\%.$$

This compact calculation is intentionally kept separate from the earlier two-detector click model because the assumptions are not identical.

Assemble the compact bit-error numerator

The dark, single-photon and multiphoton error terms are added and divided by the compact detection denominator. This gives the modeled bit QBER, distinct from physical click-conditioned QBER.

Example calculation

DPS QBER calculation
Dark error contribution = 1.999998000e-06
Signal error contribution = 4.645685919e-06
Multi error contribution = 8.810740451e-06
Error numerator = 1.545642437e-05
Event denominator = 4.859503976e-04
 $Q_{\text{bit}} = 0.031806589 = 3.181\%$

Yield and Photon-Number Gain

A **yield** answers the question:

If Alice emitted exactly n photons, what is the probability that Bob registers a detection?

For a vacuum pulse, the only event in this simplified model is a dark/background detection:

$$Y_0 = P_d.$$

For $n \geq 1$ photons:

$$Y_n = 1 - (1 - \eta)^n.$$

This is easy to interpret. Each photon independently has probability η of being detected. The probability that **none** of the n photons is detected is $(1 - \eta)^n$, so the probability of at least one detection is its complement.

For example, if $\eta = 0.1$:

$$\begin{aligned}Y_1 &= 1 - 0.9 = 0.1, \\ Y_2 &= 1 - 0.9^2 = 0.19, \\ Y_3 &= 1 - 0.9^3 = 0.271.\end{aligned}$$

A pulse with more photons therefore has more chances to trigger a threshold detector.

The Poisson probability of emitting n photons is:

$$p_n = e^{-\mu} \frac{\mu^n}{n!}.$$

The **gain** of the n -photon class is:

$$G_n = p_n Y_n.$$

This combines two probabilities:

1. Alice emits an n -photon pulse;
2. Bob detects it.

The main gain classes are:

$$G_0 = e^{-\mu} P_d,$$

$$G_1 = \mu e^{-\mu} \eta,$$

and

$$G_{\geq 2} = \sum_{n=2}^{\infty} p_n [1 - (1 - \eta)^n].$$

The multi-photon sum can be evaluated efficiently as:

$$G_{\geq 2} = 1 - e^{-\mu\eta} - G_1.$$

For the current sample point:

$$G_0 \approx 3.44 \times 10^{-6},$$

$$G_1 \approx 4.47 \times 10^{-4},$$

$$G_{\geq 2} \approx 7.22 \times 10^{-5}.$$

This exact multi-photon gain is larger than the simple approximation $\eta P_{\geq 2}$ because a pulse containing two or more photons has multiple chances to cause a detection.

The table evaluates $n = 0$ through 8 using the photon-number probability, yield and their product. The summed multiphoton gain is compared with its closed form and with the separate simplified multiphoton approximation.

Example calculation

Photon-number gains and closed-form comparison

Table 7. Photon-number gains and closed-form comparison

n	Photon probability	Yield	Photon-number gain
0	8.607079764e-01	3.999996000e-06	3.442828463e-06
1	1.291061965e-01	3.460000000e-03	4.467074398e-04
2	9.682964735e-03	6.908028400e-03	6.689019538e-05
3	4.841482367e-04	1.034412662e-02	5.008090665e-06
4	1.815555888e-05	1.376833594e-02	2.499718339e-07
5	5.446667663e-07	1.718069750e-02	9.357754951e-09
6	1.361666916e-08	2.058125229e-02	2.802481033e-10
7	2.917857677e-10	2.397004115e-02	6.994116860e-12
8	5.470983144e-12	2.734710481e-02	1.496155495e-13

Closed-form gains

$$G_0 = 3.442828463e-06$$

$$G_1 = 4.467074398e-04$$

$$G_{\geq 2} \text{ exact} = 7.215790303e-05$$

$$G \text{ total} = 5.223081713e-04$$

$$G_{\geq 2} \text{ sum } n=2..8 = 7.215790303e-05$$

$$\text{Simple } P_{\text{multi}} \text{ approximation} = 3.524296180e-05$$

Phase-Error Proxy

For the modeled SKR calculation, we also build a phase-error proxy from the vacuum, one-photon and multi-photon gain classes:

$$Q_{ph} = \frac{\frac{1}{2} G_0 + G_1 Q_{bit} + \frac{1}{4} G_{\geq 2}}{G},$$

where:

$$G = G_0 + G_1 + G_{\geq 2}.$$

The terms have the following interpretation:

- Vacuum/background events are treated as random and weighted by 1/2;
- The one-photon class uses the modeled bit error Q_{bit} ;
- The multi-photon class is weighted by 1/4 in this teaching model.

For the current operating point:

$$Q_{ph} \approx 0.06504 = 6.50\%.$$

This is a **model quantity used inside the rate calculation**. It is not a complete finite-key phase-error bound for a deployed DPS system.

The 1/4 weighting assigned to the multiphoton class and the resulting phase-error proxy are modeling assumptions adopted for the comparative engineering proxy in this project; they are not presented as a protocol security bound derived from a complete DPS-QKD security proof.

Assemble the phase-error proxy

Vacuum, single-photon and multiphoton contributions are pooled using the total compact gain. This is the phase-error quantity adopted by the teaching rate formula.

Example calculation

```
Phase-error proxy
0.5*G0      = 1.721414231e-06
G1*Q_bit    = 1.420823976e-05
0.25*G>=2   = 1.803947576e-05
G total     = 5.223081713e-04
Q_phase     = 0.065036566 = 6.504%
```

Binary Entropy and Modeled SKR

The binary entropy function is:

$$h_2(x) = -x \log_2 x - (1-x) \log_2 (1-x).$$

It converts an error probability into an information penalty. A perfectly predictable binary variable has entropy 0, while a completely random binary variable with probability 0.5 has entropy 1 bit.

The modeled per-gate key-rate proxy is:

$$r = \frac{1}{2} [G_1(1 - h_2(Q_{ph})) - G f_{EC} h_2(Q_{bit})].$$

The two main terms are:

- **Privacy term:** $G_1(1 - h_2(Q_{ph}))$,
- **Error-correction cost:** $G f_{EC} h_2(Q_{bit})$.

The factor $f_{EC} > 1$ represents practical error correction being less efficient than the Shannon limit.

If the expression becomes negative, this study clips it to zero because a negative usable key rate has no physical meaning.

For the current operating point:

$$r \approx 8.42 \times 10^{-5} \text{ modeled bits/gate.}$$

At a 10 MHz usable-gate rate:

$$R = 10^7 r \approx 842 \text{ modeled bit/s.}$$

This value is used as a **performance proxy** in the later FSO and satellite studies.

Entropy and modeled-rate evaluation

Binary entropy is evaluated with continuous endpoint handling. The privacy contribution and reconciliation penalty are shown separately before the factor one half and zero clipping are applied.

Example calculation

```
Modeled SKR proxy
h2(Q_phase)  = 0.347121921
h2(Q_bit)    = 0.203372434
privacy term  = 2.916454952e-04
EC cost term  = 1.232187775e-04
unclipped rate = 8.421335886e-05 modeled bits/gate
clipped rate  = 8.421335886e-05 modeled bits/gate
at 10 MHz     = 842.1 modeled bit/s
```

Reusable DPS QBER/SKR Function

The next function does not introduce new physics. It simply packages the calculations above so that later sections can pass arrays of η , μ or visibility values and efficiently generate QBER and modeled-rate curves. The function returns intermediate quantities as well as the final outputs so the calculation remains easy to inspect.

Reusable compact-model function

The implementation broadcasts efficiency and source/device inputs, forms all photon-number terms, and returns bit QBER, phase proxy, component gains, total gain and modeled rate per gate. The function is the shared numerical definition for later studies.

Example calculation

Reusable function check at the same example point

```
qber      0.0318065886
phase_proxy 0.0650365658
gain      0.000522308171
rate_per_gate 8.42133589e-05
```

Approximate and Exact Multi-Photon Detection

Earlier we used the simple approximation:

$$P_{multi} = \eta P_{\geq 2}.$$

This treats every multi-photon pulse as though it had only one opportunity to be detected.

A more explicit threshold-detector model uses:

$$Y_n = 1 - (1 - \eta)^n,$$

so an n -photon pulse has multiple opportunities to trigger the detector.

The difference is small when both η and μ are very small, but it becomes more visible as either quantity increases.

The next plot compares:

- The exact multi-photon gain,
- The simple approximate multi-photon term,
- The earlier two-detector click QBER,
- The compact DPS QBER/SKR model.

The two QBER curves do not need to be identical because they were built from different assumptions.

Check the multiphoton approximation

The comparison varies mean photon number and total efficiency. The exact signal multiphoton gain used in the gain partition need not equal the approximation used in the compact bit-error numerator.

Example calculation

eta=0.00346: at mu=0.5 exact $G_{\geq 2} = 0.000679$, simple approximation = 0.000312

eta=0.1: at mu=0.5 exact $G_{\geq 2} = 0.018444$, simple approximation = 0.009020

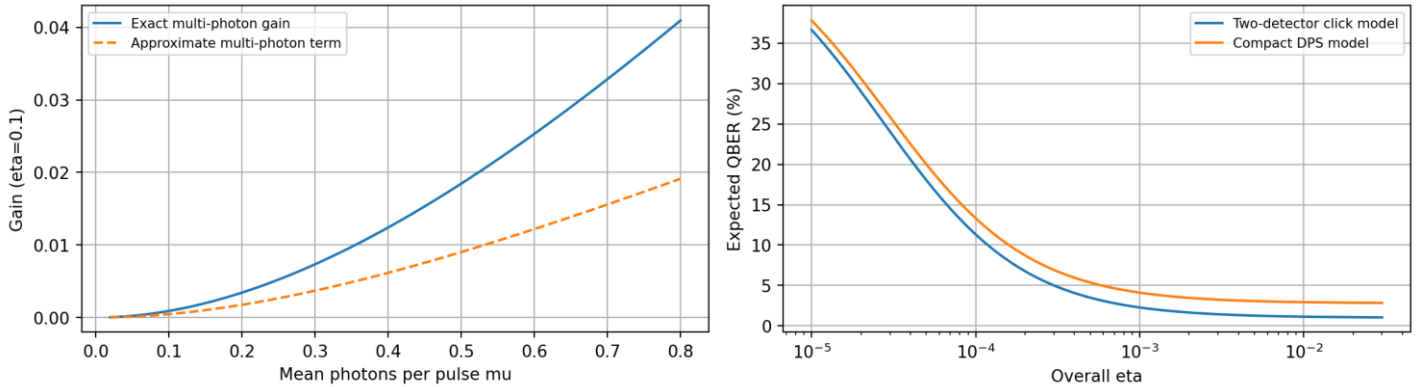


Figure 11. Exact versus approximate multiphoton gain and comparison of the two QBER models

Free-Space Optical Channel Model

We now connect the DPS receiver model to the free-space optical (FSO) channel model used for the terrestrial study.

The calculation follows this chain:

distance -> beam radius -> geometric collection -> atmospheric transmission ->
pointing/device losses -> total efficiency -> DPS QBER and modeled SKR

For the 20 km baseline we use:

Table 8. Baseline 20 km terrestrial FSO parameters.



Parameter	Value
Distance L	20 km
Initial beam radius w_0	0.05 m
Beam divergence θ	50 microrad
Receiver radius a	0.10 m
Atmospheric attenuation	0.1 dB/km
Pointing loss	1 dB
Receiver optics efficiency	0.7
Detector efficiency	0.5
DPS mean photons μ	0.15

These are **editable educational assumptions**, not measured Bahrain-KSA weather or hardware data.

Beam Expansion and Geometric Collection

We use a centered Gaussian-beam approximation:

$$w(L) = \sqrt{w_0^2 + (\theta L)^2}.$$

In this report, $w(L)$ denotes the $1/e^2$ intensity radius of the Gaussian beam at propagation distance L .
At $L = 20,000$ m:

$$\theta L = (50 \times 10^{-6})(20,000) = 1.0 \text{ m},$$

so

$$w(20 \text{ km}) \approx 1.00125 \text{ m}.$$

For a centered circular receiver of radius a , the collected power fraction is:

$$\eta_{geo} = 1 - \exp\left(-\frac{2a^2}{w(L)^2}\right).$$

With $a = 0.10$ m:

$$\eta_{geo} \approx 0.01975.$$

So only about **1.98%** of the Gaussian beam power is collected by this aperture in the simplified centered-beam model.

Geometric link calculation

The Gaussian beam radius and collection fraction are calculated first. Distance is converted to metres, divergence is a half-angle in radians, and aperture denotes receiver radius.

Example calculation

20 km beam and aperture calculation
 $\theta * L = 1.000000 \text{ m}$
 beam radius $w(L) = 1.001249 \text{ m}$
 receiver radius $a = 0.100 \text{ m}$
 geometric collection $= 0.019752438 = 1.975\%$

Atmospheric, Pointing, Optical and Detector Losses

Atmospheric attenuation is expressed in dB/km. For coefficient α and distance L_{km} :

$$Loss_{atm,dB} = \alpha L_{km}.$$

With $\alpha = 0.1$ dB/km and $L = 20$ km:

$$Loss_{atm,dB} = 2 \text{ dB}.$$

The value of α is a chosen nominal scenario, not one derived from local weather. An optional visibility-based Kim/Kruse estimate could supply an alternative α , subject to wavelength and empirical-model validity; measured attenuation is preferable for a site-specific result.

Convert a dB loss to linear transmission using:

$$\eta_{atm} = 10^{-Loss_{dB}/10}.$$

Therefore:

$$\eta_{atm} = 10^{-2/10} \approx 0.63096.$$

Similarly, a 1 dB pointing loss gives:

$$\eta_{point} = 10^{-1/10} \approx 0.79433.$$

This fixed allowance approximates average tracking/pointing impairment; it is **not** a time-varying beam-wander simulation. In an optional jitter model, zero-mean independent Gaussian horizontal and vertical displacements x, y with common standard deviation σ_s produce a Rayleigh-distributed radius $r = \sqrt{x^2 + y^2}$. The received aperture coupling would then be evaluated at each r , alongside the turbulence fade. If this is added, replace the fixed pointing allowance or calibrate it for other residual losses to avoid counting the same jitter twice.

The total efficiency is then:

$$\eta = \eta_{geo}\eta_{atm}\eta_{point}\eta_{opt}\eta_{det}.$$

For our values:

$$\eta \approx 0.003465.$$

That means roughly **0.3465%** of the transmitted optical-photon opportunity survives all modeled collection and device factors.

Complete the 20 km loss budget

Atmospheric, pointing, optical and detector factors are multiplied exactly once. The printed stepwise budget makes the distinction between geometric collection and total detection efficiency explicit.

Example calculation

```
20 km multiplicative loss budget
geometric collection eta_geo = 0.019752438
atmosphere: 2.0 dB -> 0.630957344
pointing: 1.0 dB -> 0.794328235
optics efficiency      = 0.700
detector efficiency    = 0.500
overall eta            = 0.003464884 = 0.3465%
```

Terrestrial DPS-QKD Study

Feed the FSO Efficiency into the DPS Receiver

The link budget gives η . The DPS model then uses:

$$\lambda = \mu\eta$$

as the mean detected signal photon number per gate.

For $\mu = 0.15$ and $\eta \approx 0.003465$:

$$\lambda \approx 5.20 \times 10^{-4}.$$

We now calculate two different outputs:

- **two-detector physical model:** raw detection probability and click-conditioned QBER;
- **compact DPS model:** QBER and SKR proxy.

They need not match exactly because they make different approximations. Keeping both is useful for model comparison.

Calculate the 20 km operating point

The exact optical efficiency is passed independently to the physical two-detector model and compact DPS model. Their gains and QBER definitions remain separate.

Example calculation

```
20 km DPS example
lambda = mu*eta      = 5.197326560e-04
physical-model gain/gate = 5.235955360e-04
physical-model raw detections/s = 5,236.0
physical-model QBER   = 1.374%
modeled QBER          = 3.180%
modeled proxy/gate    = 8.434848888e-05
modeled proxy bit/s    = 843.5
```

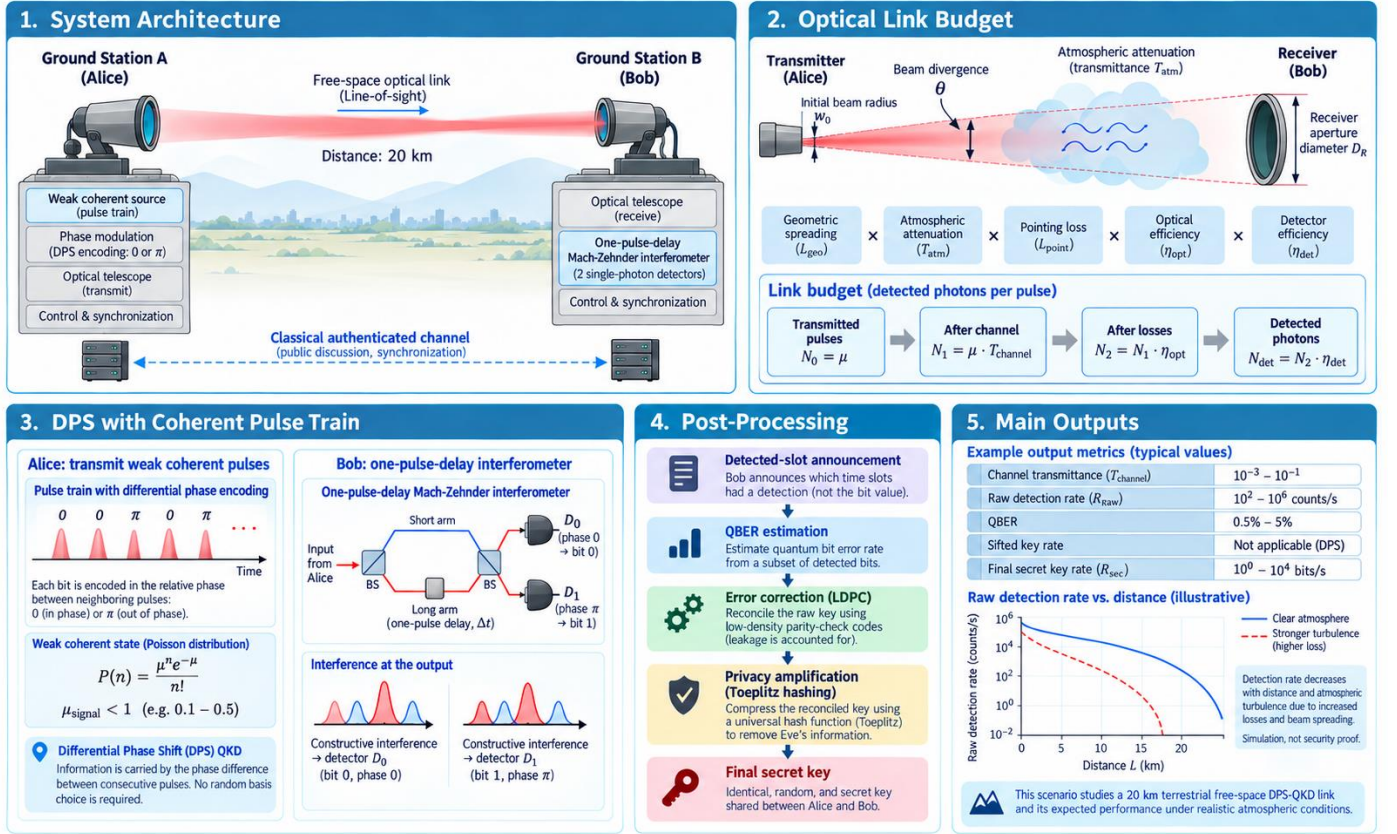


Figure 12. 20 km terrestrial DPS-QKD architecture, optical link budget and post-processing flow

Sweep Distance Instead of Evaluating Only One Point

A single 20 km number does not show the trend. We therefore evaluate the same optical model from 1 km to 50 km.

For every distance the notebook recalculates:

1. Beam radius,
2. Geometric collection,
3. Atmospheric attenuation,
4. Total η ,
5. Detector gain, and
6. Detector QBER.

The vertical line marks the 20 km example used above.

Ground distance sweep

A 250-point grid from 1 to 50 km illustrates changing raw detections and physical QBER at fixed source/device settings. The marker identifies the 20 km baseline.

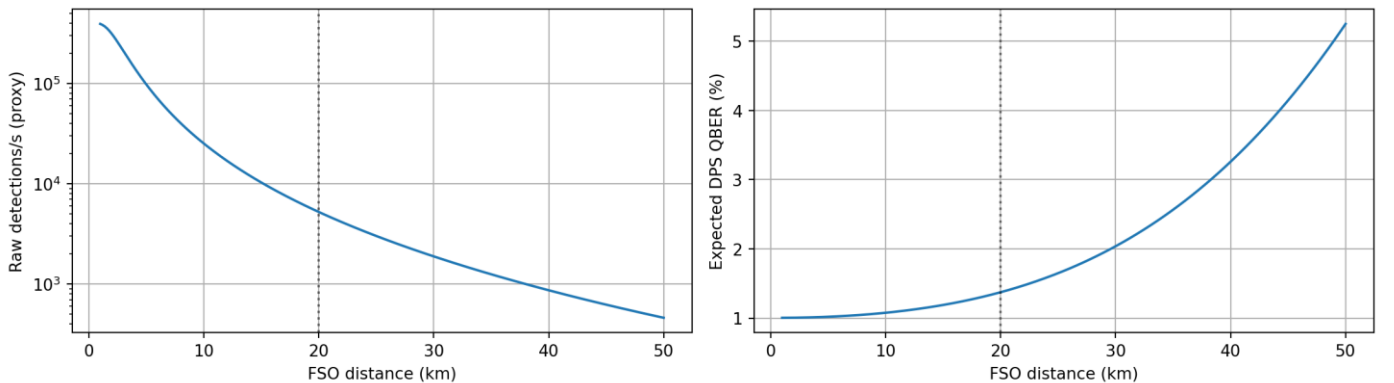


Figure 13. Ground-link distance sweep of raw detection rate and physical QBER

Distance, Visibility and Photon-Number Sweeps

Question 1: What happens if Alice changes μ ?

At very low μ , signal detections become scarce and dark/background events become more important. At larger μ , multi-photon terms grow. The model therefore has a parameter-dependent operating region rather than one universally optimal μ .

Question 2: What happens if interferometer visibility falls?

From

$$e_{vis} = \frac{1 - V}{2},$$

lower V directly increases the device-error contribution and therefore increases QBER and reduces the modeled SKR proxy.

The following cell sweeps both effects. All values remain simulation outputs under the declared assumptions.

Intensity and visibility sensitivity

The plotted distance curves compare $\mu = 0.08, 0.15, 0.30$ and 0.50 . The printed visibility study isolates the effect of interferometer quality at the 20 km point.

Example calculation

Visibility sensitivity at 20 km, $\mu=0.15$

$V=0.900$: QBER 6.86%, modeled SKR proxy 119 modeled bit/s

$V=0.950$: QBER 4.56%, modeled SKR proxy 551 modeled bit/s

$V=0.980$: QBER 3.18%, modeled SKR proxy 843 modeled bit/s

$V=0.995$: QBER 2.49%, modeled SKR proxy 1002 modeled bit/s

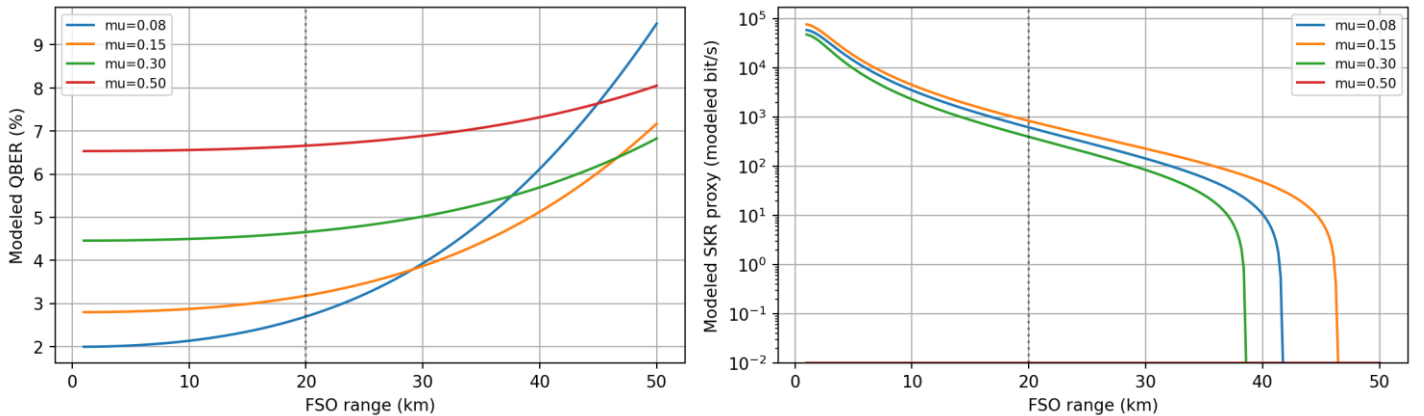


Figure 14. Modeled QBER and SKR sensitivity to distance and mean photon number

The logarithmic plot floors zero or very small rates at 0.01 for display only.

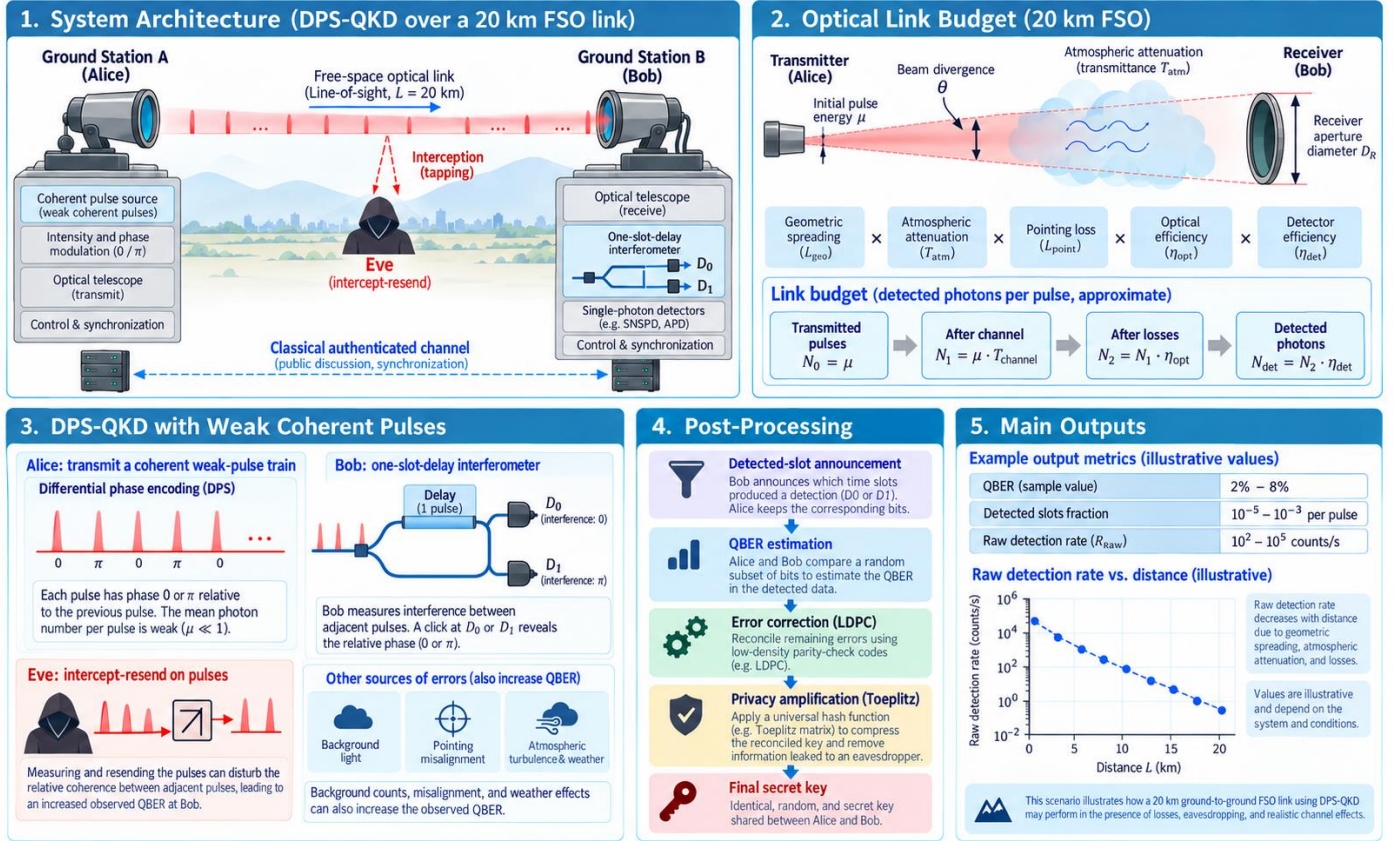


Figure 15. DPS-QKD over a 20 km FSO link: architecture, impairments, post-processing and outputs

Atmospheric Turbulence and Fading

Independent Fading and Detection-Weighted Statistics

So far, η was constant. A real FSO channel fluctuates with atmospheric turbulence.

As a first teaching step, let a positive random factor F multiply the static 20 km efficiency:

$$\eta_i = \min(\eta_{20} F_i, 1).$$

We use a log-normal power factor:

$$\ln F \sim \left(-\frac{\sigma_{log}^2}{2}, \sigma_{log}^2 \right).$$

The shift $-\sigma_{log}^2/2$ is chosen so that:

$$E[F] = 1.$$

Therefore the turbulence factor changes the **distribution** of transmission without intentionally changing its mean before clipping.

Why QBER Must Be Detection-Weighted

Suppose one weak fade produces almost no detections but a high QBER, while a strong fade produces many detections and low QBER. Giving both time bins equal weight would not represent the error fraction among the actually detected bits.

The pooled QBER is therefore:

$$Q_{pooled} = \frac{\sum_i G_i Q_i}{\sum_i G_i},$$

where G_i is the detection probability in time bin i .



Seed 8081 generates a small set of independently drawn log-normal power factors. Each is applied before recalculating detector gain and physical QBER. The two final error averages demonstrate the importance of detection weights.

Example calculation

Twelve-sample fading calculation

Table 9. Twelve-sample fading calculation

Bin	Fade F	Total η	Detections/s	Physical QBER
0	0.7321	2.536786e-03	3844.4	1.510%
1	1.4180	4.913292e-03	7407.2	1.265%
2	0.5226	1.810904e-03	2756.0	1.711%
3	1.3960	4.836886e-03	7292.7	1.269%
4	1.2687	4.395768e-03	6631.5	1.295%
5	0.9330	3.232734e-03	4887.9	1.401%
6	1.1199	3.880361e-03	5858.8	1.334%
7	1.1323	3.923186e-03	5923.0	1.331%
8	1.0133	3.510836e-03	5304.8	1.369%
9	1.3199	4.573193e-03	6897.4	1.284%
10	1.6491	5.713847e-03	8607.1	1.228%
11	1.4560	5.044739e-03	7604.2	1.258%

Arithmetic mean of per-bin QBER : 1.355%

Detection-weighted pooled QBER : 1.322%

Larger Fading Sample

The 12-row example explains the arithmetic. We now use 2,000 independent fading samples to obtain a smoother estimate.

This is still a toy fading experiment: it has no measured , weather series, scintillation spectrum or temporal correlation.

Two-thousand-sample fading calculation

An independent seeded sample gives a smoother view of efficiency and QBER distributions. This marginal fading experiment has no temporal correlation.

Example calculation

Average raw detections/s with fading: 5,298

Detection-weighted QBER: 1.37%

Mean fade factor before clipping: 1.0120

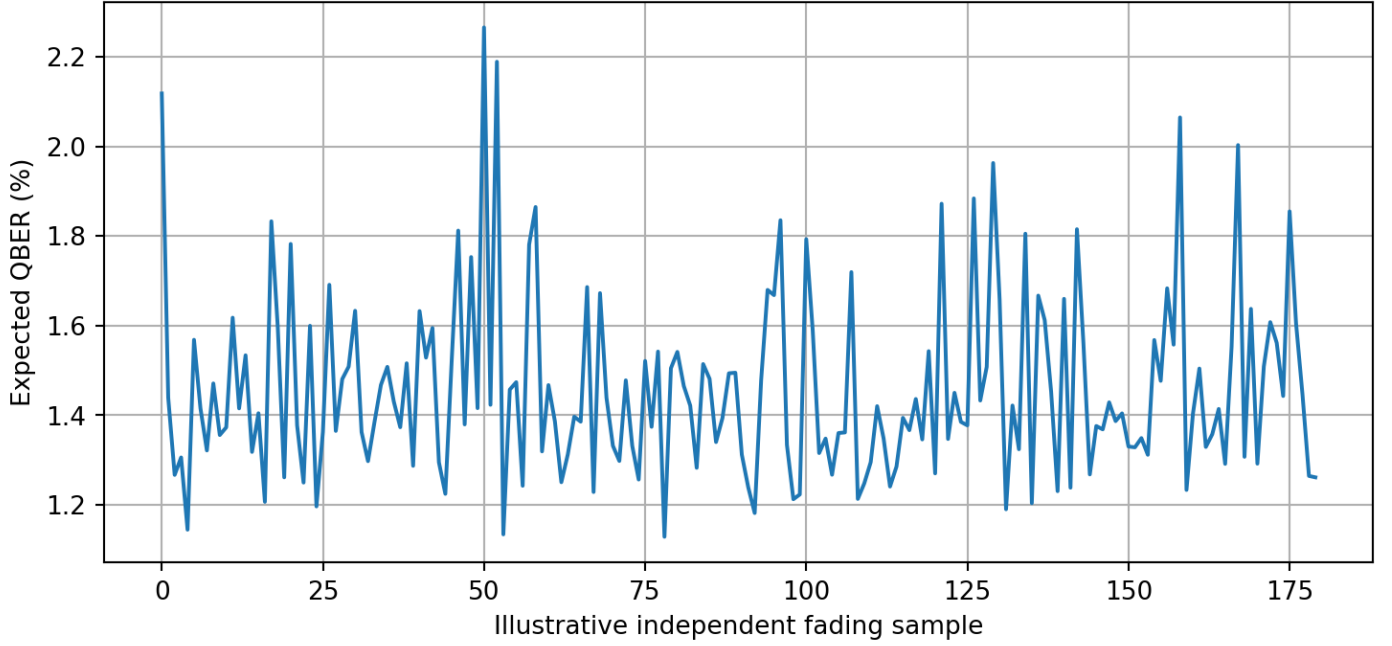


Figure 16. Independent log-normal fading sample and resulting physical QBER variation

Log-Normal Turbulence and Gauss-Hermite Integration

Weak-to-moderate atmospheric turbulence can be represented with a multiplicative log-normal transmission factor. The static FSO efficiency is multiplied by a random turbulence term. The random factor below represents turbulence alone; the separate 1 dB pointing term remains fixed in the baseline.

We transfer **the channel distribution only** into the DPS simulation:

$$\eta_{eq} = \eta_{static}\eta_t,$$

with

$$\ln\eta_t N\left(\frac{-\sigma_t^2}{2}, \sigma_t^2\right).$$

For every possible fade, we recompute the DPS detector and DPS equations.

Gauss-Hermite Quadrature for Log-Normal Fading

Atmospheric turbulence causes the received optical power to fluctuate with time.

Instead of assuming one fixed channel efficiency,

$$\eta = \eta_{static},$$

we model the instantaneous efficiency as:

$$\eta = \eta_{static}F$$

where:

- η_{static} is the FSO efficiency without turbulence
- F is a random multiplicative fading factor

For weak-to-moderate turbulence, F is commonly represented by a log-normal random variable.

We define:

$$F = \exp\left(\frac{-\sigma^2}{2} + \sigma Z\right)$$

where:

$$ZN(0,1)$$

and σ controls the strength of the fluctuations.

Why is $-\sigma^2/2$ included?

The factor

$$\frac{-\sigma^2}{2}$$

normalizes the fading distribution so that:

$$E[F] = 1.$$

This is useful because the turbulence model then represents fluctuations around the existing static link budget rather than adding another average attenuation term.

For example, if:

$$\eta_{static} = 0.003$$

and the instantaneous fade is:

$$F = 0.5,$$

then:

$$\eta = 0.003 \times 0.5 = 0.0015.$$

The received signal is temporarily weaker.

If instead:

$$F = 1.5,$$

then:

$$\eta = 0.003 \times 1.5 = 0.0045.$$

The instantaneous received signal is stronger than its long-term average.

Example with $\sigma = 0.45$

For this study we use:

$$\sigma = 0.45.$$

If $Z = 0$, which represents the center of the underlying Gaussian distribution:

$$F = \exp\left(-\frac{0.45^2}{2}\right)$$
$$F = e^{-0.10125} \approx 0.904.$$

Notice that the central fading value is slightly below 1.

This follows from the asymmetry of the log-normal distribution.

A log-normal distribution is asymmetric. Many samples are below 1, while a smaller number of large positive fades pull the overall average back to:

$$E[F] = 1.$$

Why do we need an average?

The DPS detector performance depends nonlinearly on channel efficiency.

For example:

$$G = G(\eta)$$

and:

$$Q = Q(\eta).$$

Since turbulence continuously changes η , we need quantities such as:

$$E[G]$$

rather than evaluating the detector only once at the average channel efficiency.

In principle:

$$E[g(F)] = \int_0^\infty g(F)p(F)dF.$$

Monte Carlo simulation estimates this integral by generating thousands of random fading samples.

Gauss-Hermite quadrature provides another approach: it replaces the continuous Gaussian distribution with a small number of carefully selected deterministic points.

Transforming the log-normal variable

Because:

$$F = \exp\left(\frac{-\sigma^2}{2} + \sigma Z\right)$$

with:

$$Z \sim N(0,1),$$

an expectation can be written as:

$$E[g(F)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(e^{-\sigma^2/2 + \sigma z}) e^{-z^2/2} dz.$$

Gauss-Hermite quadrature is defined using the weighting function:



$$e^{-x^2}.$$

Using the substitution:

$$z = \sqrt{2}x,$$

the expectation becomes approximately:

$$E[g(F)] \approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^N w_i g(F_i),$$

where:

$$F_i = \exp\left(\frac{-\sigma^2}{2} + \sqrt{2}\sigma x_i\right).$$

Here:

- x_i are the Gauss-Hermite nodes
- w_i are the Gauss-Hermite weights
- F_i is the fading factor represented by that node

This explains the two apparently unusual terms in the Python code:

`np.sqrt(2) * sigma * nodes`

`weights / np.sqrt(np.pi)`

Each row shows a Hermite node, equivalent standard-normal coordinate, transformed fading factor, normalized weight and weighted contribution to the mean. The weights and unit-mean fading factor are checked.

Example calculation

Five-node quadrature worked table

Table 10. Five-node quadrature worked table

Node	GH x	Equivalent Z	Fade F	Weight	Weight × F
0	-2.020183	-2.856970	0.249852	0.011257411	0.002812687
1	-0.958572	-1.355626	0.491014	0.222075922	0.109042480
2	0.000000	0.000000	0.903707	0.533333333	0.481977108
3	0.958572	1.355626	1.663264	0.222075922	0.369370829
4	2.020183	2.856970	3.268681	0.011257411	0.036796885

Checks

Sum of normalized weights = 1.000000000

Weighted mean fade factor = 0.999999989

Expected theoretical mean = 1.000000000

48 nodes will later be used for the production calculation.

Visualizing the Gaussian Nodes

Before converting the nodes into atmospheric fading values, we can look at where they lie on the underlying standard-normal distribution.

The middle node represents the most important central region of the distribution.

The two outermost nodes represent rare extreme conditions and therefore receive very small quadrature weights.

Gaussian-node interpretation

The five deterministic integration points lie on the underlying standard-normal coordinate axis. They are not five equally likely atmospheric observations.

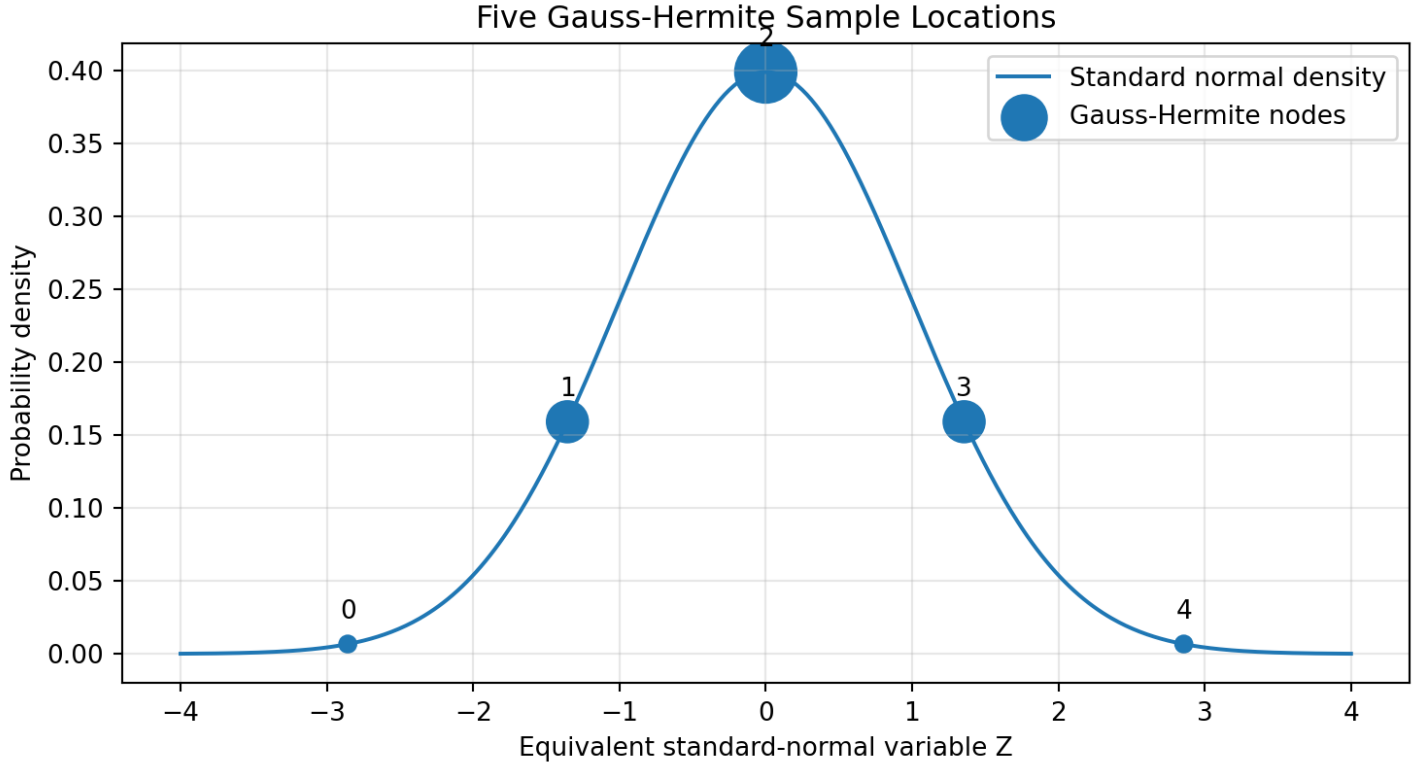


Figure 17. Five Gauss-Hermite sample locations on the equivalent Gaussian coordinate

From Gaussian Position to Fading Factor

The Gaussian nodes are converted into atmospheric fading factors using:

$$F_i = \exp\left(\frac{-\sigma^2}{2} + \sigma Z_i\right).$$

Values:

$$F < 1$$

represent weaker-than-average instantaneous transmission.

Values:

$$F > 1$$

represent stronger-than-average instantaneous transmission.

The marker size in the following plot represents the quadrature weight.

Transform the Gaussian coordinates

The exponential map produces an asymmetric set of positive fading factors. Marker size indicates the quadrature weight.

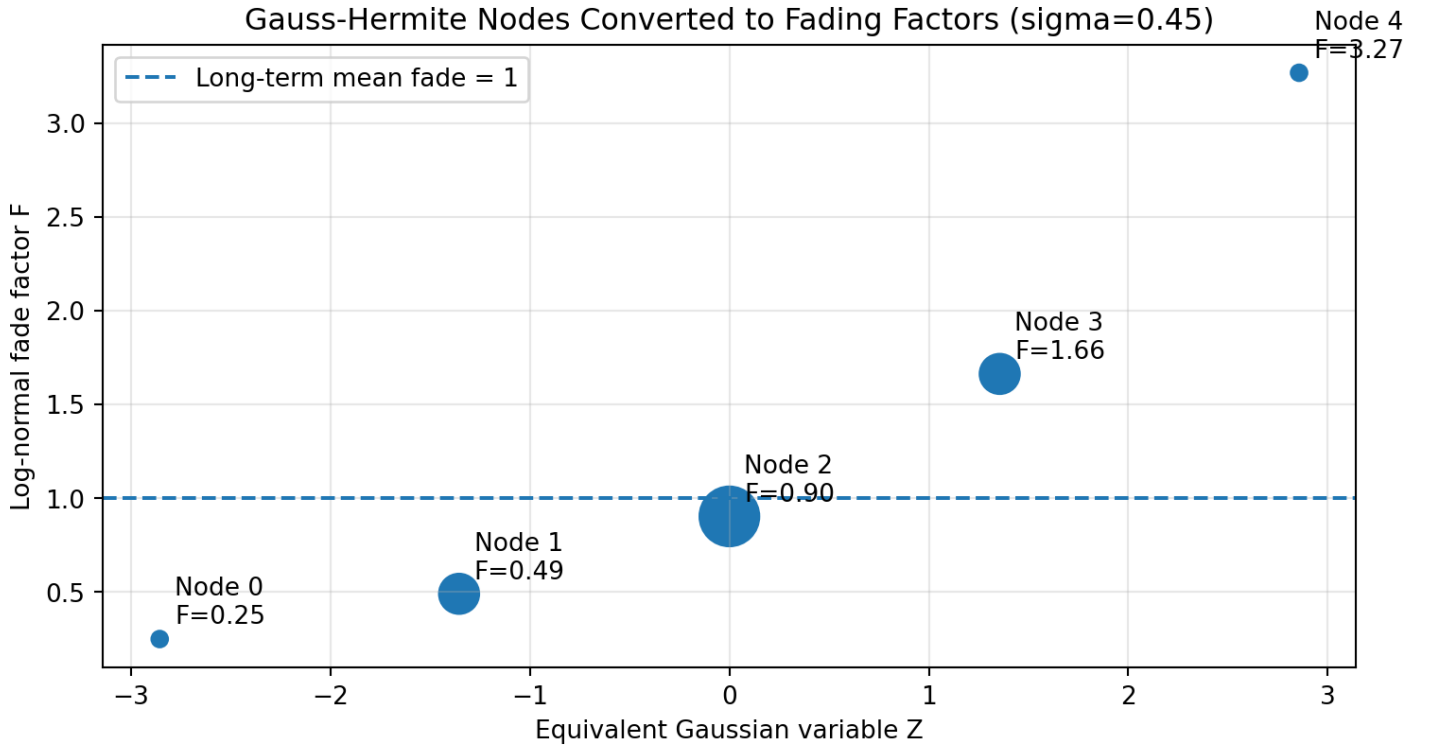


Figure 18. Gauss-Hermite nodes transformed into log-normal fading factors

Weighted Contribution of Each Node

Gauss-Hermite quadrature does not simply average the five fading factors.

Each fading state is multiplied by its quadrature weight:

$$C_i = w_i F_i.$$

The expected fading factor is then:

$$E[F] \approx \sum_i C_i.$$

For this example the result is almost exactly:

$$E[F] = 1.$$

Sum the weighted fading contributions

Multiplying each fading state by its normalized weight reconstructs the expected unit-mean power factor.

Example calculation

Total contribution = 0.9999999893283253

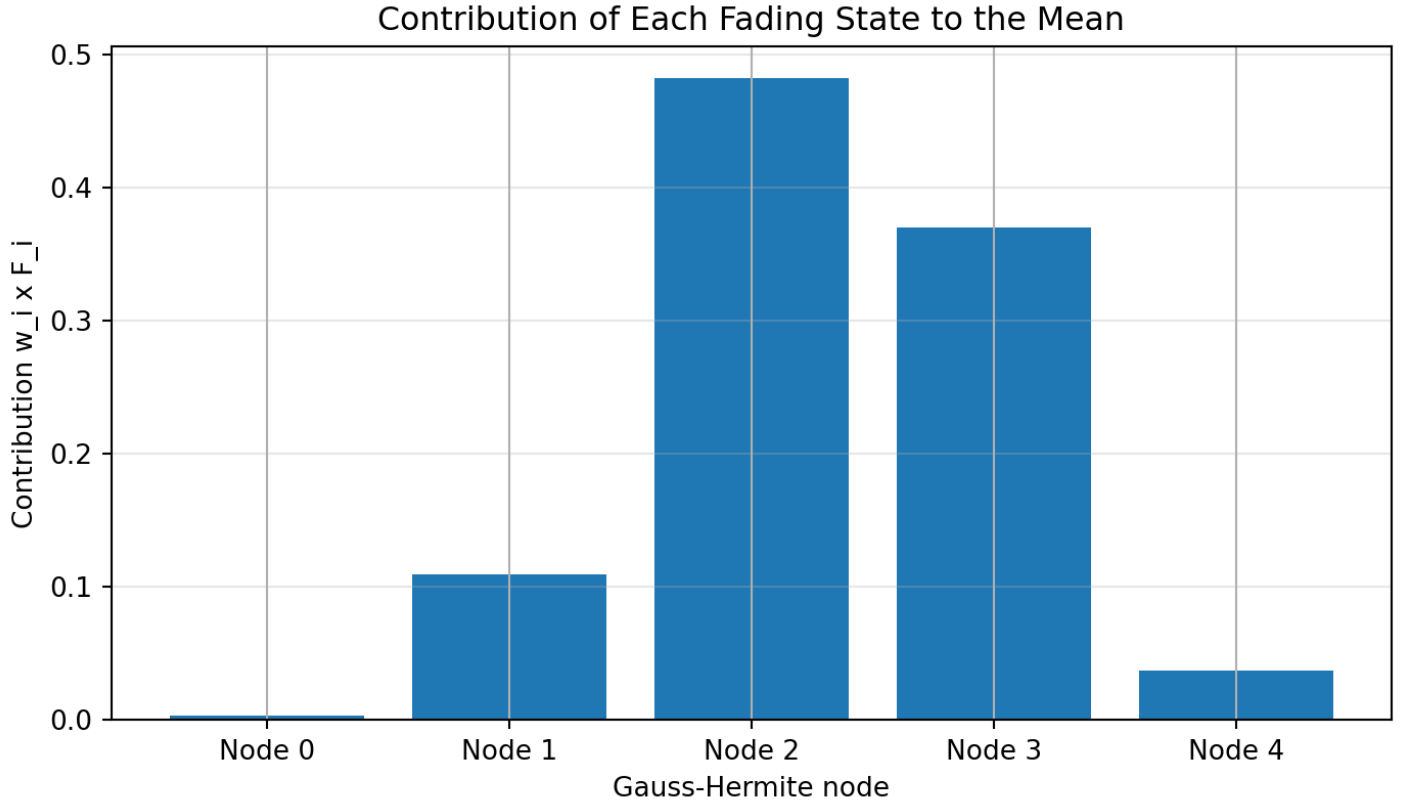


Figure 19. Weighted contribution of the five fading states to the unit-mean channel

Log-Normal Fading Distribution

The same model can also be viewed directly in the fading domain.

The distribution is asymmetric:

- Many fading states occur below $F = 1$
- Strong positive fading values are less common
- The long right-hand tail keeps the mean equal to 1

The Gauss-Hermite points provide a small deterministic representation of this continuous distribution.

Interpret the log-normal density

The central mass and right tail show why the median fading factor is below one while the mean is one.

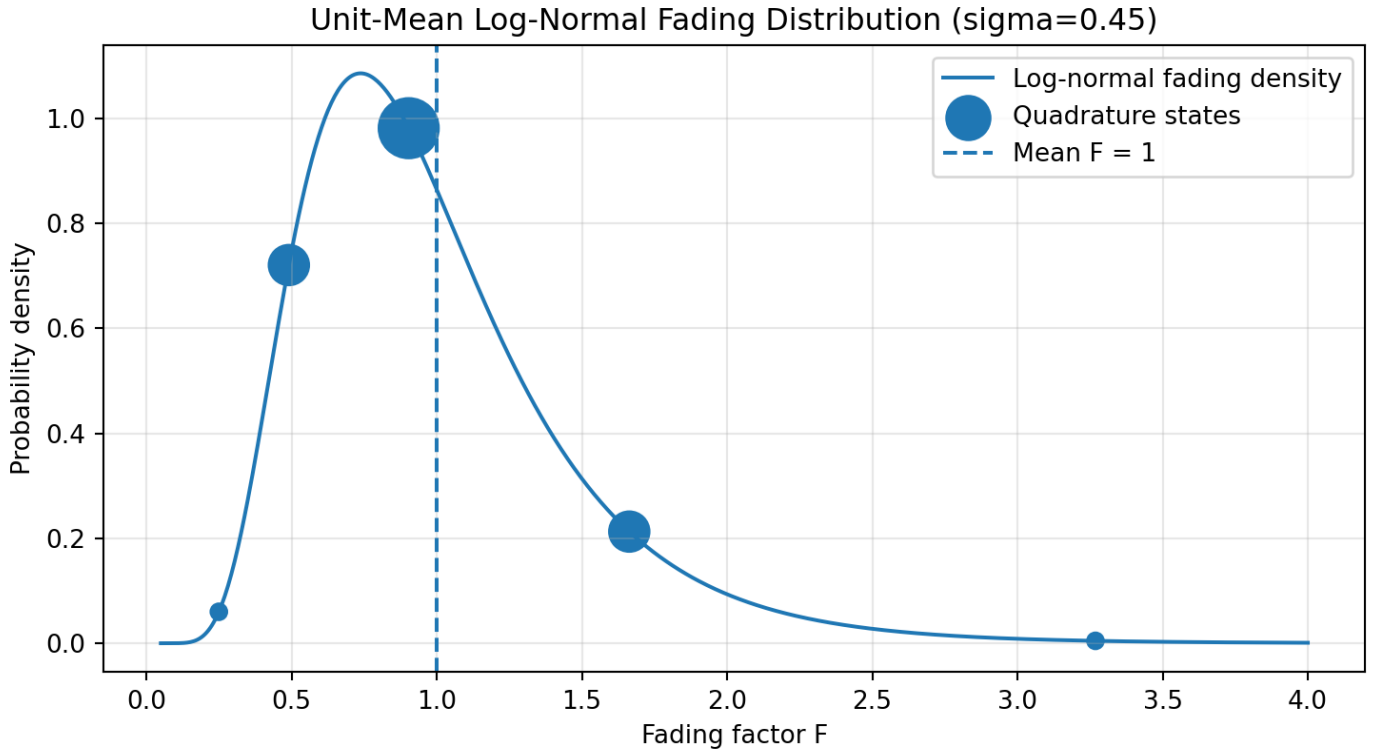


Figure 20. Unit-mean log-normal fading distribution and representative quadrature nodes

Convergence with Number of Gauss-Hermite Nodes

For the simple quantity $E[F]$, even a small number of Gauss-Hermite points provides a very accurate result. However, the DPS model contains nonlinear quantities such as:

- Detector click probability
- QBER
- Phase-error proxy
- Modeled SKR

These functions can change strongly with transmission.

For that reason, the production calculation uses more quadrature nodes.

The following test shows how the calculated mean fading factor approaches its theoretical value of 1 as the quadrature order increases.

Quadrature-order convergence

The test increases node count and compares the estimated mean fading factor with its exact value. It tests a simple known moment before applying quadrature to the DPS response.

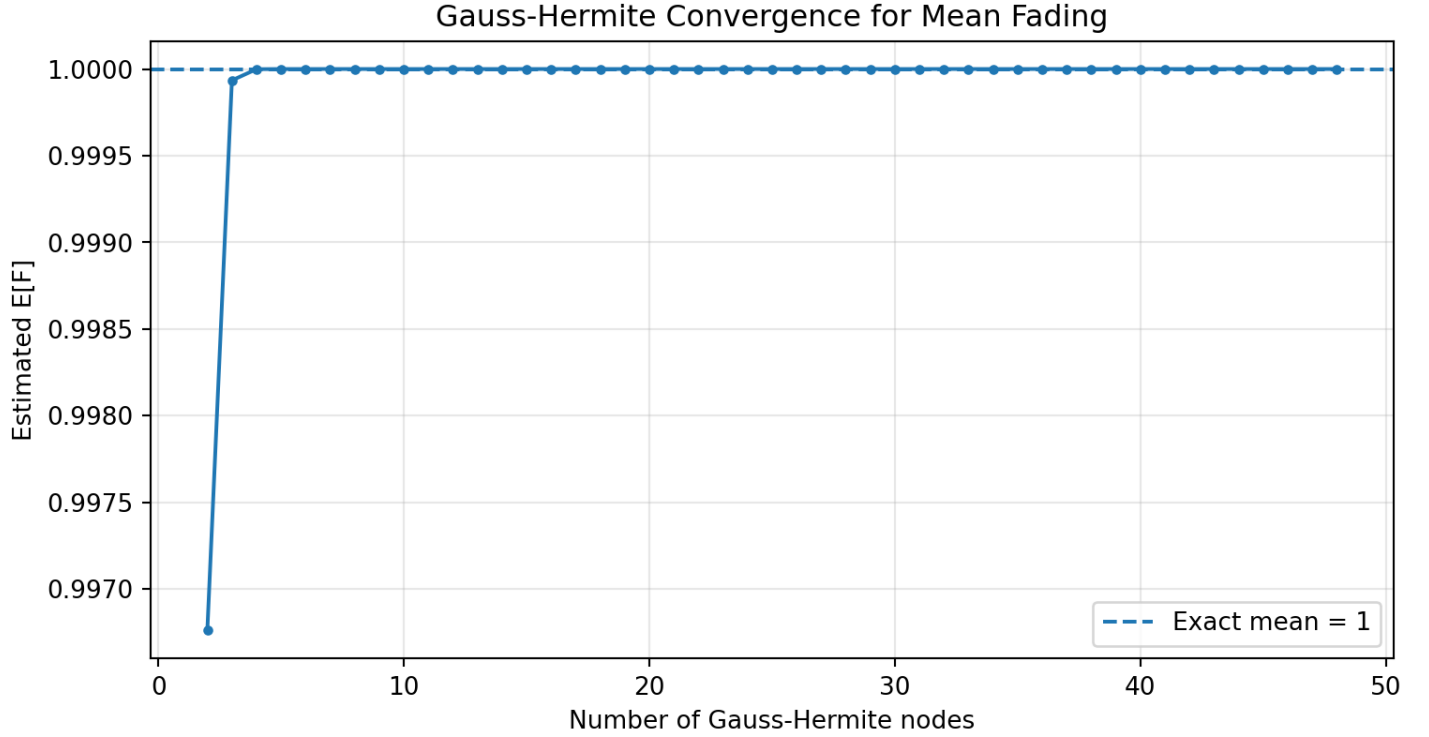


Figure 21. Gauss-Hermite convergence of the mean fading factor

Absolute convergence error

The second convergence figure exposes small integration errors on a logarithmic scale.

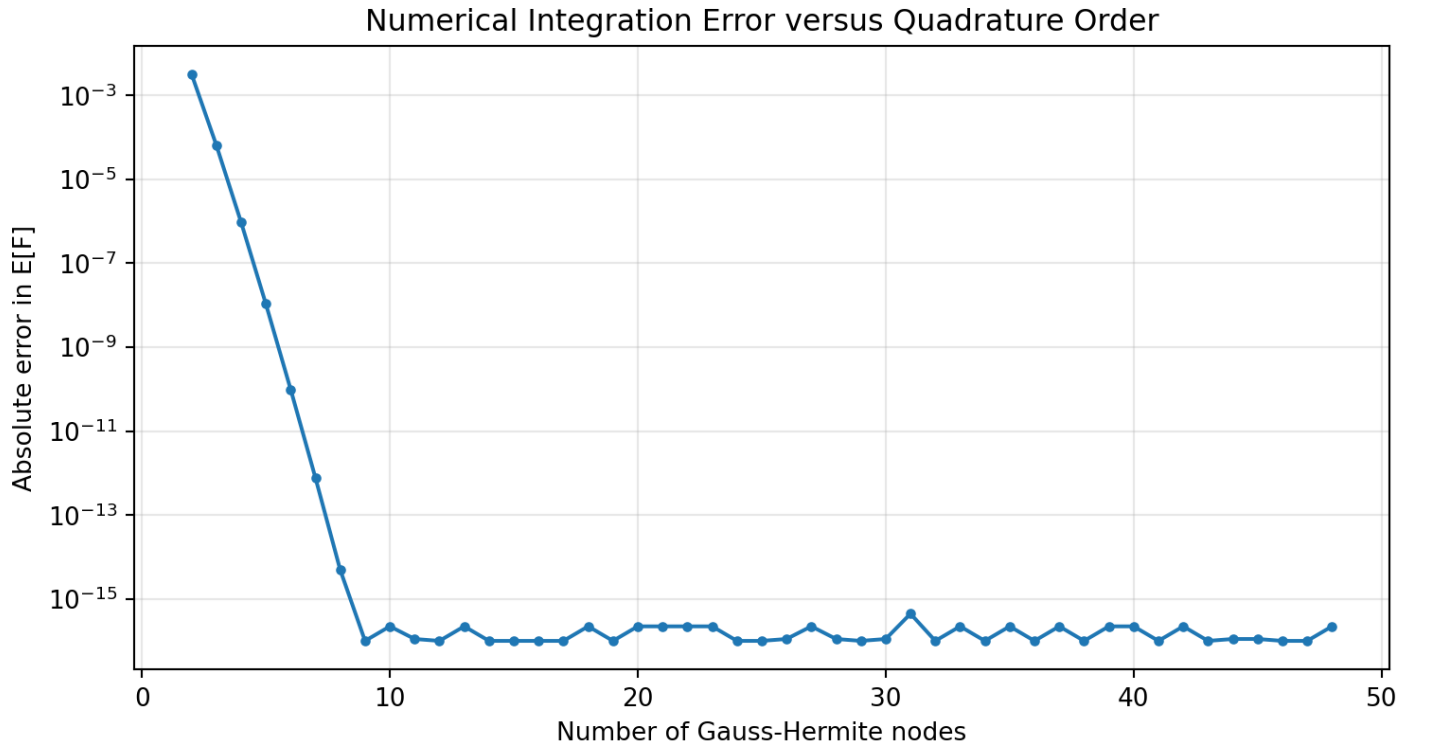


Figure 22. Absolute numerical-integration error versus Gauss-Hermite quadrature order

Applying the Five Fading States to the DPS Link

The purpose of Gauss-Hermite quadrature is not merely to reproduce the average fading factor.

For every fading state, the complete DPS channel model is recalculated.

At quadrature node i :

$$\eta_i = \eta_{static} F_i.$$

The DPS detector then calculates:

$$G_i = G(\eta_i)$$

and:

$$Q_i = Q(\eta_i).$$

This means that a weak fading state produces a lower received signal, while a strong fading state produces a larger detection probability.

Because the number of detections is different for each fading state, the overall QBER should not normally be calculated as a simple arithmetic average of the individual QBER values.

Instead, errors are pooled according to the number of detections:

$$Q_{pooled} = \frac{\sum_i w_i G_i Q_i}{\sum_i w_i G_i}.$$

This is equivalent to:

$$Q_{pooled} = \frac{\text{expected erroneous detections}}{\text{expected total detections}}.$$

The following five-node example applies this idea directly to the 20 km DPS FSO link.

At each node, the current total efficiency produces its own physical gain and QBER and compact modeled rate. The reported integrated physical QBER weights errors by detections.

Example calculation

Apply all five states to the detector and rate model

Table 11. Apply all five states to the detector and rate model

Node	Weight	Fade F	Total η	Physical gain	QBER	Modeled bit/s
0	0.011257	0.2499	8.657083e-04	1.338473e-04	2.464%	130.2
1	0.222076	0.4910	1.701308e-03	2.591626e-04	1.756%	357.1
2	0.533333	0.9037	3.131241e-03	4.735739e-04	1.414%	751.2
3	0.222076	1.6633	5.763017e-03	8.680755e-04	1.226%	1480.3
4	0.011257	3.2687	1.132560e-02	1.701391e-03	1.115%	3025.7

Five-node integrated result

Mean detector gain = 5.23565236e-04

Pooled QBER = 1.374%

Physical QBER across the five states

Weak states raise the relative dark-count contribution, whereas strong states approach the optical-error floor.

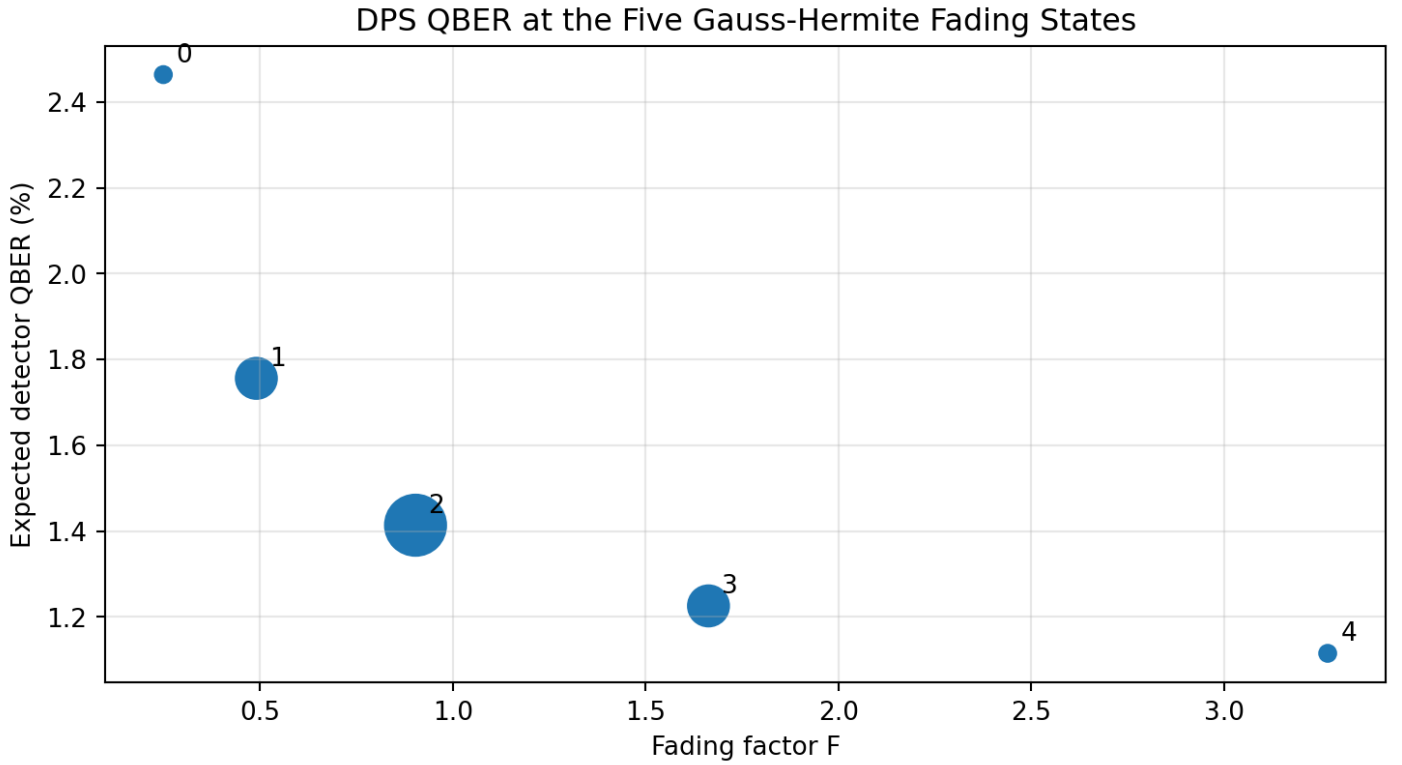


Figure 23. Physical QBER across the five Gauss-Hermite fading states

This is physical click-conditioned QBER, not the compact modeled QBER.

Modeled rate across the five states

The compact rate is recomputed at every state before any averaging. This curve illustrates its nonlinear response to received signal.

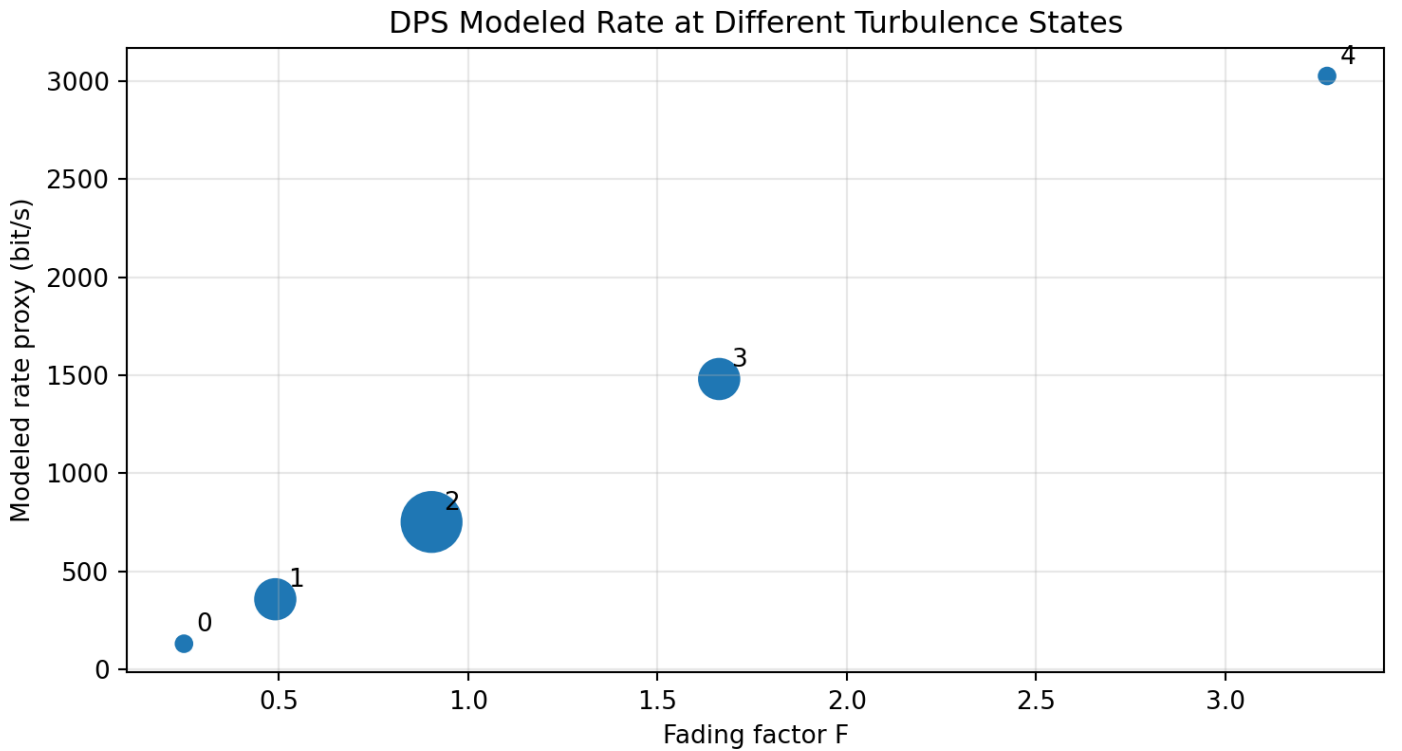


Figure 24. Modeled DPS rate across the five Gauss-Hermite fading states

Modeled rates and phase errors are engineering proxies under the stated compact model.

Accurate Turbulence Integration with 48 Gauss-Hermite Nodes

In earlier sections we used only five Gauss-Hermite nodes so that every fading state and weight could be inspected manually. We now use 48 nodes for the numerical turbulence calculation. The value 48 has no special physical meaning. It is simply a sufficiently high quadrature order for this smooth weak-to-moderate log-normal example while remaining computationally inexpensive. For each quadrature node i :

$$F_i = \exp\left(\frac{-\sigma^2}{2} + \sqrt{2}\sigma x_i\right)$$

and the instantaneous channel efficiency is:

$$\eta_i = \min(\eta_{static} F_i, 1).$$

The cap at 1 enforces the physical requirement that total efficiency cannot exceed 100%.

The production calculation therefore follows this chain:

Gauss-Hermite node $\rightarrow$ fading factor $\rightarrow$ instantaneous η $\rightarrow$ DPS detector $\rightarrow$ gain/QBER/rate $\rightarrow$ weighted average

The important point is that we do **not** replace turbulence by one average fading number. Detector gain, QBER, binary entropy, and the modeled rate are nonlinear functions of transmission. In general:

$$E[g(\eta)] \neq g(E[\eta]).$$

So the nonlinear DPS model is evaluated at every quadrature state before the results are combined.

Construct the production 48-node grid

Normalized weights sum to one. Extreme transformed factors receive vanishingly small weights and must not be interpreted as typical weather samples.

Example calculation

48-node Gauss-Hermite integration grid
Number of nodes : 48
Sum of normalized weights : 1.0
Weighted mean fading factor : 1.0000000000000002
Fading range represented by quadrature:
Minimum fade factor : 0.002988135866742993
Maximum fade factor : 273.309688387189
Extreme-node weights:
Lowest-node weight : 4.4771554738758594e-36
Highest-node weight : 4.4771554738758594e-36

Inspecting Representative Quadrature Nodes

A 48-node calculation contains many points, so the next table prints a representative subset from the low tail, central region, and high tail.

Some quadrature nodes lie extremely far into the mathematical tails of the Gaussian distribution. For $\sigma = 0.45$, the most extreme transformed fading factors can be very small or very large. These are **integration points, not equally probable atmospheric measurements**.

Their weights can be as small as roughly 10^{-36} , so the extreme states contribute essentially nothing to the final expectation. A fading value should therefore never be interpreted without its quadrature weight.

The printed subset spans the far tails and central nodes so both the transformed factor and its probability weight can be inspected.

Example calculation

Inspect representative integration states

Table 12. Inspect representative integration states

Node	GH x	Equivalent Z	Fade F	Weight
0	-8.97532	-12.69301	0.002988	4.477155e-36
8	-5.19629	-7.34866	0.033102	3.975806e-13
16	-2.41676	-3.41782	0.194122	5.395866e-04
20	-1.11881	-1.58224	0.443411	5.180518e-02
23	-0.15949	-0.22556	0.816482	1.754635e-01
24	0.15949	0.22556	1.000251	1.754635e-01
27	1.11881	1.58224	1.841826	5.180518e-02
31	2.41676	3.41782	4.207073	5.395866e-04

39	5.19629	7.34866	24.671464	3.975806e-13
47	8.97532	12.69301	273.309688	4.477155e-36

Weight Carried by the 48 Fading States

The normalized quadrature weights satisfy:

$$\sum_i w_i = 1.$$

The graph below uses a logarithmic horizontal axis because the transformed fading factors span several orders of magnitude. The central fading region carries almost all the numerical weight, while the extreme tails carry vanishingly small weight.

Visualize the 48 weights

The logarithmic fading axis resolves the wide range of transformed integration factors.

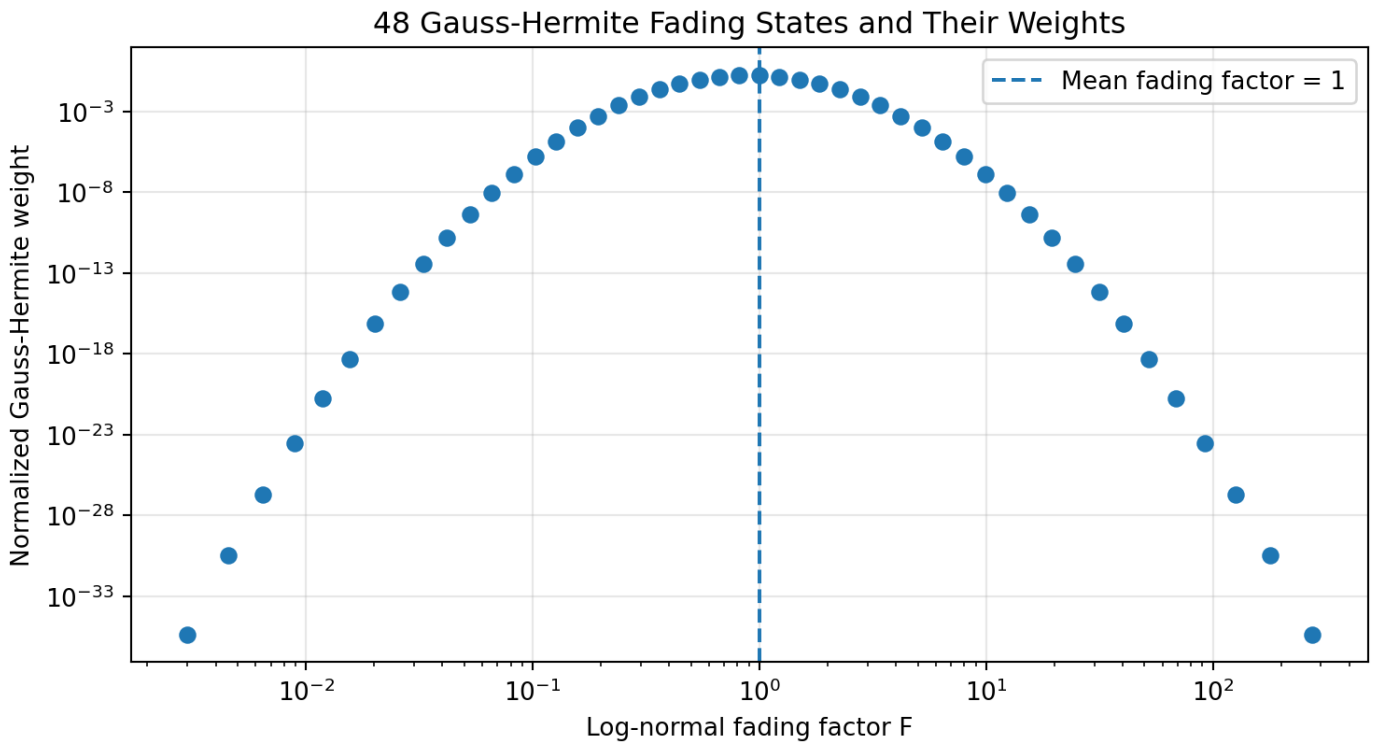


Figure 25. Weights of the 48 Gauss-Hermite fading states

Converting Fading into Instantaneous Link Efficiency

The static 20 km FSO link has an efficiency close to:

$$\eta_{static} \approx 0.003465.$$

Turbulence multiplies this value by each fading state:

$$\eta_i = \eta_{static} F_i.$$

For example, if $F = 0.5$:

$$\eta \approx 0.003465 \times 0.5 = 0.00173.$$

If $F = 2$:

$$\eta \approx 0.00693.$$

Because the log-normal model is normalized so that $E[F] = 1$, the weighted mean efficiency remains close to the original static efficiency unless the physical cap $\eta \leq 1$ becomes relevant.

Convert factors to total efficiencies

The cap at total efficiency one is applied per state. Minimum, maximum and weighted mean values are printed.

Example calculation

20 km static efficiency : 0.0034648843733626287

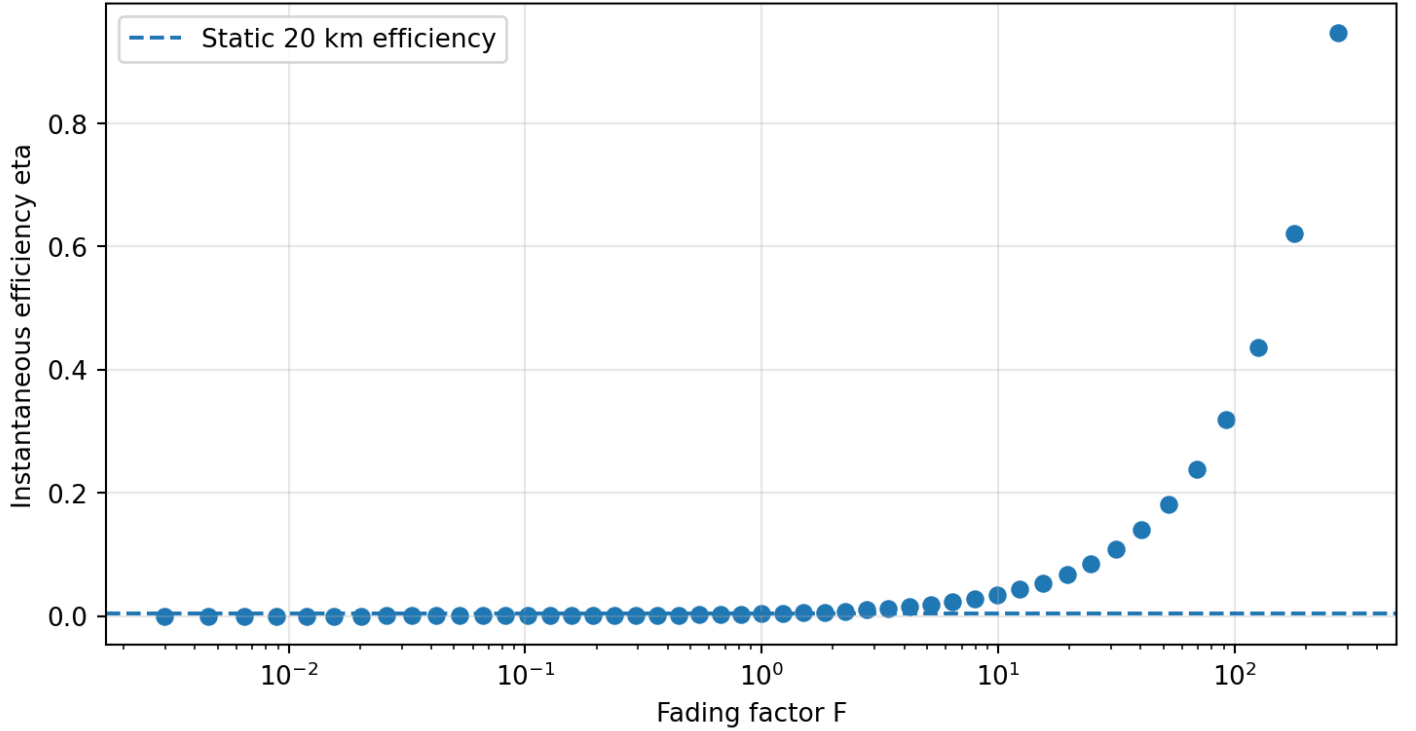
Weighted mean effective efficiency: 0.003464884373362629

Static efficiency: 0.0034648843733626287

Efficiency at each state

The figure makes the nonlinear exponential transform and physical efficiency cap visible.

Effect of Turbulence on the 20 km FSO Efficiency





39	24.6715	8.548377e-02	3.976e-13	127446.6	1.015%
47	273.3097	9.469865e-01	4.477e-36	1324238.3	1.003%

Detector gain response

This is physical probability of at least one click per usable gate.

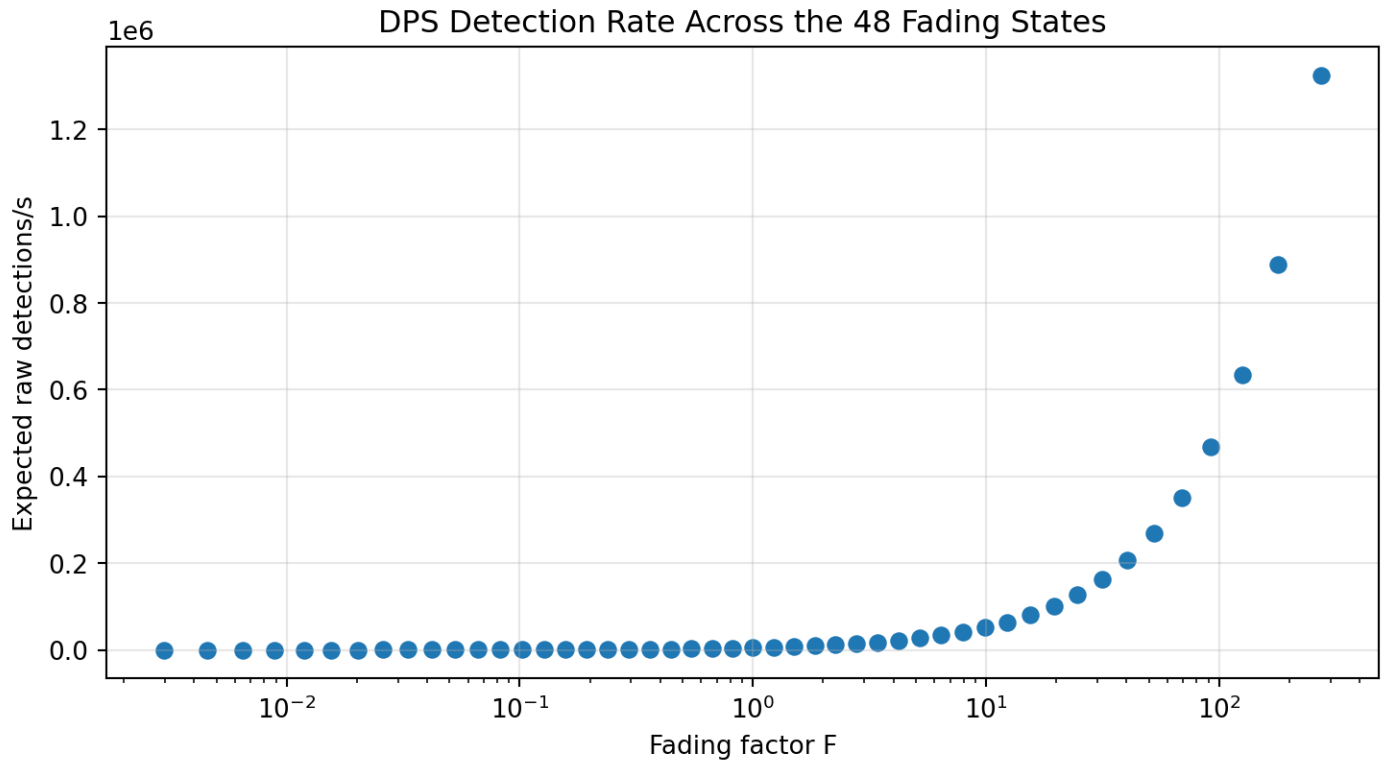


Figure 27. DPS physical detection rate across the 48 fading states

Detector QBER response

The physical error fraction approaches background domination at very low received signal.

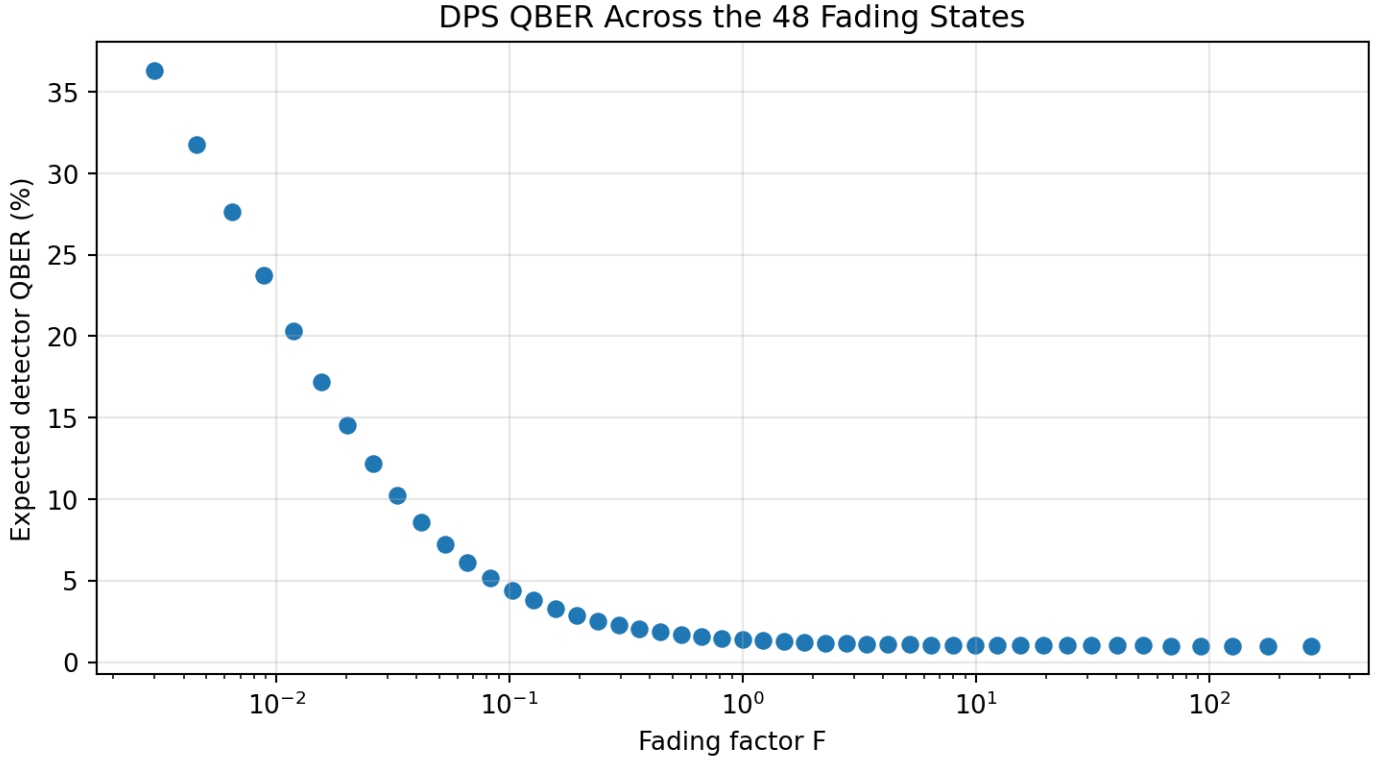


Figure 28. DPS physical QBER across the 48 fading states

Detection-Weighted QBER

A simple average of the individual QBER values is not the correct way to combine a fading channel because different fading states produce different numbers of detections.

For node i :

$$C_{det,i} = w_i G_i$$

is its expected contribution to total detections, while:

$$C_{err,i} = w_i G_i Q_i$$

is its expected contribution to erroneous detections.

Therefore:

$$\dot{G} = \sum_i w_i G_i$$

and the pooled QBER is:

$$Q_{pooled} = \frac{\sum_i w_i G_i Q_i}{\sum_i w_i G_i}.$$

Small example: if one state produces 1,000 detections at 2% QBER and another produces only 100 detections at 10% QBER, the arithmetic average is 6%, but the actual pooled result is:

$$\frac{1000(0.02) + 100(0.10)}{1100} = 2.73\%.$$

The study therefore reports both the simple weighted average for comparison and the physically more meaningful detection-weighted pooled QBER.

Pool detections and errors

The calculation separately accumulates weighted total detections and weighted erroneous detections. Their ratio defines pooled physical QBER.

Example calculation

48-node physical detector integration
 Expected gain per gate = 5.235652399e-04
 Expected raw detections/s = 5,235.7
 Simple weighted mean QBER = 1.457%
 Detection-weighted pooled QBER = 1.374%

Contribution to total detections

This figure identifies which fading states dominate the pooled detector count.

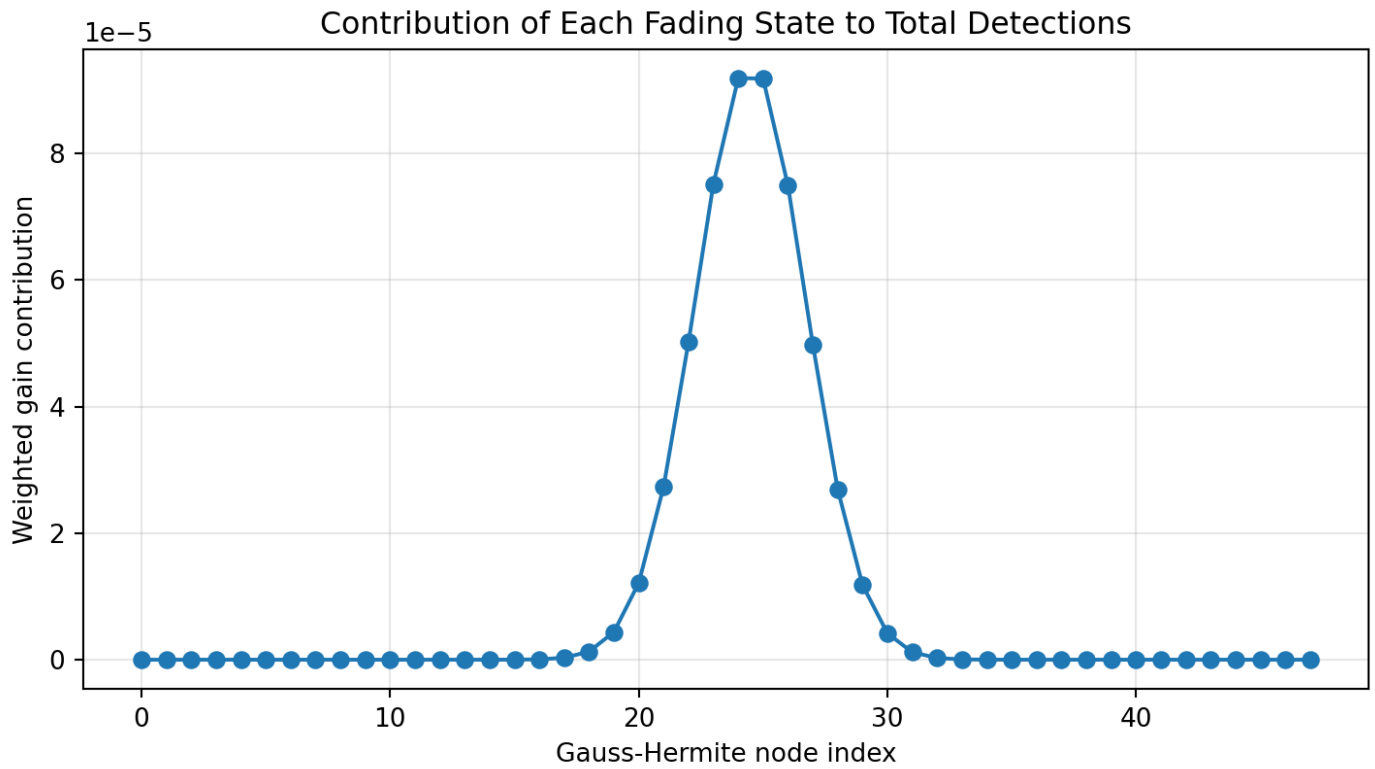


Figure 29. Contribution of each fading state to total detections

Contribution to erroneous detections

Error-count weights need not peak at the same positions as simple node weights.

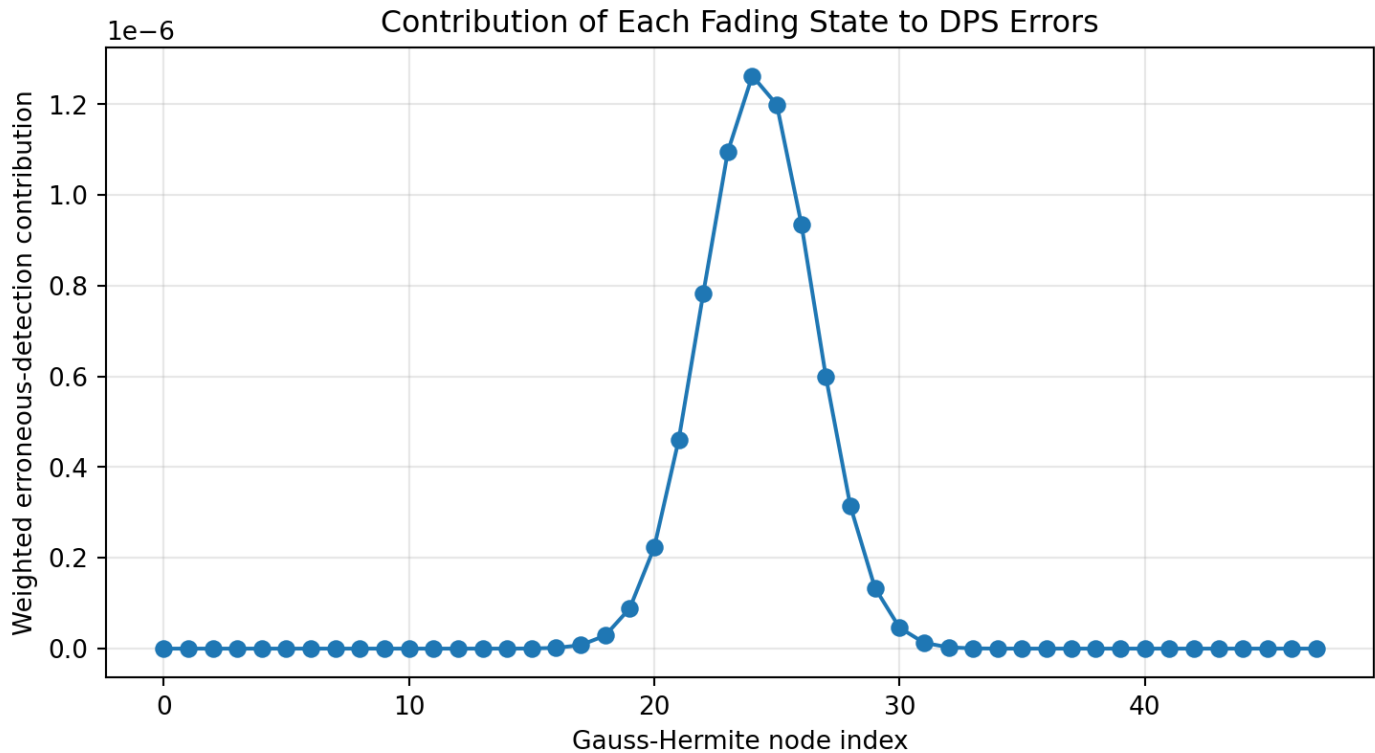


Figure 30. Contribution of each fading state to total detected errors

Compare normalized detection and error contributions

The two normalized distributions explain why pooled QBER differs from an arithmetic mean of node percentages.

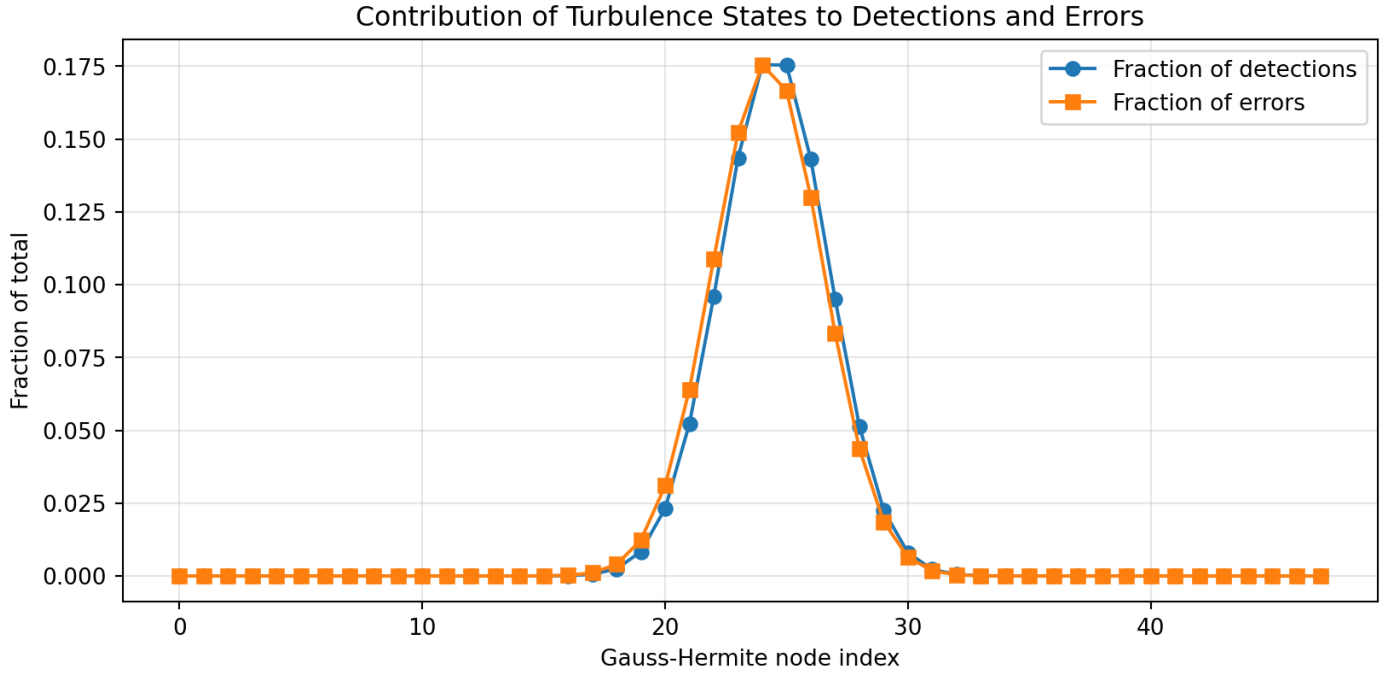


Figure 31. Normalized detection and error contributions across fading states

The physical click model above evaluates detector gain and click-conditioned QBER. The notebook also carries the compact DPS model introduced earlier.

At every fading state it evaluates:

$$F_i \rightarrow \eta_i \rightarrow G_i, G_{1,i}, Q_{bit,i}, Q_{ph,i}, R_i.$$

The total and single-photon gains are averaged directly:

$$\hat{G} = \sum_i w_i G_i, \hat{G}_1 = \sum_i w_i G_{1,i}.$$

The bit-error and phase-error quantities are pooled by detected gain:

$$Q_{bit,pooled} = \frac{\sum_i w_i G_i Q_{bit,i}}{\hat{G}},$$

$$Q_{ph,pooled} = \frac{\sum_i w_i G_i Q_{ph,i}}{\hat{G}}.$$

Those pooled quantities are then inserted into the same educational modeled-rate expression used in earlier sections. The result remains a modeled performance proxy, not a complete finite-key security proof.

Pool the compact model

The compact model is evaluated independently at the same 48 efficiencies. Its own gains weight its bit and phase error proxies before evaluating the pooled rate.

Example calculation

48-node compact DPS integration
Mean total gain = 5.230101506e-04
Mean single-photon gain = 4.473380426e-04
Pooled modeled QBER = 3.180%
Pooled phase-error proxy = 6.501%
Pooled modeled rate = 843.7 modeled bit/s
Mean per-fade modeled rate = 844.2 modeled bit/s

Compact bit-QBER response

This follows the photon-number approximation rather than the physical correct/wrong-port formula.

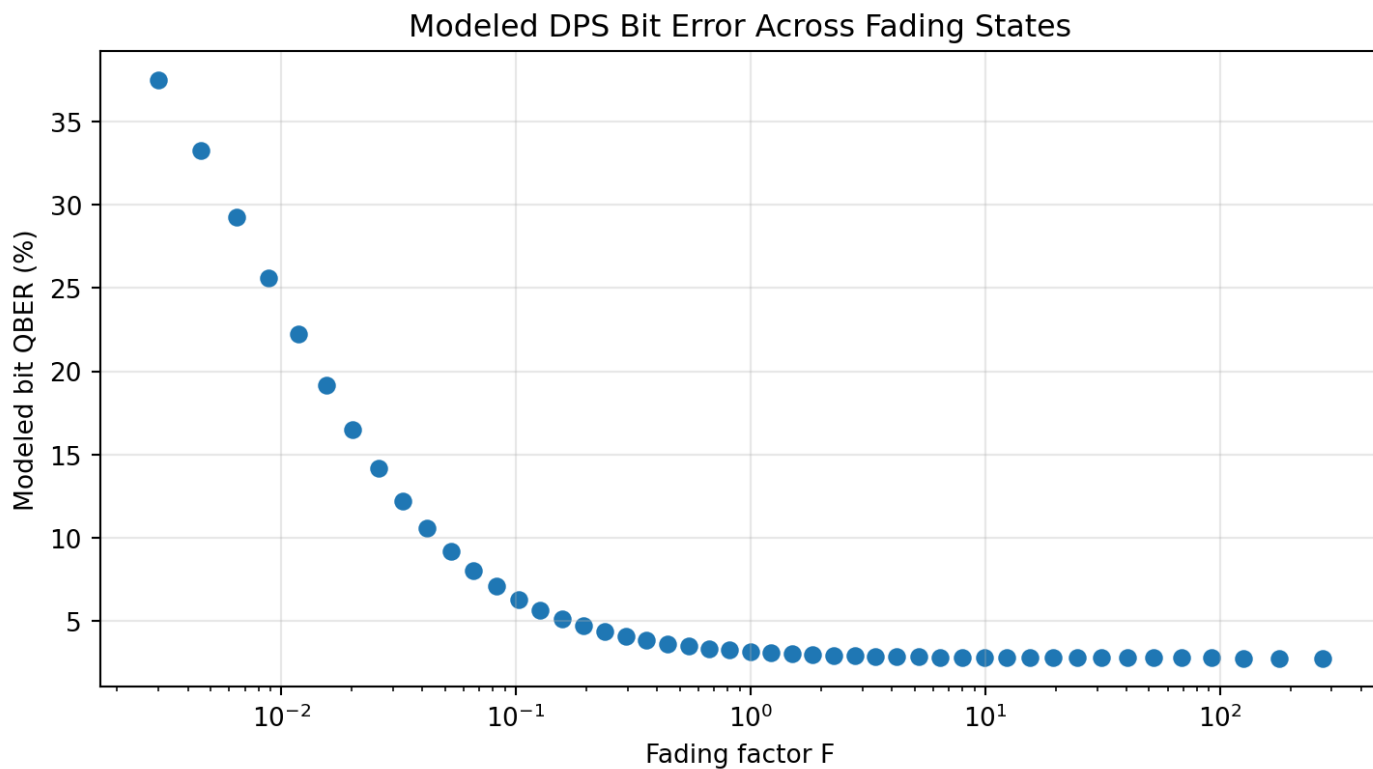


Figure 32. Modeled DPS bit QBER across fading states

Phase-error response

The plotted quantity is the compact model's phase-error proxy.

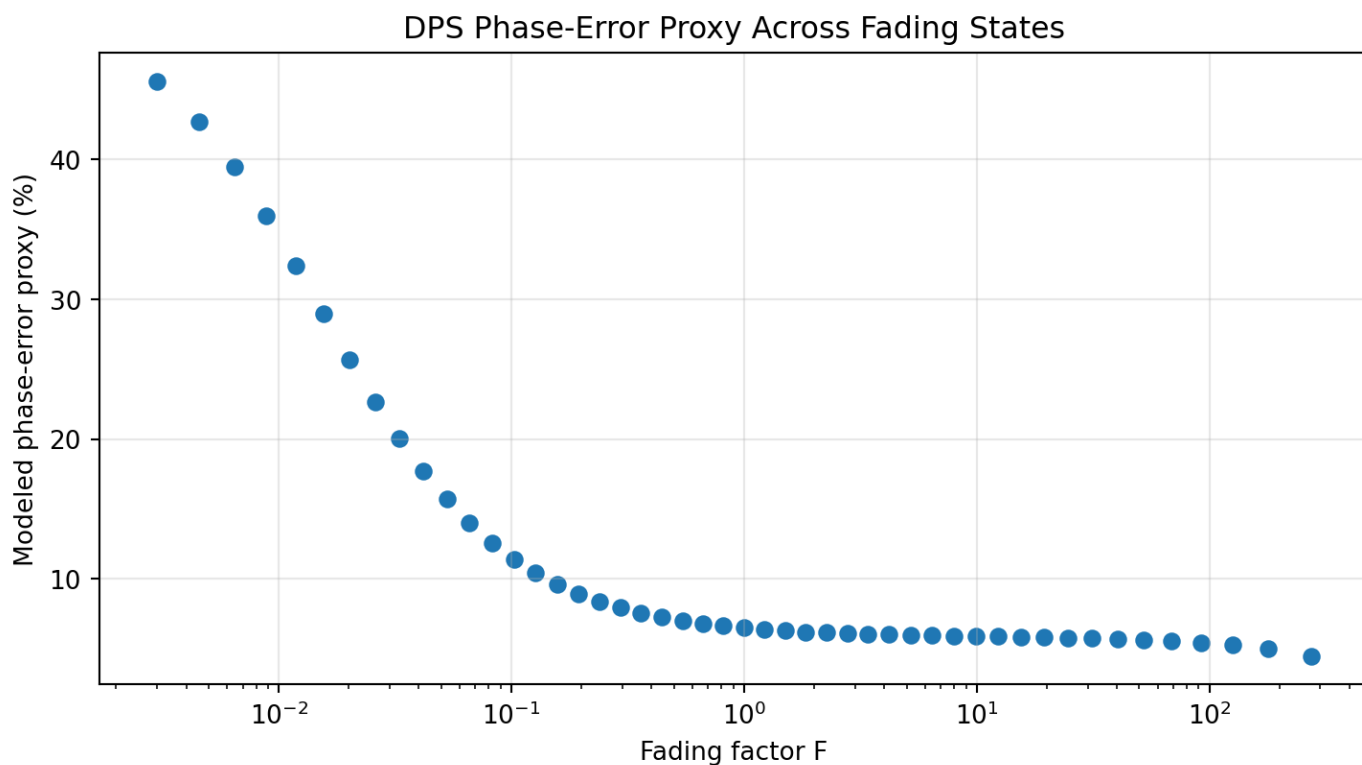


Figure 33. DPS phase-error proxy across fading states

Conditional modeled rate

Every fading state has an individually evaluated and zero-clipped rate; its weighted mean is distinct from the rate of pooled statistics.

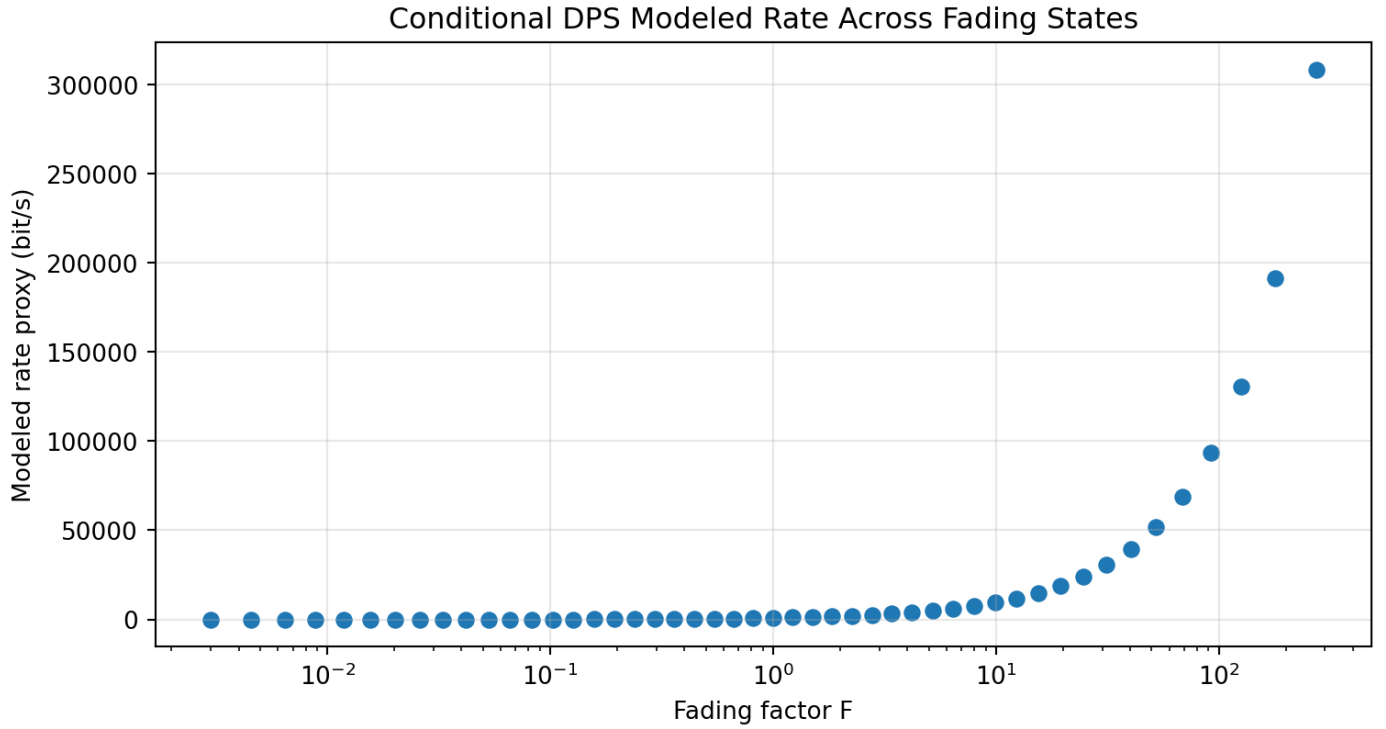


Figure 34. Conditional modeled DPS rate across fading states

Pooled Rate versus Mean Per-Fade Rate

There are two different ways to summarize the modeled rate over turbulence.

Pooled processing: accumulate gains and error statistics across all fading states first, then calculate one modeled rate.

Per-fade processing: calculate a modeled rate R_i for each fading state first and then average:

$$\dot{R}_{conditional} = \sum_i w_i R_i.$$

Because the entropy terms and zero-clipping are nonlinear, the two results need not be identical:

$$R(E[x]) \neq E[R(x)].$$

For the present weak-turbulence operating point they are expected to be close; stronger fading or different parameters can increase the difference.

Compare pooled and conditional rate processing

The small difference at the baseline arises because entropy and zero clipping are nonlinear operations.

Example calculation

Rate averaging comparison
Pooled modeled rate = 843.7 modeled bit/s
Mean per-fade modeled rate = 844.2 modeled bit/s
Difference = 0.532 modeled bit/s

Reusable 48-Node Turbulence Function

The preceding cells deliberately expose every intermediate quantity. Later sections need the same calculation repeatedly, so the complete quadrature procedure is packaged into `fading_moments()`. The function accepts the static efficiency, mean photon number, turbulence strength, visibility, dark-count probability, and error-correction efficiency. It returns the averaged physical detector statistics and both modeled-rate summaries.

No new physics is introduced by the function; it only automates the calculation demonstrated above.

Reusable turbulence algorithm

`fading_moments` broadcasts the input arrays across 48 Hermite states, applies clipped efficiencies, evaluates both receiver models, averages gains and weighted error numerators, and returns physical statistics plus pooled and mean-conditional modeled rates.

km Turbulence-Averaged Example

We now apply the reusable integrator to the same 20 km ground link using:



$$\mu = 0.15, \sigma = 0.45.$$

The output reports the long-term physical detector gain, detection-weighted QBER, pooled compact-model QBER, pooled phase-error proxy, and the two modeled-rate summaries.

Complete 20 km turbulence result

The function reports the physical detector statistics and compact pooled statistics separately at $\mu = 0.15$ and log-irradiance $\sigma = 0.45$.

Example calculation

20 km DPS link with log-normal fading, $\sigma=0.45$
Average physical gain/gate = 5.235652399e-04
Average raw detections/s = 5,235.7
Detection-weighted click QBER = 1.374%
Pooled modeled QBER = 3.180%
Pooled phase-error proxy = 6.501%
Pooled modeled rate proxy = 843.7 modeled bit/s
Mean per-fade modeled rate proxy = 844.2 modeled bit/s

Sanity Check: Zero Turbulence Must Recover the Fixed Link

If:
 $\sigma = 0$,
then every log-normal fading factor becomes exactly 1. The quadrature calculation must therefore reproduce the fixed 20 km channel.
The next cell checks this explicitly with numerical assertions. If either assertion fails, the turbulence implementation is inconsistent with the fixed-link model.

Zero-turbulence consistency assertions

At σ zero, all factors become one. The asserted gain and rate agreement verify recovery of the fixed-link calculation.

Example calculation

No-turbulence sanity check
Fixed-link gain: 5.235955360e-04
Quadrature gain: 5.235955360e-04
Fixed-link QBER: 1.374237%
Quadrature QBER: 1.374237%

Fixed Link versus Turbulence-Averaged Link

A useful final check is to place the fixed-link result and the $\sigma = 0.45$ turbulence-averaged result side by side. Because the fading factor has unit mean, the **long-term average** can remain close to the fixed-link result even though the instantaneous link fluctuates strongly. This does not mean turbulence is irrelevant: temporary deep fades can still create low detection rate, high QBER, and operational outage periods.

Fixed and fading results side by side

The baseline averages are close, while individual fades can still cause low-rate and high-QBER intervals.

Example calculation

Table 14. Fixed-link and turbulence-averaged results

Quantity	Fixed link	$\sigma_{\log} = 0.45$
Physical gain/gate	5.235955360e-04	5.235652399e-04
Raw detections/s	5,236.0	5,235.7
Physical detector QBER	1.374%	1.374%
Modeled DPS QBER	3.180%	3.180%
Phase-error proxy	6.503%	6.501%
Modeled rate proxy (bit/s)	843.5	843.7

Interpretation of the 48-Node Result

The 48-node calculation shows four important ideas.

First, the log-normal model is normalized so that:

$$E[F] = 1.$$

Second, instantaneous conditions still vary strongly. During a deep fade:

$$\eta \downarrow, G \downarrow, QBER \uparrow.$$

During a favorable fade:

$$\eta \uparrow, G \uparrow, QBER \downarrow.$$

Third, QBER must be combined according to the number of detections, not by treating every fading state as equally important.

Finally, a long-term mean close to the fixed-link result does **not** imply that turbulence has no operational effect. Time-varying fading can still produce temporary high-QBER intervals, bursty key generation, and outages.

Validate Quadrature with an Independent Monte Carlo Simulation

A numerical integration method should be checked against a second method.

We therefore generate 100,000 independent log-normal fades using a separate seeded random generator and compare:

- Mean detector gain,
- Detection-weighted QBER.

Small differences are expected because Monte Carlo has finite-sample noise. The assertions allow a modest tolerance rather than requiring bit-for-bit equality.

After validation, we sweep σ_{log} from 0 to 0.9 to show how increasing turbulence changes the pooled quantities.

Independent Monte Carlo validation

One hundred thousand independent fades at seed 913 test physical gain and pooled physical QBER. The allowed relative gain tolerance is 2%, and absolute QBER tolerance is 0.002. A separate sweep varies log-irradiance sigma from zero to 0.9.

Example calculation

Quadrature gain = 0.0005235652; Monte Carlo gain = 0.0005226168

Quadrature QBER = 1.374%; Monte Carlo QBER = 1.375%

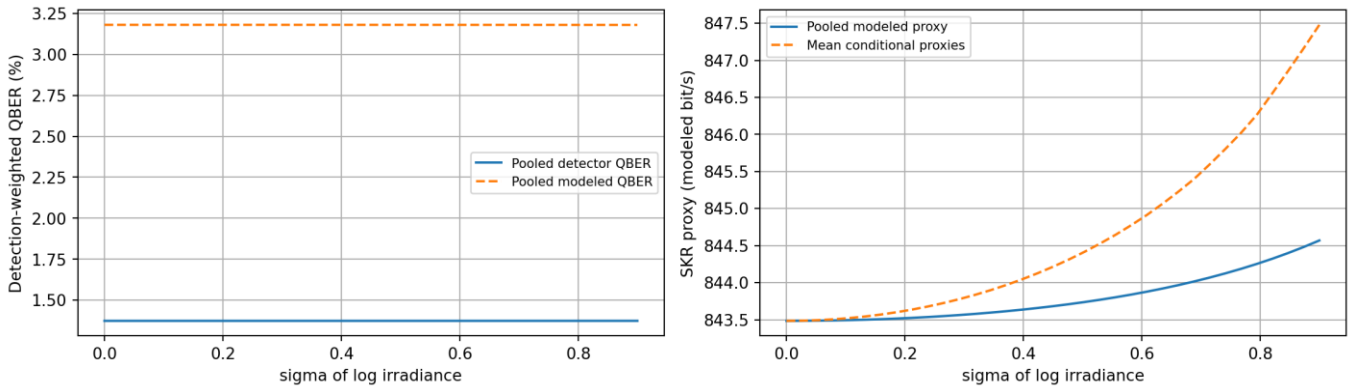


Figure 35. Turbulence-strength sweep of pooled QBER and modeled-rate summaries

Turbulence strength: pooled QBER and two rate summaries

Long-Time Correlated Fades and Outage

The log-normal model above describes the **marginal distribution** of transmission, but independent samples jump unrealistically from one fade to another. To demonstrate slowly varying conditions, we add an explicit AR(1) Gaussian process:

$$Z_t = \rho Z_{t-1} + \sqrt{1 - \rho^2} \epsilon_t,$$

where $\epsilon_t \sim N(0,1)$.

We then convert the correlated Gaussian variable to a unit-mean log-normal factor:

$$F_t = \exp\left(\sigma Z_t - \frac{\sigma^2}{2}\right).$$

and

$$\eta_t = \min(\eta_{20} F_t, 1).$$

For this teaching experiment:

- $\sigma = 0.45$ controls fade strength,

- $\rho = 0.96$ gives strong correlation between adjacent samples,
- Samples are interpreted every 10 seconds,
- An engineering threshold of 2,000 raw detections/s defines an illustrative outage.

The **2,000/s threshold is not a QKD security threshold**. It is only an availability criterion chosen for this example. The AR(1) correlation is an additional time-series assumption used only for this teaching experiment.

Generate the correlated process

The AR(1) process starts from a standard-normal draw and uses $\rho = 0.96$. Seed 991 makes this NumPy realization reproducible.

Example calculation

First 12 correlated fade factors:

[1.6886 1.7082 1.8097 1.6365 1.7893 1.6451 1.5745 1.2942 1.4303 1.2451
1.578 1.6267]

Adjacent values tend to remain similar when ρ is close to 1.

Sampled outage and long-time pooled QBER

There are 2,400 ten-second bins, interpreted as 400 minutes; timestamps run from zero to 399.833 minutes. Outage counts raw physical detections below 2,000/s. This definition differs from the current Octave modeled-bit-rate outage.

Example calculation

Study duration: 400 minutes

Fraction of 10-second sampled bins below 2,000 raw detections/s: 1.92%

Detection-weighted QBER over the study: 1.35%

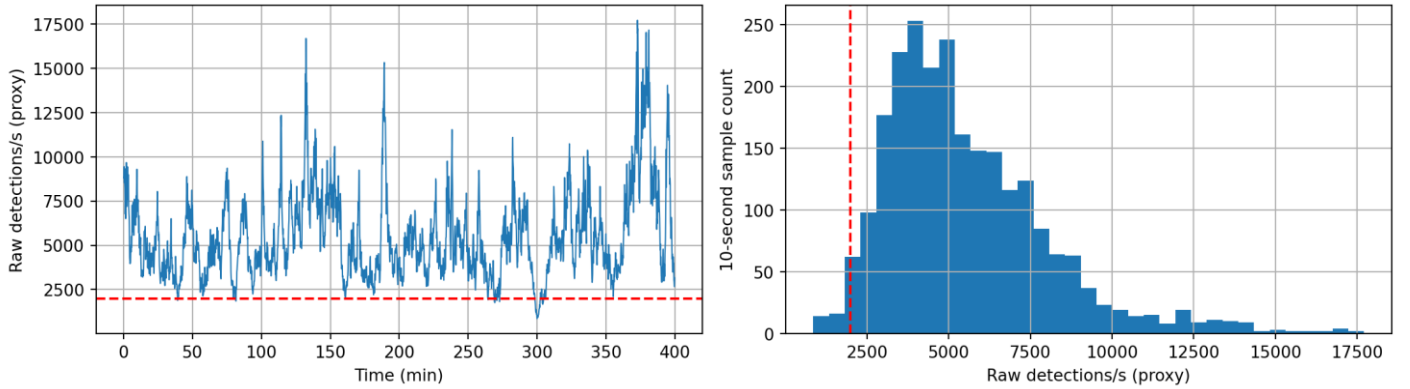


Figure 36. Correlated fading over 400 minutes and raw-detection outage

Machine-Learning Validation

The physics model can be evaluated directly, so machine learning is **not required** for correctness. Here ML is used as a fast surrogate for exploring a larger design space and then proposing an operating point. Every proposed optimum is rechecked with the original DPS-FSO physics model.

The workflow is:

physics model -> training data -> ML surrogate -> constrained search -> physics recheck

The ML model never replaces the physical equations or the QKD security assumptions.

Physics Model as the Ground Truth

Each input vector contains five parameters:

$$X = [\mu, a, \sigma_{log}, V, \log_{10}(d)],$$

where:

- μ = mean photons per pulse;
- a = receiver radius;
- σ_{log} = log-normal turbulence strength;
- V = interferometer visibility;

- d = dark/background click probability per detector per gate.

The dark probability is represented by $\log_{10}(d)$ because detector background can span orders of magnitude. The physics model converts the feature vector into the 20 km FSO efficiency, performs the 48-node turbulence integration, and returns two targets:

$$R_{model} = f_{gate} r_{pooled}$$

and

$$Q_{model} = Q_{bit,pooled}$$

Thus the surrogate learns **the project's modeled performance surface**. It does not learn directly from measured atmospheric or hardware data.

Construct physics labels for machine learning

For each five-feature vector, the function recomputes the 20 km optical efficiency and 48-node pooled response. The returned labels are simulated modeled rate and modeled bit QBER.

Example calculation

Example ML feature vector

[mu, receiver radius, sigma_log, visibility, log10(dark)]

[0.15 0.1 0.45 0.98 -5.69897]

Physics-model rate = 843.68 modeled bit/s

Physics-model QBER = 3.180%

Five-Parameter RBF Kernel-Ridge Surrogate

We sample the following parameter ranges:

Table 15. Five-feature training and validation domain

Feature	Minimum	Maximum	Role
μ	0.03	0.38	transmitter operating parameter
receiver radius a	0.06 m	0.18 m	optical design parameter
σ_{log}	0	0.85	atmospheric condition
visibility V	0.94	0.995	interferometer/device quality
dark probability d	10^{-7}	about 2×10^{-5}	detector/background condition

The features are scaled into [0,1]. The radial-basis-function kernel is:

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right).$$

Kernel ridge regression solves:

$$(K + \lambda I)W = Y.$$

Here Y has **two outputs**: modeled rate and modeled QBER. We use the same kernel matrix for both targets. The model is trained on 520 randomly sampled parameter combinations and evaluated on 180 independent hold-out combinations that were not used during fitting.

Fit the two-output RBF model

Seed 3602 supplies 520 training and 180 independent test rows. Features scale to [0,1]; the RBF length is 0.42 and ridge coefficient 0.001. Rate and QBER targets are scaled separately before solving the shared linear system.

Example calculation

Training samples: 520

Independent test samples: 180

Input dimensions: 5

Predicted outputs: modeled rate and modeled QBER

Hold-Out Validation

A surrogate should not be trusted merely because it fits its training points. We therefore evaluate it on the 180 independent test combinations.

For modeled rate we calculate:

- **MAE** - mean absolute error in modeled bit/s;
- R^2 - fraction of held-out variance reproduced by the surrogate.

For QBER we report:

- MAE in **percentage points**;
- held-out R^2 .

The scatter plots compare the original physics-model outputs against ML predictions. Perfect predictions would fall on the dashed diagonal.

Evaluate on independent test inputs

The printed MAE and R-squared metrics describe prediction of the assumed physics surface. QBER MAE is reported in percentage points, not percent relative error.

Example calculation

Hold-out validation

Rate MAE = 35.2 modeled bit/s

Rate $R^2 = 0.9921$

QBER MAE = 0.124 percentage points

QBER $R^2 = 0.9755$

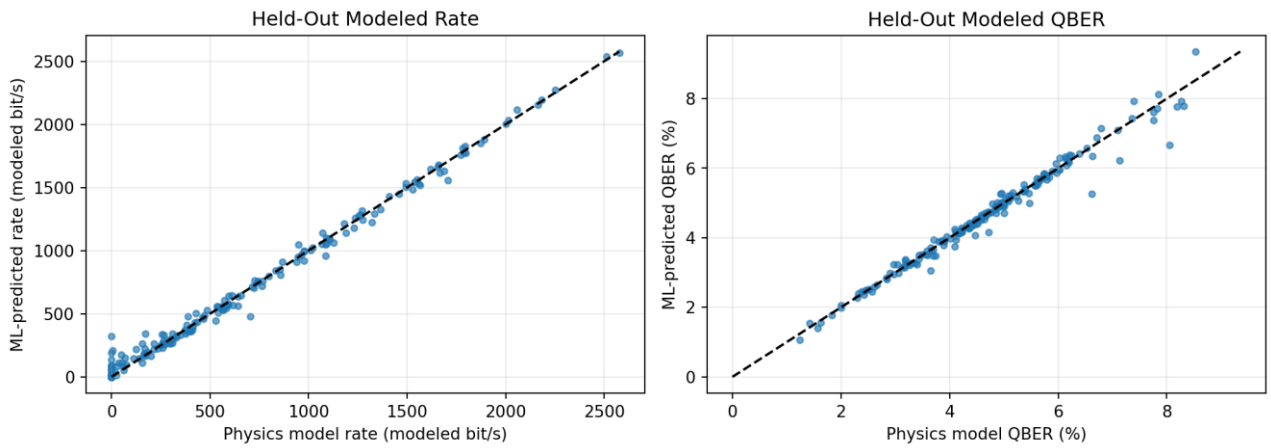


Figure 37. Independent hold-out validation of the RBF surrogate

Constrained ML-Assisted Parameter Optimization

We now use the surrogate for an actual design search rather than only prediction.

For the optimization example, the environmental/device conditions are fixed at:

$$\sigma_{log} = 0.45, V = 0.98, d = 2 \times 10^{-6}.$$

We search two design/operating variables:

$$0.03 \leq \mu \leq 0.38$$

and

$$0.06 \leq a \leq 0.18 \text{ m.}$$

The objective is:

maximize modeled SKR

subject to the illustrative engineering constraint:

$$QBER < 5\%.$$

The ML surrogate first scans the two-dimensional design grid. The best feasible ML point is then **re-evaluated with the original 48-node physics model**. Finally, we also evaluate the same grid directly with the physics model as a verification benchmark.

This is important: the ML output is a **proposal**, not the final answer. The physics-model recheck remains authoritative within this study.

Scan 14,000 designs and recheck the proposal

The finite search combines 140 μ values and 100 receiver radii at fixed $\sigma = 0.45$, visibility 0.98 and dark probability $2e-6$. Feasible points satisfy modeled QBER below 5%. The selected proposal is re-evaluated by physics and compared with the direct optimum on that same finite grid.

Example calculation

ML-assisted constrained optimization

Candidates searched by surrogate = 14,000

Constraint = QBER < 5.0%
 Fixed sigma_log = 0.45
 Fixed visibility = 0.980
 Fixed dark probability = 2.0e-06
 ML-proposed design
 mu = 0.1660
 receiver radius = 0.1800 m
 ML predicted rate = 3003.8 modeled bit/s
 ML predicted QBER = 3.053%
 Physics recheck of ML proposal
 modeled rate = 2940.0 modeled bit/s
 modeled QBER = 3.085%
 Direct physics optimum on the same finite grid
 mu = 0.1609
 receiver radius = 0.1800 m
 modeled rate = 2942.2 modeled bit/s
 modeled QBER = 3.031%
 Difference between ML proposal and direct-grid optimum
 delta mu = +0.00504
 delta receiver radius = +0.00000 m

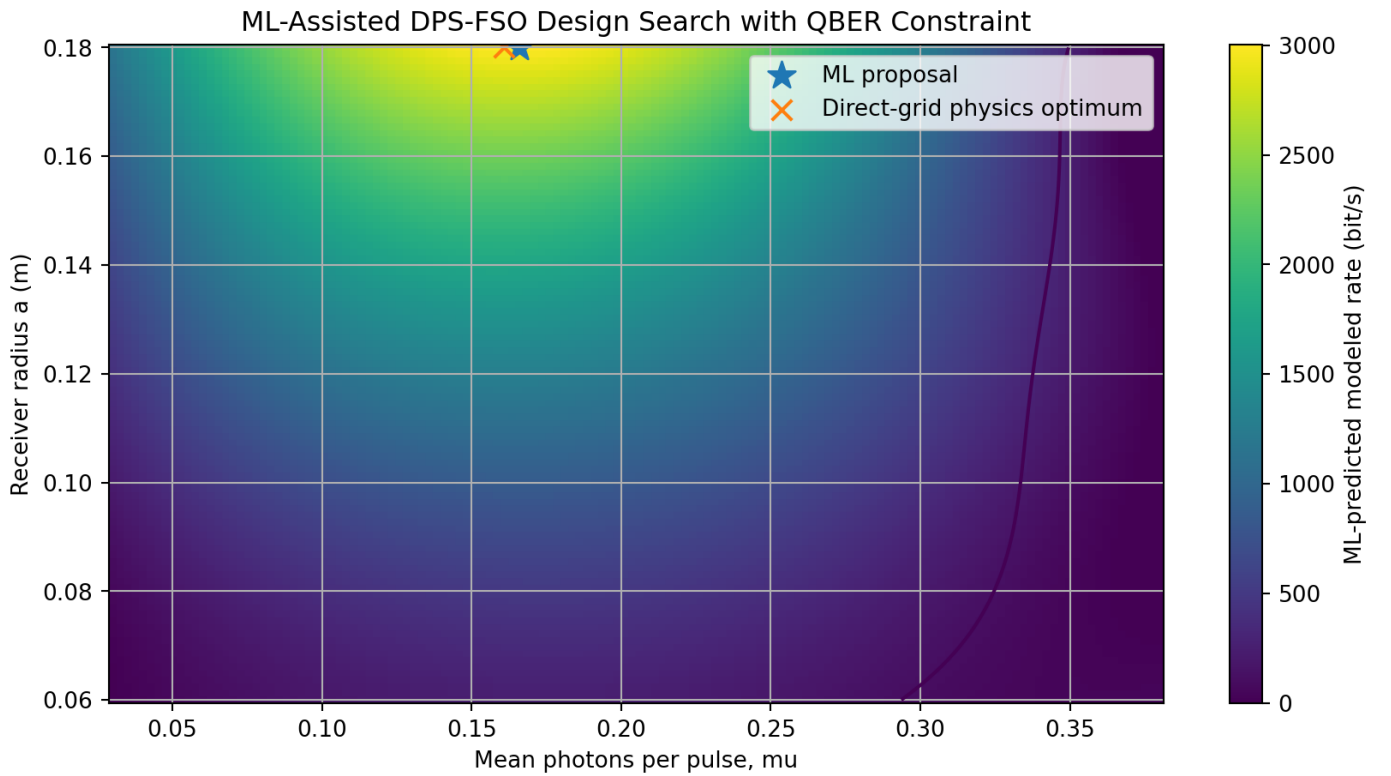


Figure 38. Constrained ML design surface and physics-checked operating points

Interpretation of the Optimization Result

The optimizer demonstrates the intended ML contribution to the project:

1. The physics model generates reference simulation labels under the assumptions of this study;
2. The surrogate learns the multidimensional DPS-FSO performance surface;
3. The surrogate rapidly searches a dense design grid;
4. A QBER constraint filters unacceptable points;
5. The proposed optimum is rechecked using the original physical model.

For this unconstrained aperture-cost example, the optimizer naturally tends toward the largest allowed receiver aperture because a larger aperture collects more of the beam. That is a useful engineering observation rather than an ML failure. A deployment study would normally add practical constraints such as aperture size, cost, mass, pointing capability or hardware availability. A high held-out R^2 only means the surrogate reproduces this smooth simulation over the sampled parameter range. It does **not** validate the physical assumptions, predict unmeasured weather, or strengthen the QKD security proof. The final operating point must still be checked with the physics model and, for deployment, with measured channel and device data. The hold-out metrics therefore demonstrate interpolation quality within the sampled five-parameter domain; they do not establish extrapolation accuracy outside that domain.

LEO Satellite-to-Ground DPS Downlink

We now replace the fixed 20 km horizontal link with a moving 500 km LEO satellite. The model is intentionally simple so that each geometric effect can be followed:

1. Place the ground station on a spherical Earth,
2. Place the satellite on a circular 500 km orbit,
3. Move the satellite with constant angular rate,
4. Calculate satellite-ground slant range,
5. Calculate elevation angle,
6. Calculate how much of the ray lies inside a 20 km atmosphere shell,
7. Apply beam spreading over the **full slant range**,
8. Apply atmospheric loss only to the **in-atmosphere path**,
9. Close the link below 10 degrees elevation,
10. Calculate DPS gain, QBER and modeled SKR proxy.

Assumptions used here:

Table 16. LEO satellite-to-ground teaching-model assumptions.

Parameter	Value
Earth radius R	6371 km
Satellite altitude h	500 km
Earth gravitational parameter μ_E	$398600.4418 \text{ km}^3/\text{s}^2$
Minimum elevation	10 degrees
Atmospheric shell height	20 km
Zenith atmospheric loss	2 dB
Beam divergence	10 microrad
Receiver radius	0.5 m

This is a **planar overhead-pass teaching geometry**. Earth rotation, orbital inclination, clouds, sunlight constraints and detailed pointing dynamics are omitted.

Circular-Orbit Angular Speed

The orbital radius is:

$$r_{orb} = R + h = 6371 + 500 = 6871 \text{ km.}$$

For a circular orbit, the angular rate is:

$$\omega = \sqrt{\frac{\mu_E}{r_{orb}^3}}.$$

For the chosen values:

$$\omega \approx 1.1085 \times 10^{-3} \text{ rad/s.}$$

The corresponding orbital period is:

$$T = \frac{2\pi}{\omega} \approx 94.47 \text{ min.}$$

The satellite angle relative to the overhead point is then:

$$\theta_s(t) = \omega t.$$



Calculate the circular-orbit motion

The angular speed and period follow from a 6,871 km orbital radius. The sampled time vector extends from -950 to +950 seconds in five-second steps.

Example calculation

Circular-orbit teaching model
orbital radius = 6871.0 km
angular rate ω = 1.108508340e-03 rad/s
orbital period = 94.47 min
time sampling = 5 s

Range, Elevation and Atmospheric Slant Path

The ground station and satellite are represented by 2-D position vectors.
For each time sample:

$$v = s - g$$

is the line-of-sight vector, and the slant range is:

$$L = \|v\|.$$

The local vertical unit vector at the ground station is:

$$\hat{u} = \frac{g}{R}.$$

The elevation angle is obtained from the projection of the line of sight onto the local vertical:

$$\sin(\text{elevation}) = \frac{v \cdot \hat{u}}{\|v\|}.$$

For atmospheric attenuation we intersect the line of sight with a spherical shell of radius $R + 20$ km. This produces the **in-atmosphere path length**.

At zenith:

- Slant range = 500 km,
- Elevation = 90 degrees,
- In-atmosphere path = 20 km.

Near the 10 degree cutoff, the optical path through the 20 km-high atmosphere becomes much longer than 20 km.

Evaluate two geometric checkpoints

The zenith and near-cutoff cases distinguish full optical range from in-atmosphere slant length.

Example calculation

$t = 0$ s: range = 500.0 km, elevation = 90.00 deg, atmosphere path = 20.00 km

$t = 220$ s: range = 1685.4 km, elevation = 10.14 deg, atmosphere path = 108.54 km

Convert Satellite Geometry into Optical Efficiency

Beam expansion uses the **full satellite-ground range** because diffraction occurs over the entire optical path.

Atmospheric attenuation is applied only to the path inside the 20 km atmospheric shell. If the zenith 20 km path has 2 dB loss, the simple scaled atmospheric loss is:

$$Loss_{atm,dB}(t) = 2 \frac{L_{atm}(t)}{20}.$$

and

$$\eta_{atm}(t) = 10^{-Loss_{atm,dB}(t)/10}.$$

At zenith, $L_{atm} = 20$ km, so the loss is exactly 2 dB. Near the 10 degree elevation cutoff in this sampled pass, the atmospheric path exceeds 100 km and the simple scaled loss becomes more than 10 dB.

Below 10 degrees the receiver gate is treated as closed; even dark counts are excluded from the modeled contact because no key-acquisition window is open.

Calculate the physical downlink response

The elevation cutoff gates the raw count contribution. Under the discrete sample-count convention used by the code, 89 centered-pass samples above 10 degrees correspond to a sample-counted contact budget of 445 s; this should not be interpreted as an exact continuous-visibility duration.

Example calculation

Contact above 10 deg: 7.42 min

x

t= 0 s: elevation= 90.00 deg, eta=3.473e-03, raw detections/s=5,248, QBER=1.37%

t=220 s: elevation= 10.14 deg, eta=4.016e-05, raw detections/s=100, QBER=20.55%

Satellite-Pass QBER and Modeled SKR Proxy

The next cell applies the DPS QBER/SKR model to every visible 5-second sample.

For each sample:

$$\text{geometry} \rightarrow \eta(t) \rightarrow Q_{bit}(t), Q_{ph}(t) \rightarrow r_{model}(t).$$

The instantaneous modeled rate is:

$$R(t) = f_{gate} r_{model}(t).$$

We also form an illustrative integrated pass budget:

$$B \approx \sum_i R(t_i) \Delta t.$$

The units of B are modeled proxy bits. This numerical integration is useful for comparing link opportunities, but it is **not** a finite-key proof: a practical implementation must define security blocks, parameter-estimation confidence bounds, authentication consumption and finite-key privacy amplification.

Calculate the compact modeled pass budget

The independent compact model is evaluated at the deterministic downlink efficiencies. A rectangular sum of rate times five seconds produces the integrated proxy budget; no stochastic satellite fading is applied in this version.

Example calculation

Zenith modeled QBER: 3.18%

Zenith SKR proxy: 846 modeled bit/s

Integrated pass budget: 95,195 modeled proxy bits (no finite-key proof)

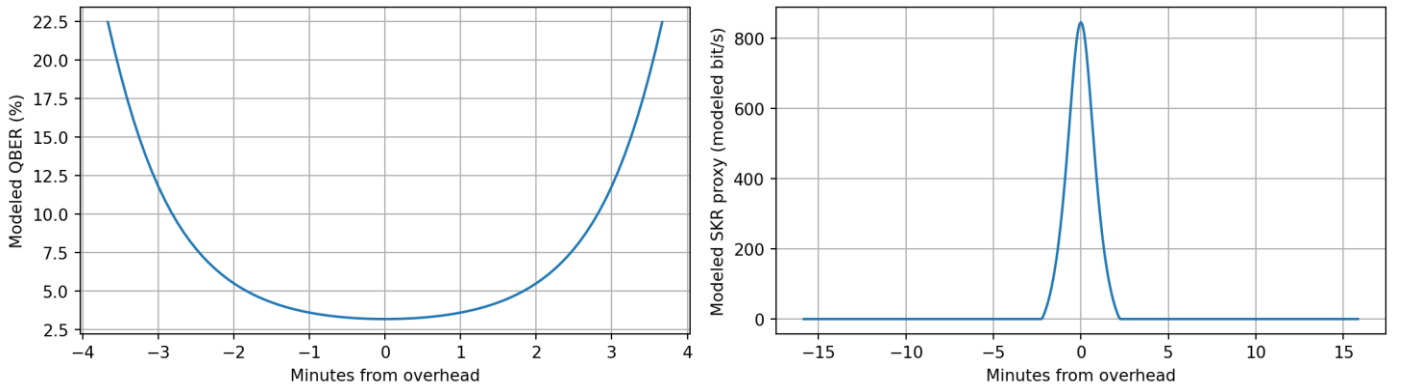


Figure 39. LEO pass: compact modeled QBER and modeled rate

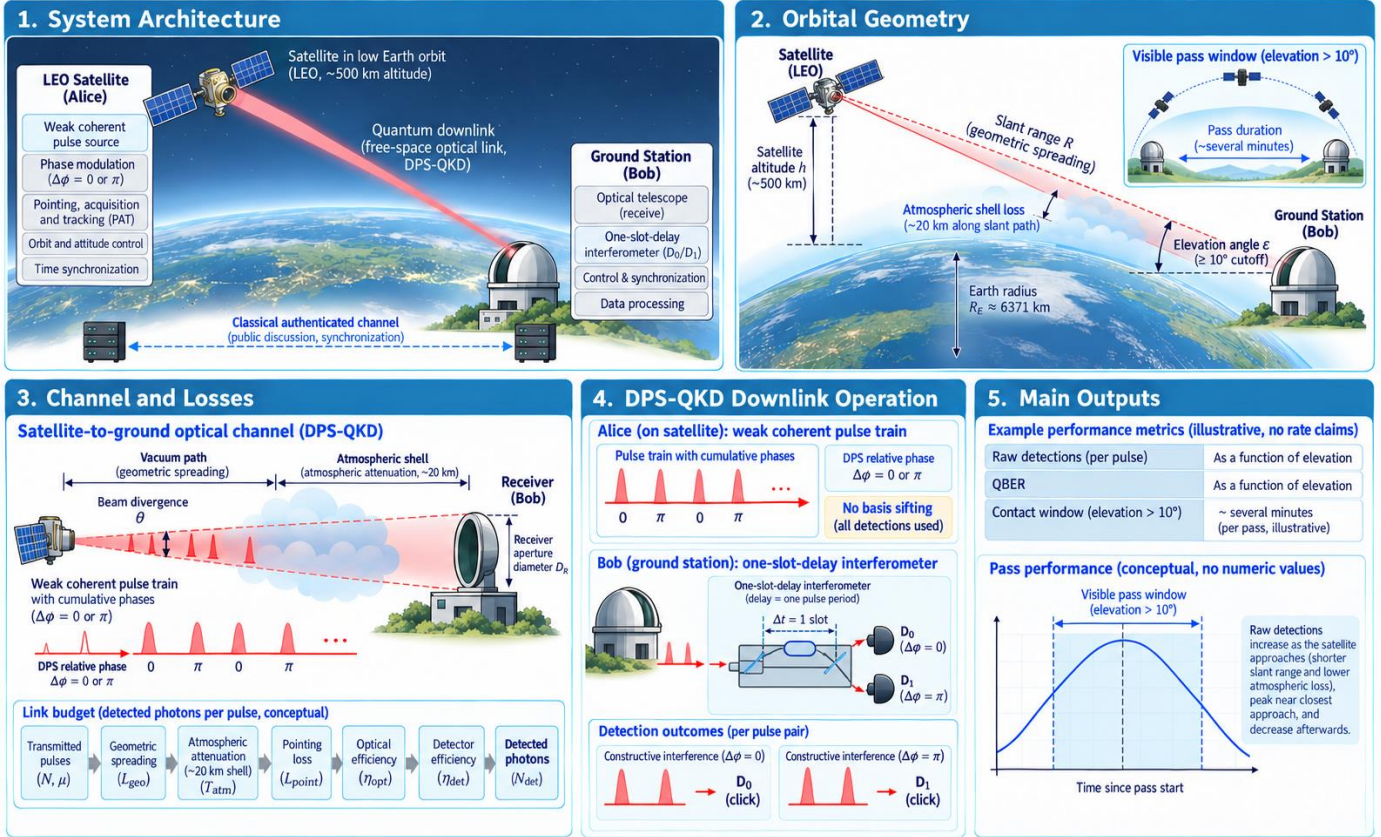


Figure 40. DPS LEO satellite-to-ground architecture

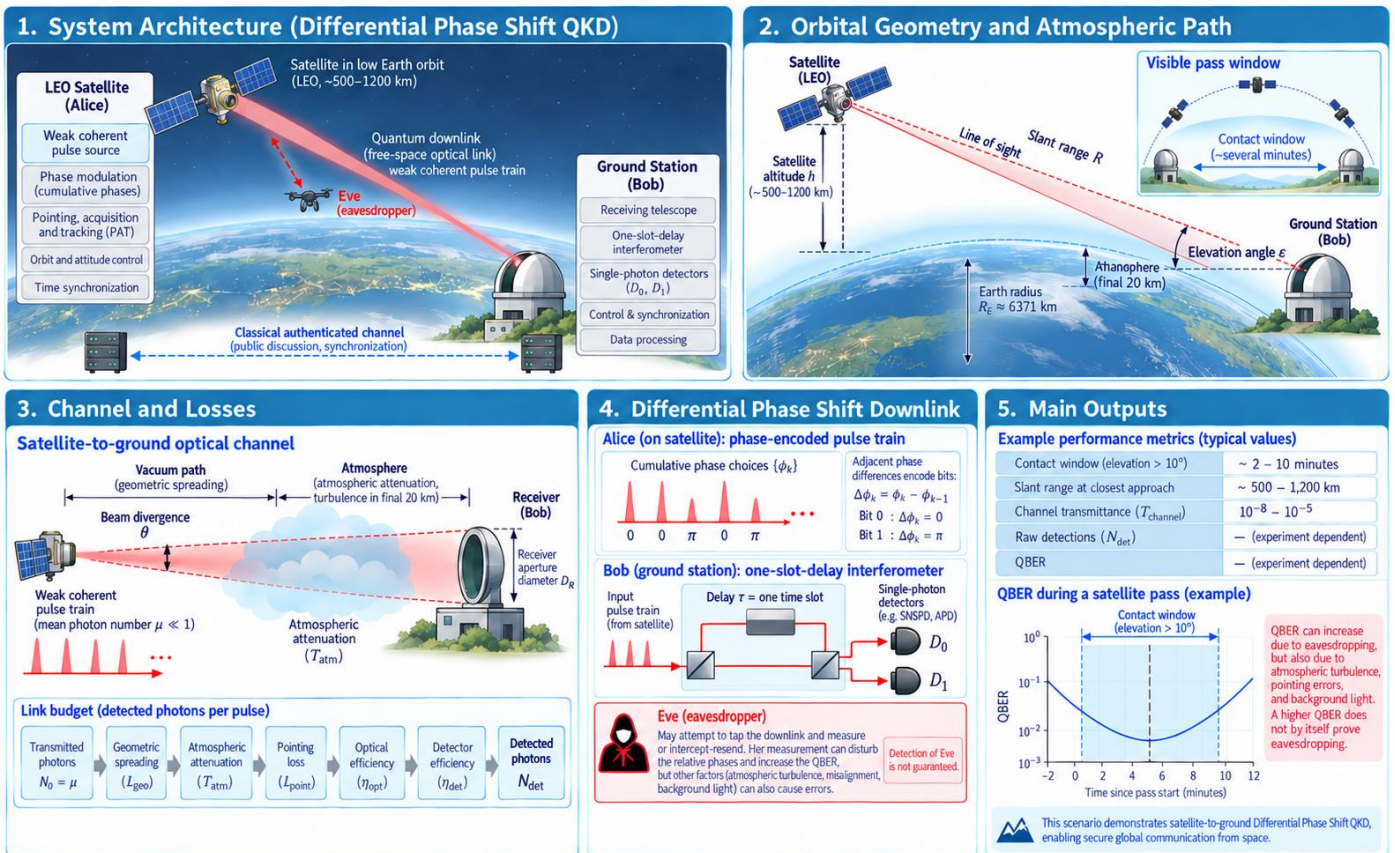


Figure 41. Satellite optical-link threats and non-adversarial error sources



Two Ground Stations and One Trusted Satellite Relay

A single satellite cannot be treated as a passive optical mirror in this model. Instead, it establishes two **independent QKD links** at different times:

$$A \leftrightarrow S$$

and

$$S \leftrightarrow B.$$

After full security processing, suppose the satellite shares:

- secure hop key K_{AS} with ground station A,
- secure hop key K_{SB} with ground station B.

The contact windows do **not** need to overlap because the trusted satellite can store the two keys.

How the Trusted-Key Relay Works

The satellite generates a new session key K and publishes:

$$C_A = K \oplus K_{AS}$$

and

$$C_B = K \oplus K_{SB}.$$

The published relay ciphertexts must be carried over an authenticated classical channel so that an adversary cannot substitute or modify the relay messages without detection.

A recovers:

$$K = C_A \oplus K_{AS},$$

while B recovers:

$$K = C_B \oplus K_{SB}.$$

The satellite knows K , so this architecture **requires trust in the relay** and secure key storage on the satellite.

If A-S produces B_A secure bits and S-B produces B_B secure bits, the maximum equal-length relayed key is bounded by:

$$B_{\text{relay}} \leq \min(B_A, B_B).$$

The weaker hop is the bottleneck because one one-time-pad bit is consumed on each hop for every relayed session-key bit.

Two Independent Satellite Contact Windows

For this example, ground station A and B are placed at Earth central angles of -17 degrees and +17 degrees relative to the satellite's midpoint pass.

The satellite therefore reaches A's useful elevation window first and B's later. We calculate each raw-detection opportunity separately.

A key point is that **simultaneous visibility is not required** in this trusted-store-and-forward architecture. What matters is how many secure hop-key bits each link can accumulate and store.

Two separate ground-station contact windows

Ground angles are -17 and +17 degrees in the synthetic plane. Each station has 88 visible samples and no simultaneous contact samples. Raw-detection budgets are integrated independently.

Example calculation

A contact samples: 88

B contact samples: 88

A and B simultaneous samples: 0

A raw-detection budget: 0.734 Mbit-equivalent detections

B raw-detection budget: 0.734 Mbit-equivalent detections

Raw bottleneck upper illustration: 0.734 Mbit

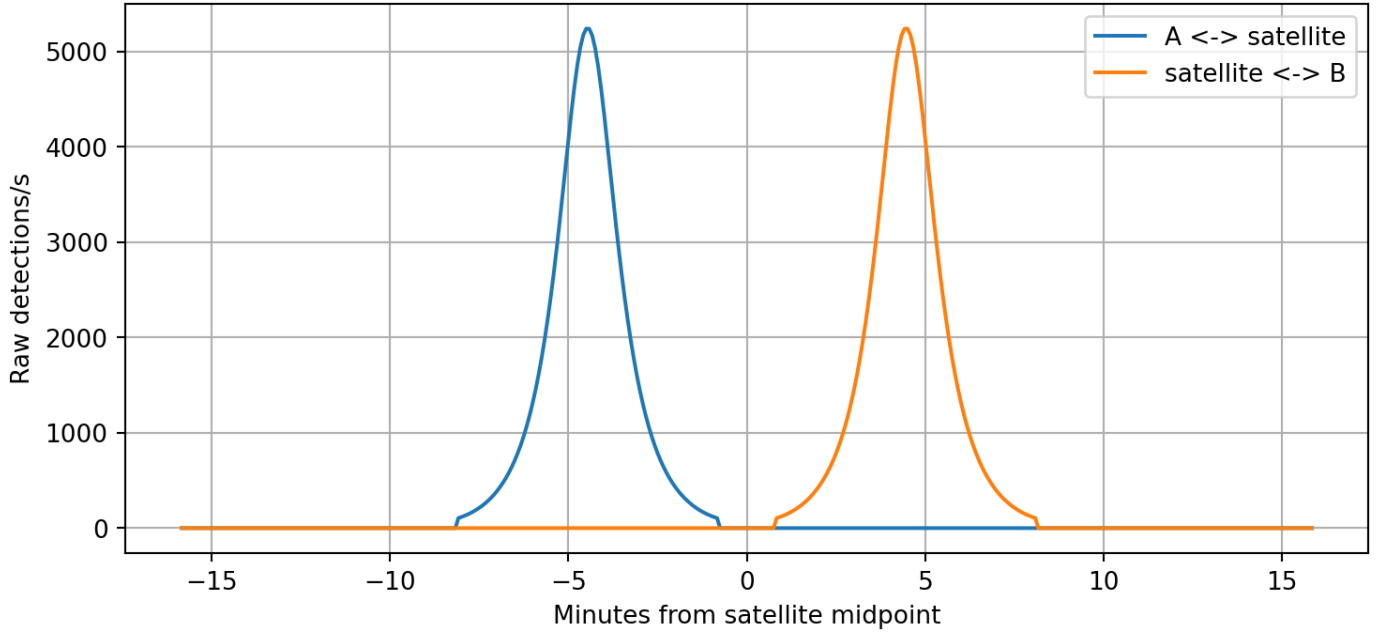


Figure 42. Nonoverlapping physical detection windows of the two relay ground links

Tiny One-Time-Pad Relay Example

To make the relay arithmetic visible, use fixed 8-bit toy keys.

Assume:

$$K_{AS} = 10110010,$$

$$K_{SB} = 01101100,$$

and the trusted satellite wants A and B to share:

$$K = 01010101.$$

The satellite sends two different ciphertexts because the hop keys are different. A and B each XOR their own ciphertext with their own hop key and recover the same K .

This demonstration is only the **relay arithmetic**. It does not generate K_{AS} or K_{SB} securely; those hop keys must first come from completed QKD post-processing.

Trusted relay XOR arithmetic

The relay encrypts the same session string with independent hop keys, one for each ground station. Both stations recover the session string, and the relay also knows it.

Example calculation

```
K_AS : 10110010
K_SB : 01101100
Session K : 01010101
C_A=K^K_AS : 11100111
C_B=K^K_SB : 00111001
A recovered : 01010101
B recovered : 01010101
A == B == K : True
Trusted relay also knows K: 01010101
```

Modeled Key Budget for Two Ground Stations

Raw detections are only an upper opportunity count. We therefore repeat the two-hop calculation using the modeled SKR proxy on each link separately.

For each hop:

$$B_A^{proxy} = \sum_i R_A(t_i) \Delta t,$$

$$B_B^{proxy} = \sum_i R_B(t_i) \Delta t.$$

The trusted-relay proxy budget is:

$$B_{\text{relay}}^{\text{proxy}} = \min(B_A^{\text{proxy}}, B_B^{\text{proxy}}).$$

This is an **asynchronous trusted-relay proxy**. The contact windows can be disjoint, but in a real deployment the satellite must store two genuinely secure, authenticated hop-key inventories.

Compact modeled two-hop budget

The minimum of the two integrated hop budgets is a trusted resource bound. There is no causal buffer, expiry or scheduling simulation.

Example calculation

A modeled proxy budget: 95,190 modeled bits
B modeled proxy budget: 95,190 modeled bits
Trusted relay bottleneck: 95,190 modeled proxy bits
Overlapping contact samples: 0

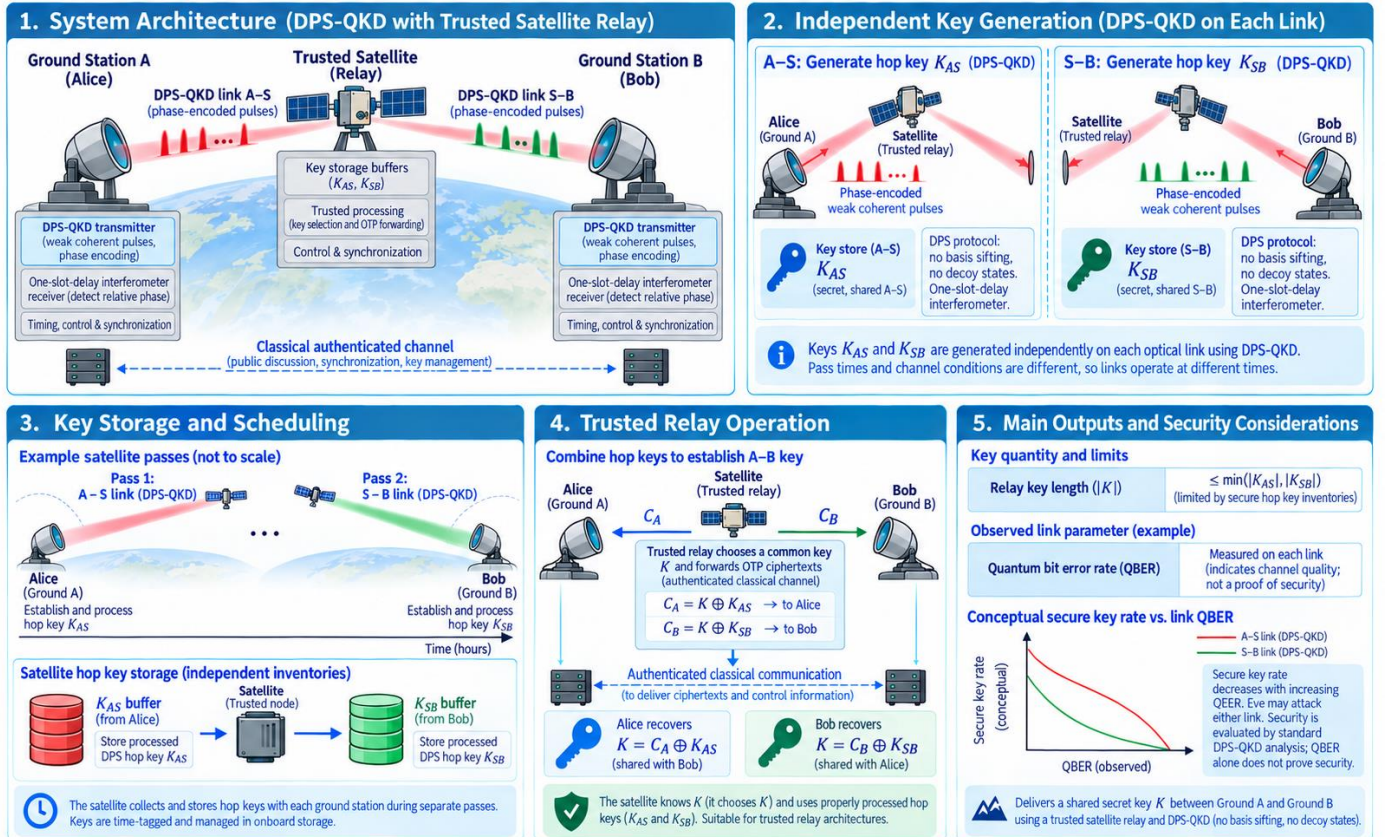


Figure 43. Two independent DPS links through a trusted satellite

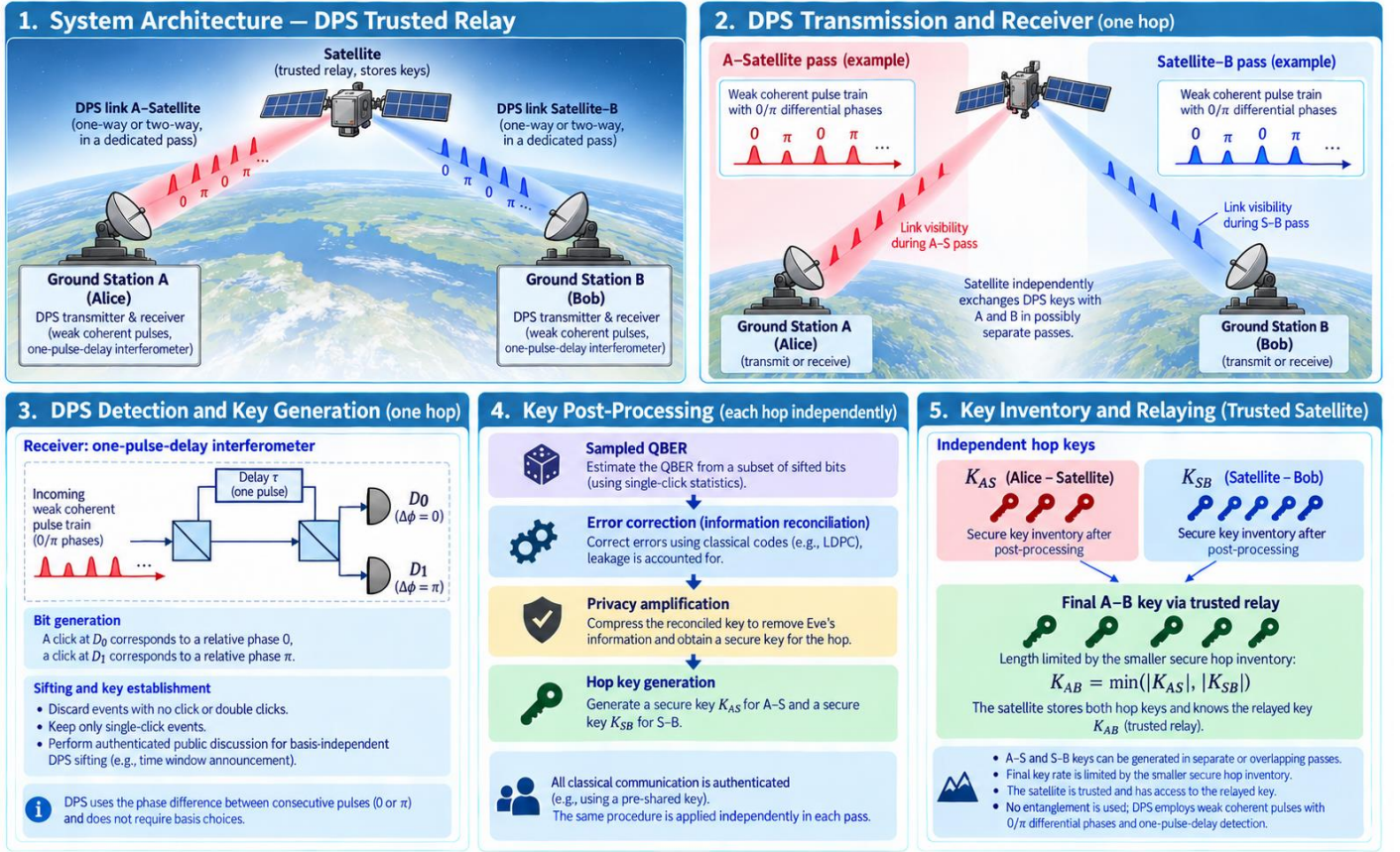


Figure 44. Independent ground contact windows and trusted key storage

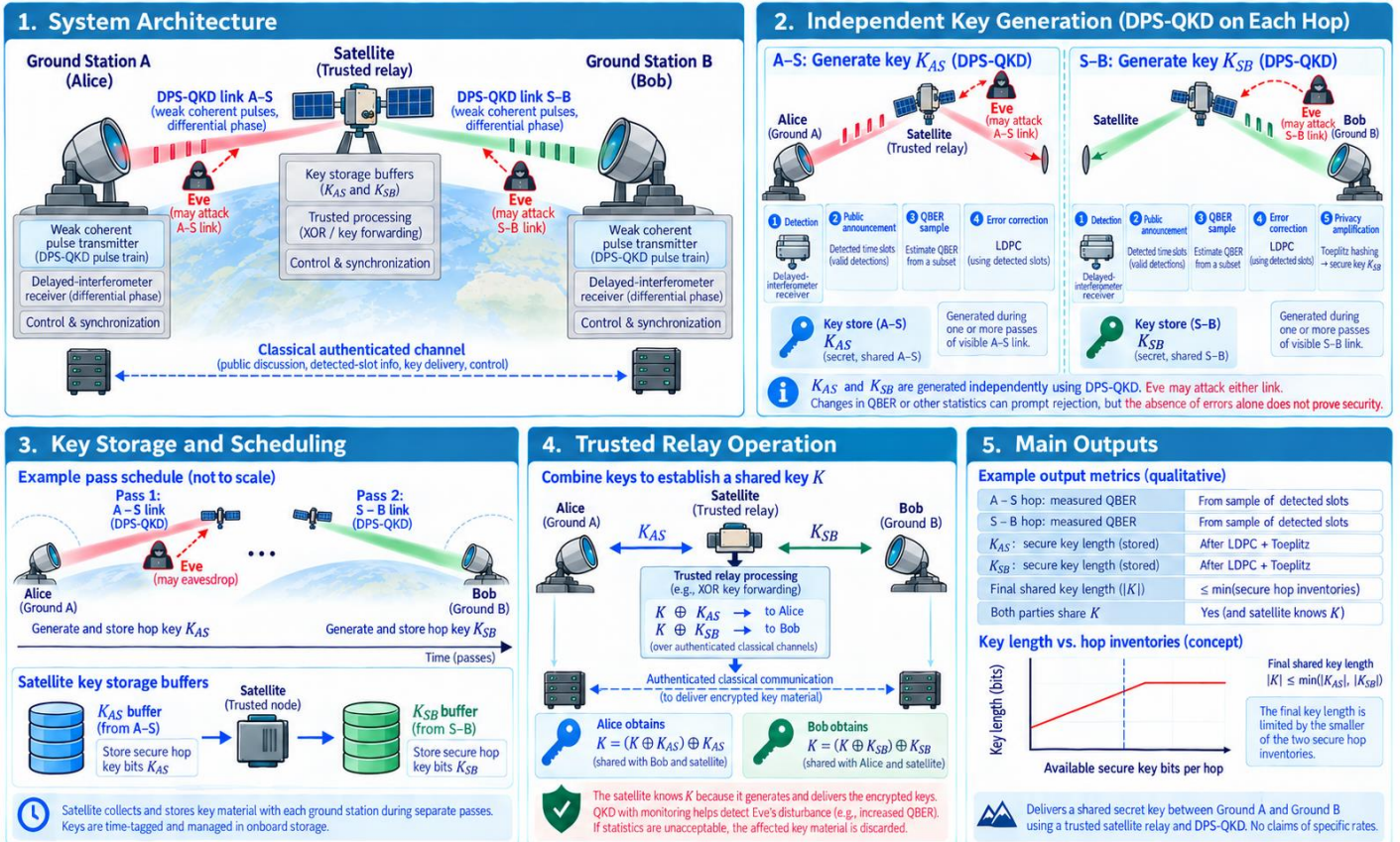


Figure 45. Independent security checks on the two trusted-relay hops

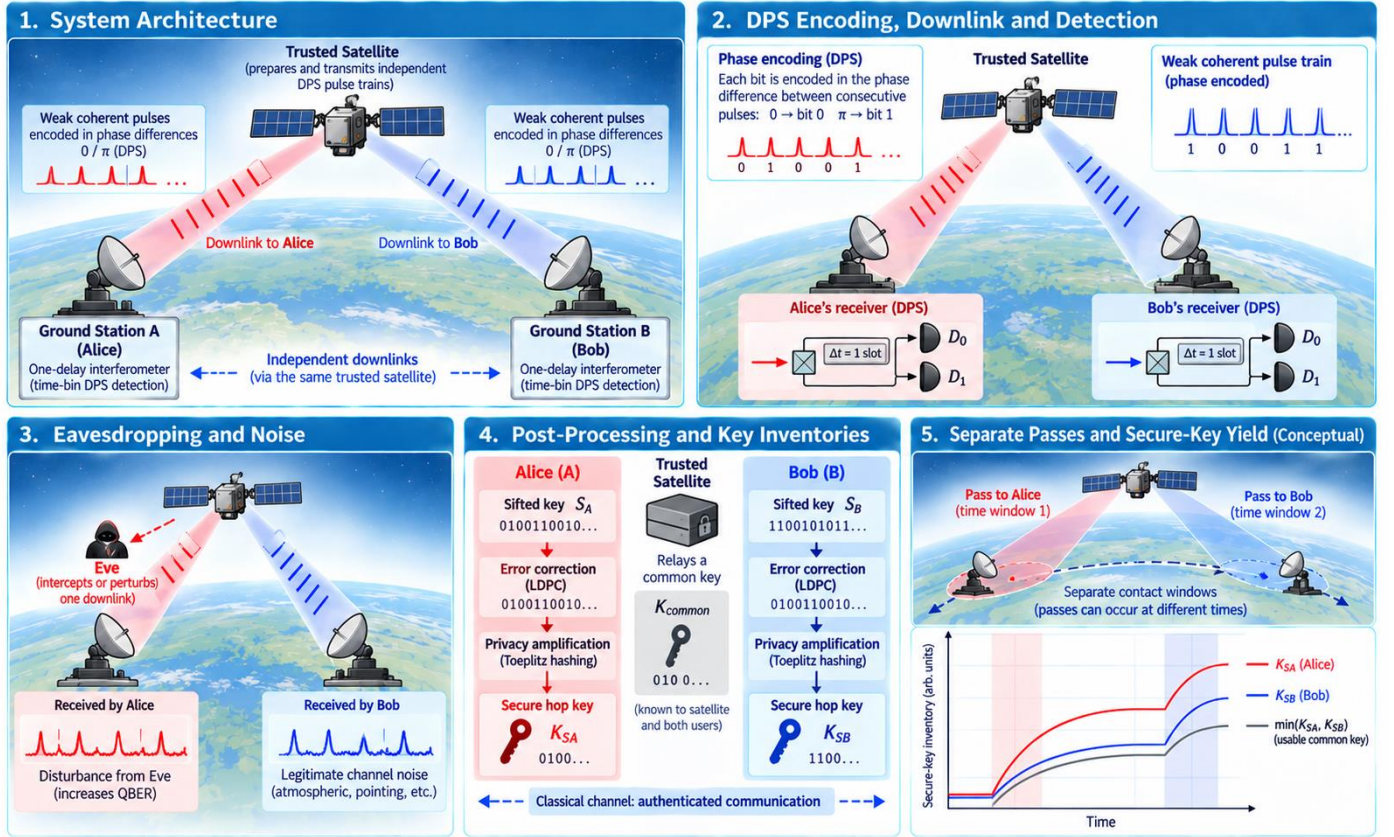


Figure 46. Separate DPS passes between two ground terminals in the trusted-relay scenario

Three-Satellite Trusted Network and Route Budgets

The final topology extends the trusted-relay idea to several satellites. Each optical edge accumulates its own key inventory:

$$A \leftrightarrow S_0, S_0 \leftrightarrow S_1, S_1 \leftrightarrow S_2, S_2 \leftrightarrow B.$$

Every satellite is treated as a **trusted key-storage and forwarding node**. This section does **not** model quantum repeaters, entanglement swapping or untrusted end-to-end QKD.

For an m -edge trusted path, every delivered one-time-pad bit consumes one secure bit on **each** edge. Therefore:

$$B_{path} \leq \min(B_1, B_2, \dots, B_m).$$

The path is constrained by its weakest accumulated hop inventory.

We compare three conceptual routes using satellites at simple coplanar phase offsets of -12, 0 and +12 degrees. These offsets are teaching geometry, not a real constellation ephemeris.

Check Inter-Satellite Line of Sight

For two satellites on the same circular orbit with angular separation $\Delta\theta$, the straight-line chord distance is:

$$L_{ISL} = 2r_{orb} \sin\left(\frac{|\Delta\theta|}{2}\right).$$

The closest point of that chord to Earth's center is:

$$r_{min} = r_{orb} \left| \cos\left(\frac{\Delta\theta}{2}\right) \right|.$$

The optical link clears Earth when:

$$r_{min} > R_{Earth}.$$

For our 500 km orbit:

- 12 degree separation gives about 1,436 km inter-satellite distance;
- 24 degree separation gives about 2,857 km;
- Both chords still clear Earth in this simplified geometry.

The inter-satellite optical link has **no atmospheric attenuation** in this model, but still has geometric spreading, pointing loss, optics efficiency and detector efficiency.

Check inter-satellite geometry explicitly

At both 12 and 24 degree separation, the chord clears Earth at the 500 km orbit. The 24 degree S0–S2 link is geometrically visible; a zero modeled key rate later is not evidence of Earth obstruction.

Example calculation

Static inter-satellite geometry examples

separation=12 deg: distance= 1436.4 km, Earth clearance= 462.4 km, LOS=True

separation=24 deg: distance= 2857.1 km, Earth clearance= 349.9 km, LOS=True

Build Raw-Detection Budgets for Every Edge

The three satellite orbital phases are sampled across the same 32-minute study interval used earlier.

For ground-satellite edges we reuse the atmospheric downlink model.

For satellite-satellite edges we use:

$$\eta_{atm} = 1$$

because the path is modeled in vacuum, and then apply geometric collection, 1 dB pointing loss, 0.7 optics efficiency and 0.5 detector efficiency.

For each edge we integrate the modeled raw detection rate over time:

$$B_{edge}^{raw} \approx \sum_i R_{edge}(t_i) \Delta t.$$

For each candidate route, the raw upper illustration is the minimum edge budget.

Integrate raw-detection budgets by edge

Six ground edges and three candidate inter-satellite edges form three named paths. Raw-detection-equivalent bottlenecks are not secret key budgets and routes cannot be summed without shared-resource accounting.

Example calculation

one hop through S0 edges [0.73, 0.73] Mbit-equivalent; bottleneck <= 0.73 Mbit

two satellites S0,S2 edges [0.73, 0.45, 0.73] Mbit-equivalent; bottleneck <= 0.45 Mbit

three satellites S0,S1,S2 edges [0.73, 1.55, 1.55, 0.73] Mbit-equivalent; bottleneck <= 0.73 Mbit

Multi-Hop Modeled Key Budgets

Raw detection counts do not include privacy and error-correction penalties. The final study therefore replaces each raw edge rate with the same modeled SKR proxy used earlier.

For every route:

- Calculate each edge's time-dependent η ,
- Evaluate the compact DPS model,
- Integrate the modeled proxy rate for that edge,
- Take the minimum accumulated edge budget.

A route can therefore have many raw detections but still have a **zero modeled proxy** on one weak edge if the modeled rate becomes negative and is clipped to zero.

Routes are compared but not summed because different routes can share the same satellite nodes or key inventories. A real network scheduler would have to track inventory consumption explicitly.

Integrate compact modeled budgets by edge

The longer direct S0–S2 link has a positive raw count budget but zero compact modeled rate. The adjacent two-link chain retains positive modeled rate. Each path is the minimum independently integrated edge budget, not a time-dependent widest-path solution.

Example calculation

one hop through S0 : bottleneck 95,191 modeled proxy bits

edge budgets: {'A-S0': 95192, 'S0-B': 95191}

two satellites S0,S2 : bottleneck 0 modeled proxy bits

edge budgets: {'A-S0': 95192, 'S0-S2': 0, 'S2-B': 95192}

three satellites S0,S1,S2 : bottleneck 74,009 modeled proxy bits

edge budgets: {'A-S0': 95192, 'S0-S1': 74009, 'S1-S2': 74009, 'S2-B': 95192}

No route value here certifies a finite end-to-end key.

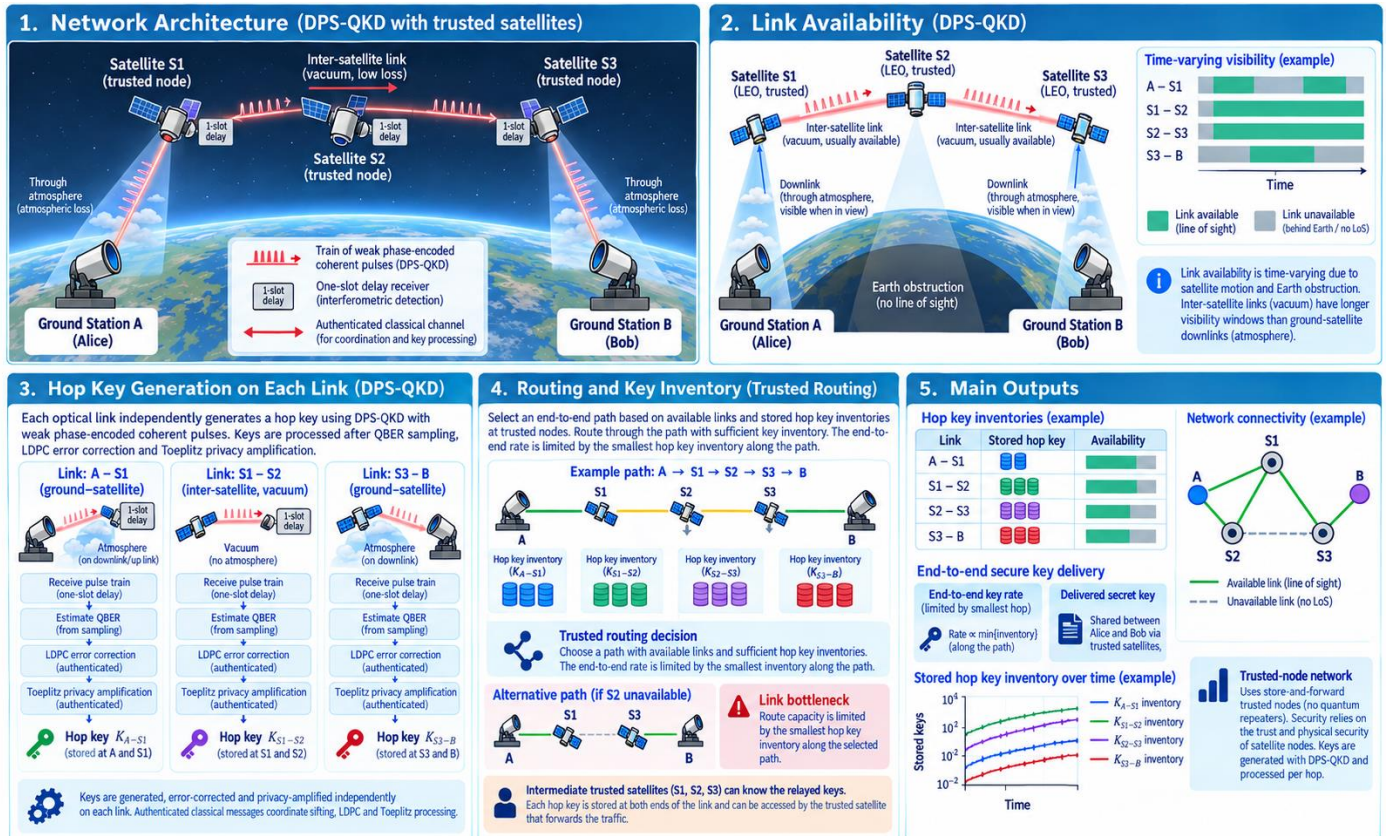


Figure 47. Three-satellite trusted DPS network and link-key budgets

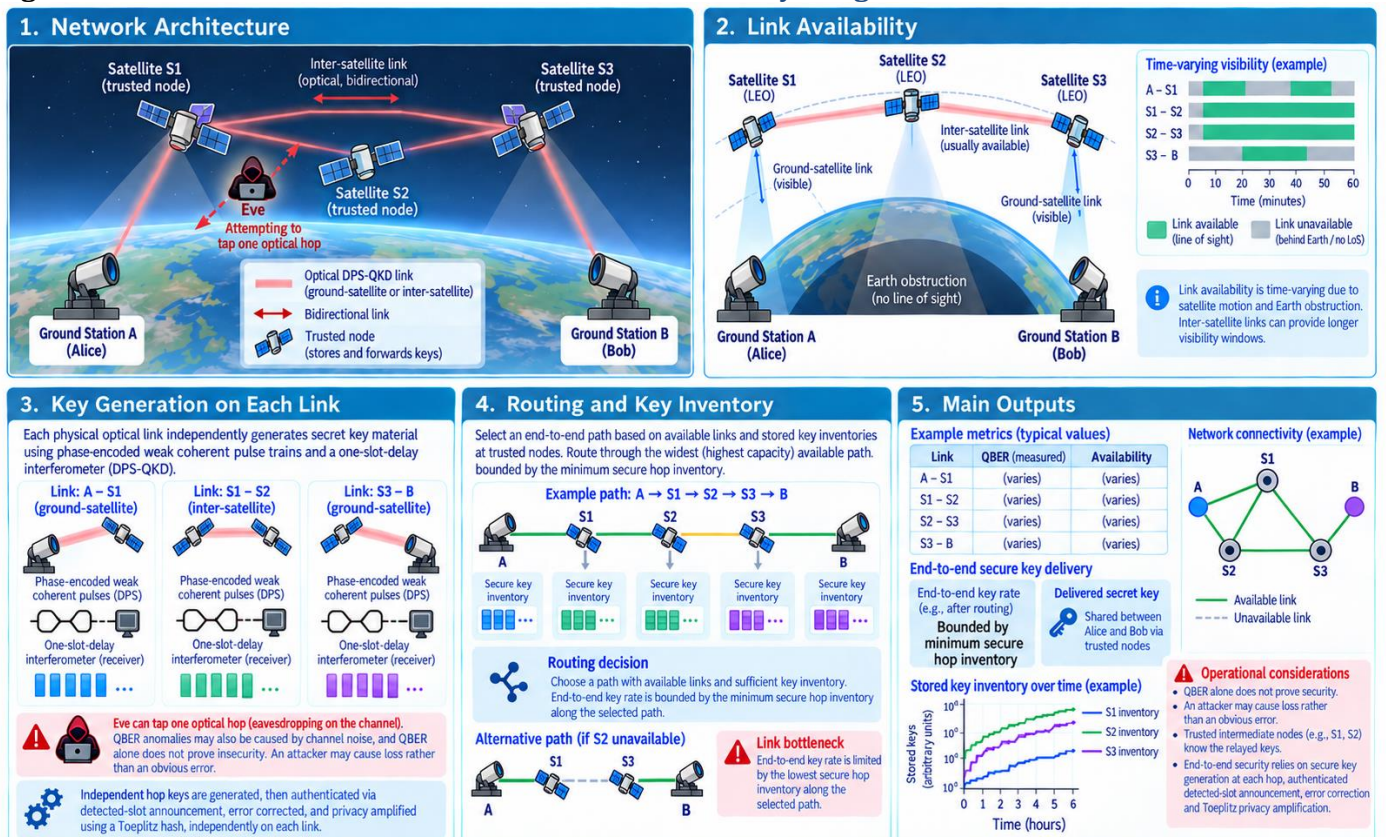


Figure 48. Threats on ground and inter-satellite optical links



Banking Application Case Study: Bahrain Sample Bank

The following application case study places the four DPS-FSO simulation scenarios in the context of Bahrain Sample Bank (BSB), a fictional institution. The site locations, workloads, connectivity and continuity requirements are planning assumptions from the original report, not an audited customer network or a deployed quantum-security claim. No telecom-route survey, bank equipment inventory or measured site-specific optical model is available.

Case study environment and existing connectivity

BSB operates a primary data center at its headquarters in Seef, Bahrain, a disaster recovery (DR) site in Amwaj, Bahrain, and an offshore computing site in Riyadh, Saudi Arabia. Core banking workloads run on IBM i (AS/400), while the mobile banking environment uses Oracle. The case assumes that these workloads and their data reside on IBM storage infrastructure, with data synchronized from headquarters to the other computing sites. The precise IBM storage models, replication software, database replication topology, bandwidth and encryption termination points are not specified and therefore are not assumed in my analysis. The existing HQ-to-DR data connection is a dedicated telecom-provider fiber service. Traffic from Bahrain to Riyadh is carried over site-to-site VPNs on the public internet. Numerous branches in Bahrain and Saudi Arabia also use site-to-site VPN connectivity. The approximately 14-15 km Seef-Amwaj and 420 km Bahrain-Riyadh figures are straight-line separations, not surveyed fiber route lengths or free-space satellite optical path lengths.

Table 17. Assumed BSB sites and connectivity

Location / traffic path	Distance or role	Existing service / project assumption
Seef headquarters (HQ)	Primary production data center	IBM i core banking, Oracle mobile banking and IBM storage; primary replication source.
Amwaj disaster recovery (DR)	Approximately 14-15 km geodesic separation from HQ	Dedicated carrier fiber; receives replicated data and assumes production during planned BCP drills.
Riyadh offshore site	Approximately 420 km geodesic separation from Bahrain HQ	Site-to-site VPN over public internet; offshore replicated computing and storage.
Bahrain and KSA branch offices	Multiple sites; individual distances not specified	Site-to-site VPNs; branch traffic is separate from the HQ-DR storage replication stream.



Figure 49. Banking case-study overview, geographic context and existing connectivity assumptions.



Operational context and quantum-related security problem

BSB performs two planned business continuity planning (BCP) drills each year. During each drill, production workloads operate from Amwaj DR for approximately two weeks before a controlled return to Seef HQ. Encryption and key management must therefore function when replication direction and production endpoints change. The design must protect HQ-to-DR synchronization, DR-to-HQ reverse synchronization, Bahrain-to-Riyadh transfers and relevant branch traffic without making banking availability dependent on clear weather or a satellite pass.

The application motivation is long-term confidentiality of archived traffic. QKD is considered as an additional potential source of key material; it does not encrypt the replication payload. Authenticated symmetric encryption and a validated key-establishment and authentication design remain necessary. Post-quantum or hybrid key establishment on conventional VPN infrastructure is a separate candidate workstream; the simulation does not implement it or assess current vendor support.

Application objective and specific research questions

My application objective is to evaluate a phased, quantum-safe key-establishment and data-in-transit encryption architecture for BSB that protects the replication of IBM i core banking and Oracle mobile banking information among Seef HQ, Amwaj DR and Riyadh, while preserving two-week DR operation and allowing selective expansion to branch connections.

- Map the assumed bank links to the four current terrestrial and trusted satellite scenarios while distinguishing generic geometry from real bank locations.
- Evaluate optical efficiency, physical and modeled QBER, rate proxies, sampled outage definitions and integrated link budgets under explicit assumptions.
- Distinguish QKD key delivery from AES-256 authenticated encryption of the actual replication stream and identify practical KMS, HSM, encryption-gateway and authentication interfaces.
- Assess trusted-relay and satellite-operator exposure explicitly; the current international-link models use DPS trusted nodes rather than an entangled-pair BBM92 link.
- Define key continuity, reverse-replication and fallback requirements for the bank's two annual, two-week DR production exercises.
- Compare dedicated QKD at selected data centres with conventional post-quantum VPN upgrades at branches, without claiming unmeasured performance, cost savings or regulatory compliance.

Cryptographic protection boundary and integration model

The bank's storage and application replication flows remain conventional data traffic. In the proposed architecture, supported encryption gateways or compatible application/transport security endpoints apply authenticated symmetric encryption, for example a correctly configured AES-256-based authenticated mode. A key-management layer identifies authorized endpoint pairs, handles key identifiers, rotation, access control, expiry, monitoring and audit, and transfers key material to the compatible encryption endpoints. I do not assume that IBM i, Oracle, IBM storage controllers or existing VPN appliances natively ingest QKD keys; those interfaces and vendors must be examined before a proof of concept.

Where compatible QKD equipment exists, separate optical quantum links could supply key material to authorized key-management entities. The original report identifies ETSI GS QKD 014 as a possible application-interface reference; actual device conformance and integration would need to be demonstrated. Existing fiber or IP paths would continue carrying encrypted replication traffic. A carrier data service does not automatically expose a transparent optical path suitable for QKD, and a public-internet VPN is not a quantum optical channel.

QKD-derived key material and a supported post-quantum key-establishment result could be incorporated under an explicitly validated hybrid key-derivation and authentication design. Merely concatenating keys or claiming a fallback is quantum-safe without a security analysis would be insufficient. Key generation, key usage and rekey intervals must be sized using measured application traffic, equipment limits and outage requirements. The present simulation does not demonstrate full-rate one-time-pad encryption, production replication throughput or an approved hybrid cryptographic protocol.

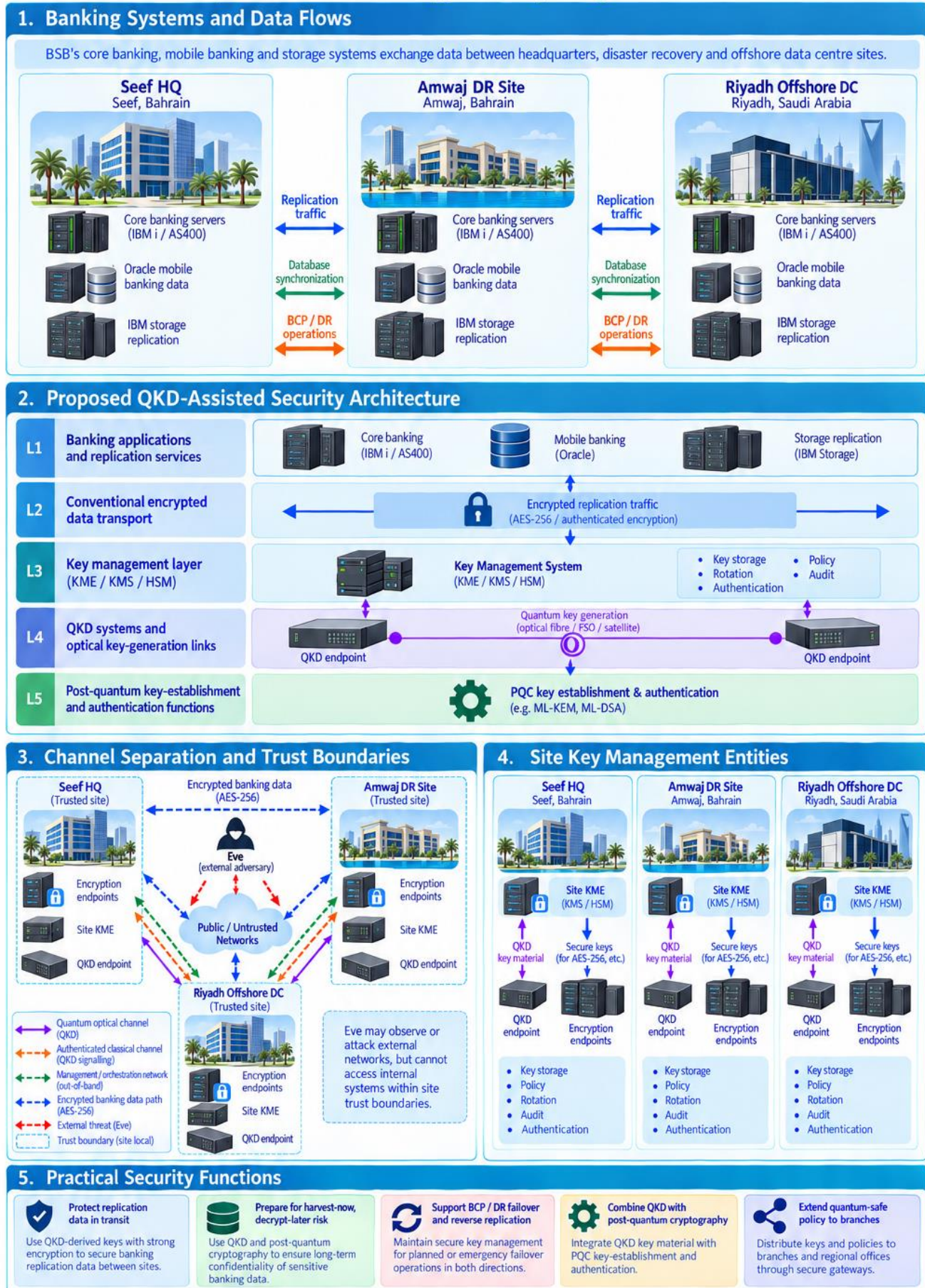


Figure 50. Proposed QKD-assisted banking security architecture and trust boundaries.



Mapping the four current DPS scenarios to BSB infrastructure

Scenario 1: Seef headquarters to Amwaj disaster recovery

It represents the short terrestrial connection between HQ and DR. My existing 20 km terrestrial example supplies a bounding educational comparison for the assumed 14-15 km separation, but I have not labeled that baseline as a direct 15 km BSB site simulation. The Gaussian optical model and DPS/log-normal turbulence extension enable evaluation of a possible rooftop FSO quantum channel, subject to a verified unobstructed optical line of sight, building permissions, alignment, dust, haze, weather and daylight background. The bank's existing telecom fiber provides a second potential physical-QKD route only if the provider can supply a compatible optical path and the fiber-specific model is separately evaluated. Encryption of IBM i/Oracle replication remains on the existing high-capacity transport, not on the quantum-photon stream.

Scenario 2: Satellite link with one Bahrain terminal

It models a LEO satellite communicating with one Bahrain ground terminal at a time. I use changing elevation, slant range, finite atmospheric-shell loss, telescope collection, QBER and contact-window integration to study whether a satellite service could replenish a local key pool. A single downlink establishes a key between that satellite and its ground receiver; it does not alone establish a shared Bahrain-Riyadh key. Actual ground coordinates, satellite ephemerides, weather availability, provider equipment and a service contract would be needed for deployment. Intermittent visibility makes authenticated local key storage and outage policy central design parameters.

Scenario 3: Bahrain-Riyadh trusted satellite relay

The trusted-relay concept establishes independent link keys with two ground terminals and assumes trusted satellite storage between separate contacts. The Bahrain-Riyadh VPN continues to carry encrypted replication traffic. Operator trust, key custody, authentication and deployment arrangements require separate assessment. The current $\pm 17^\circ$ station geometry is a generic demonstration, not a simulation of these actual sites. Its approximately 95,190 modeled proxy bits are a minimum integrated hop budget, not a scheduled or certified end-to-end key delivery result.

Scenario 4: Multiple trusted satellites

The current network studies three satellites and three fixed routes. It computes deterministic ground-link windows, Earth clearance for inter-satellite links and minimum integrated edge budgets. It does not implement the earlier report's simultaneous widest-path selection or a scheduled constellation with key inventories. Its banking relevance is a possible future path for replenishing authorized site keys, contingent on actual service access and trusted-node arrangements. Branches connected by ordinary VPN do not automatically obtain end-to-end QKD.

Table 18. Current DPS scenario mapping for the bank

Project stage	BSB use case	Deployment condition / evidence boundary
Scenario 1 — ground	Seef-Amwaj key-establishment study	Generic 20 km FSO baseline; needs surveyed site optics or a separate fiber model.
Scenario 2 — LEO	One-terminal satellite replenishment	Actual ephemeris and terminal required; one downlink does not establish a Riyadh key.
Scenario 3 — trusted relay	Two-site trusted key relay	Generic $\pm 17^\circ$ geometry; integrated bottleneck, no causal inventory simulation.
Scenario 4 — network	Fixed routes through trusted satellites	Three satellites; integrated edge minima, no schedule or resource allocation.

Phased implementation concept

I distinguish the four numerical research stages from proposed bank implementation phases. The following phases are a forward-looking plan: none is represented as a completed BSB production deployment.



Decisions about capital expenditure, supplier selection and regulatory approval remain outside the numerical simulation.

Table 19. Phased implementation and acceptance conditions

Phase	Scope of work	Acceptance condition
0 - baseline	Inventory IBM i, Oracle, storage replication, VPN crypto, certificate lifecycle, traffic and data-retention needs.	Document existing encryption endpoints, key-establishment exposure, replication directions and business requirements.
1 - PQC pilot	Evaluate supported quantum-resistant or hybrid VPN key establishment between Bahrain and Riyadh and selected branches.	Validated interoperability, authenticated handshakes and acceptable VPN availability.
2 - local QKD PoC	Survey Seef-Amwaj fiber suitability and FSO line of sight; test physical QKD where practical.	Measured optical loss, QBER, usable key rate, uptime and environmental availability.
3 - encryption integration	Test vendor-supported KMS/HSM and encryption-gateway delivery to the replication transport.	Confirmed key lifecycle, encryption correctness and replication throughput/lag within targets.
4 - BCP exercise	Run planned failover, two-week DR operation, reverse replication and restoration with monitored keys.	No unapproved downgrade; agreed recovery objectives and key-continuity requirements met.
5 - satellite pilot	Investigate available trusted DPS-relay suppliers, terminal siting and satellite passes.	Proven access, security assumptions, measured rates and operator trust/compliance assessments.
6 - expansion	Assess multiple trusted satellites and a branch-oriented quantum-safe connectivity strategy.	Demonstrated operating model and justified total cost of ownership.

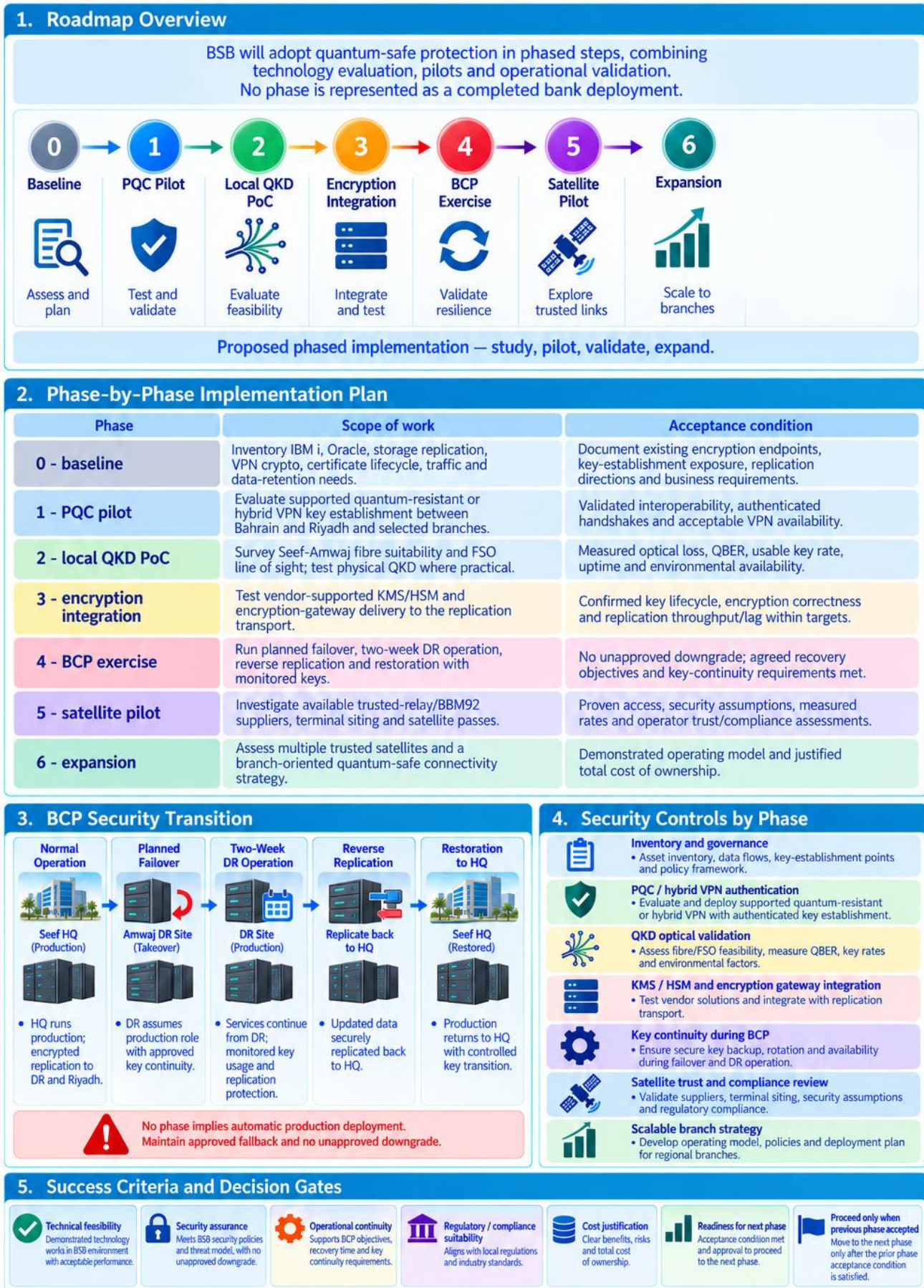


Figure 51. Proposed phased implementation roadmap and BCP security transition.



Branch-office extension and cost boundaries

For BSB branches, extending quantum-safe cryptography through supported VPN software and compatible gateways may reuse existing IP connectivity and avoid optical terminals at every location. Such branch tunnels may use a properly supported post-quantum or hybrid key-establishment mechanism, but they are not end-to-end QKD links. A branch can use a centrally governed KMS only if delivery, authorization and communication with that KMS remain suitably authenticated and protected. A central QKD link cannot automatically make ordinary public-internet branch traffic quantum-key-distributed.

Table 20. Branch quantum-safe architecture options and cost boundaries.

Architecture	Appropriate case-study scope	Additional cost drivers and limitations
PQC/hybrid VPN at branches	Many existing Bahrain/KSA VPN branches	Gateway support/licensing, firmware, certificates, monitoring, interoperability and managed migration.
QKD-assisted central sites	HQ, DR and potentially Riyadh	Optical terminals, suitable fiber or telescope sites, KMS/HSM, encryption appliances and operations.
Dedicated QKD to every branch	Only branches with justified high-security needs and a real optical path	Additional quantum endpoints, circuits, alignment/service contracts, redundancy and support; ordinary VPN is insufficient.

No BSB-specific cost estimate or claim of branch savings can be established from the available case-study inputs. Before costing, the bank would need branch counts, VPN platform inventory, actual carrier service types, optical-route surveys, required key-refresh policy, encryption throughput, storage replication licensing, satellite service availability and support pricing. The bank's Bahrain and Saudi legal, privacy, banking supervisory and outsourcing obligations also require a jurisdiction- and licence-specific assessment rather than an assumed compliance conclusion.

The case study is a phased engineering proposal. The current calculations cannot demonstrate that QKD has protected bank traffic or sustained a two-week disaster-recovery exercise. Conventional post-quantum protection and any additional terrestrial or satellite QKD service need separate implementation and validation. All DPS rate integrals reported here remain illustrative proxy quantities.

GNU Octave Dashboards

The following figures show the GNU Octave GUI dashboards developed for the four numerical QKD scenarios. The codebase is designed to remain MATLAB-compatible, although the project validation reported here was performed in GNU Octave. Choose a scenario, edit a parameter group, press Run simulation, and select a plot page to display the corresponding dashboard.

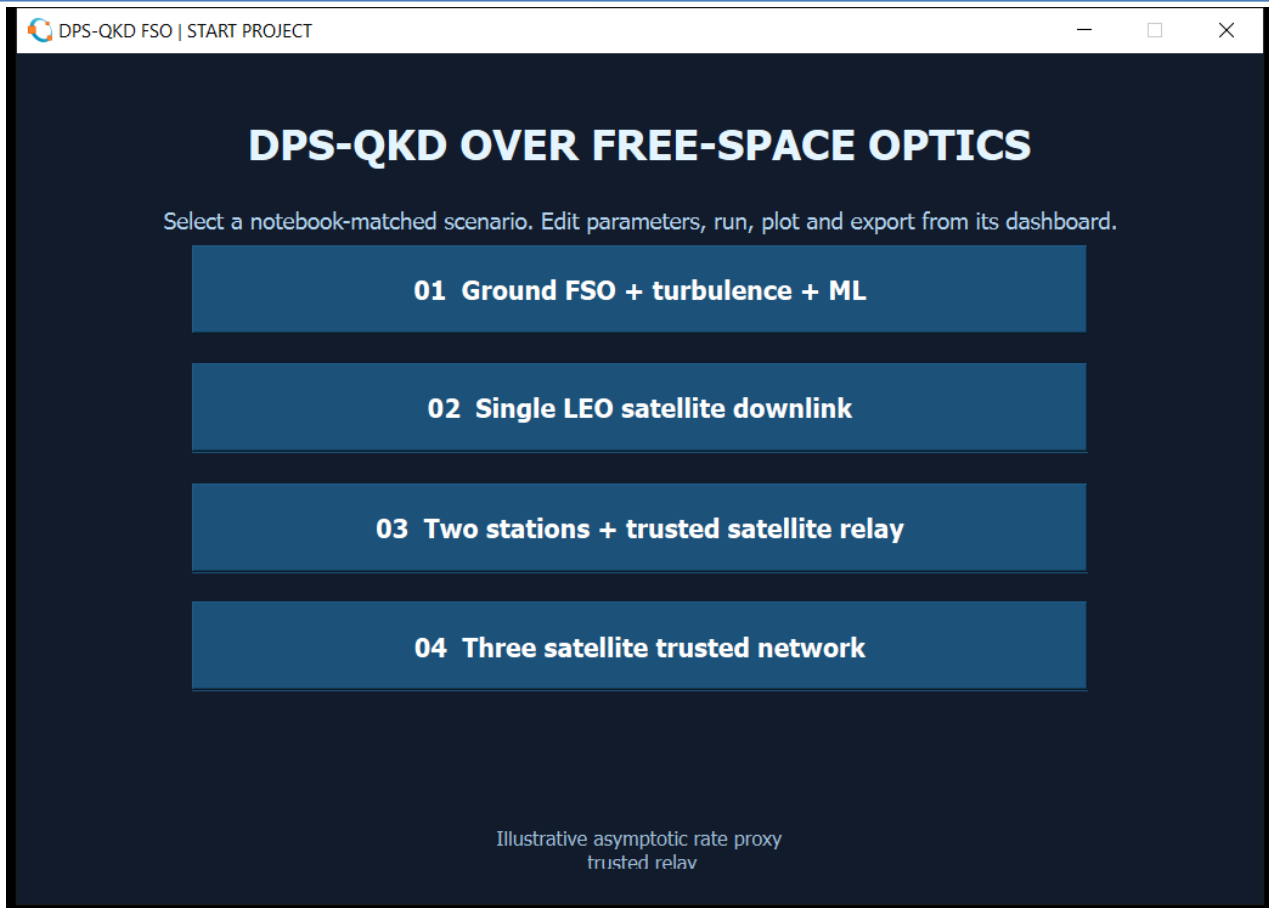


Figure 52. Project launcher: four numerical scenarios

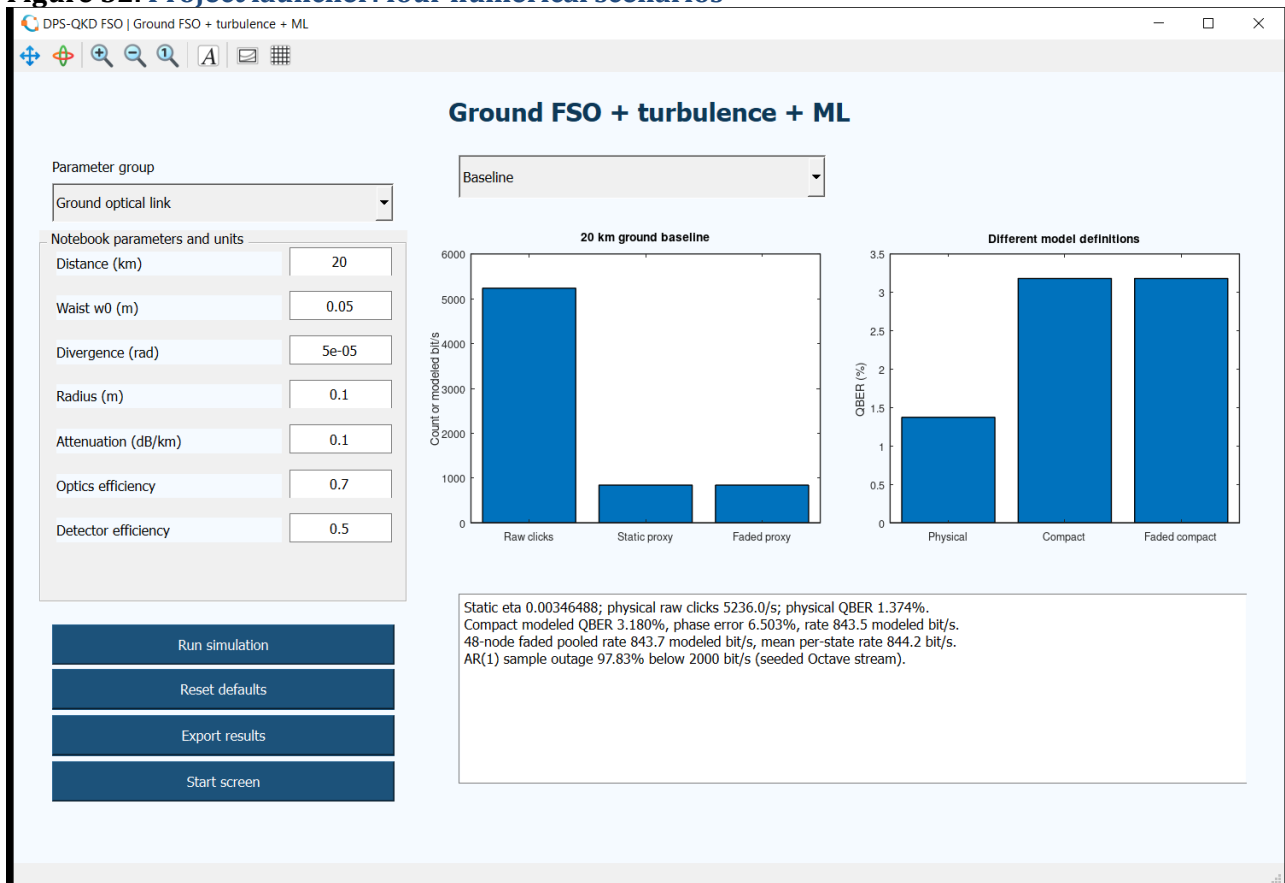


Figure 53. Ground dashboard: physical and compact baseline results

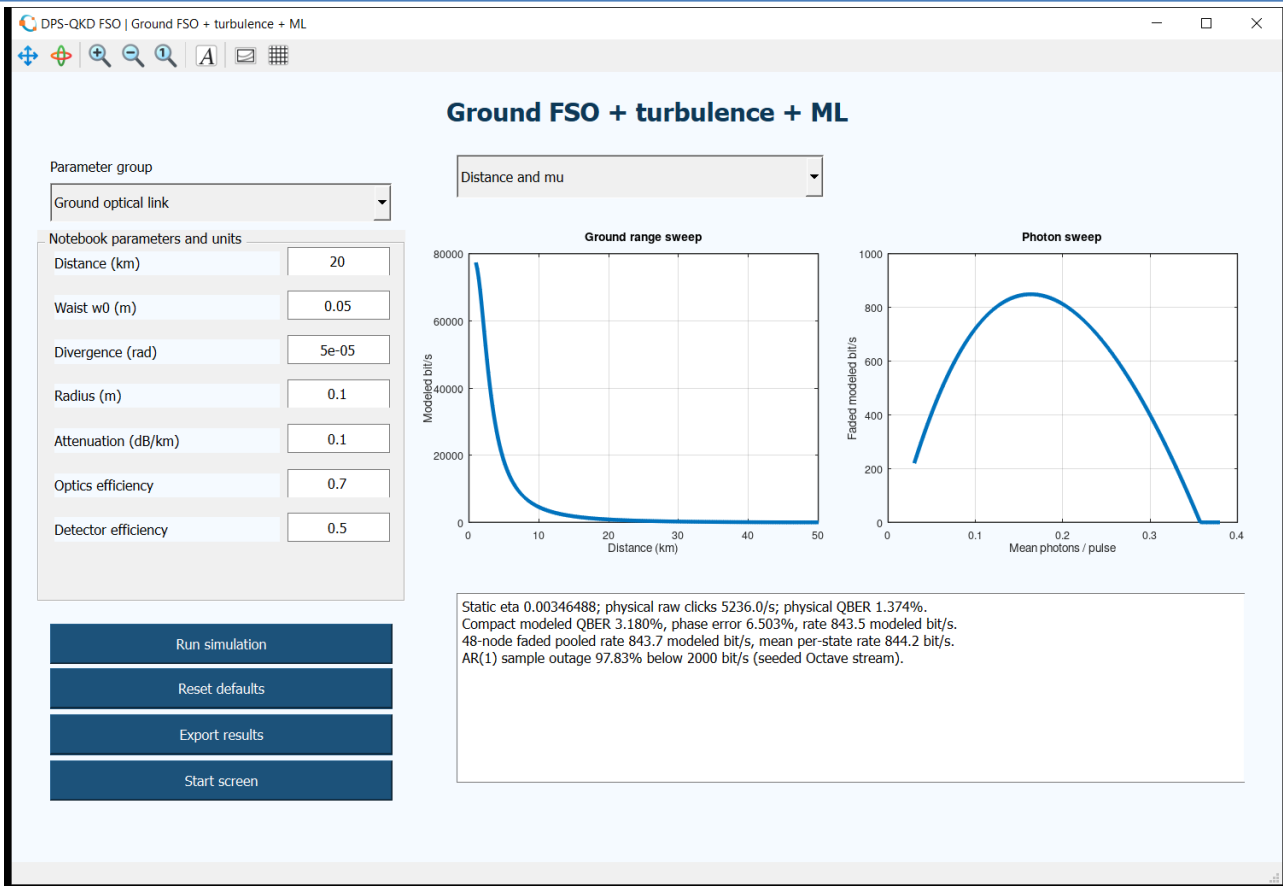


Figure 54. Ground dashboard: distance and intensity sweeps

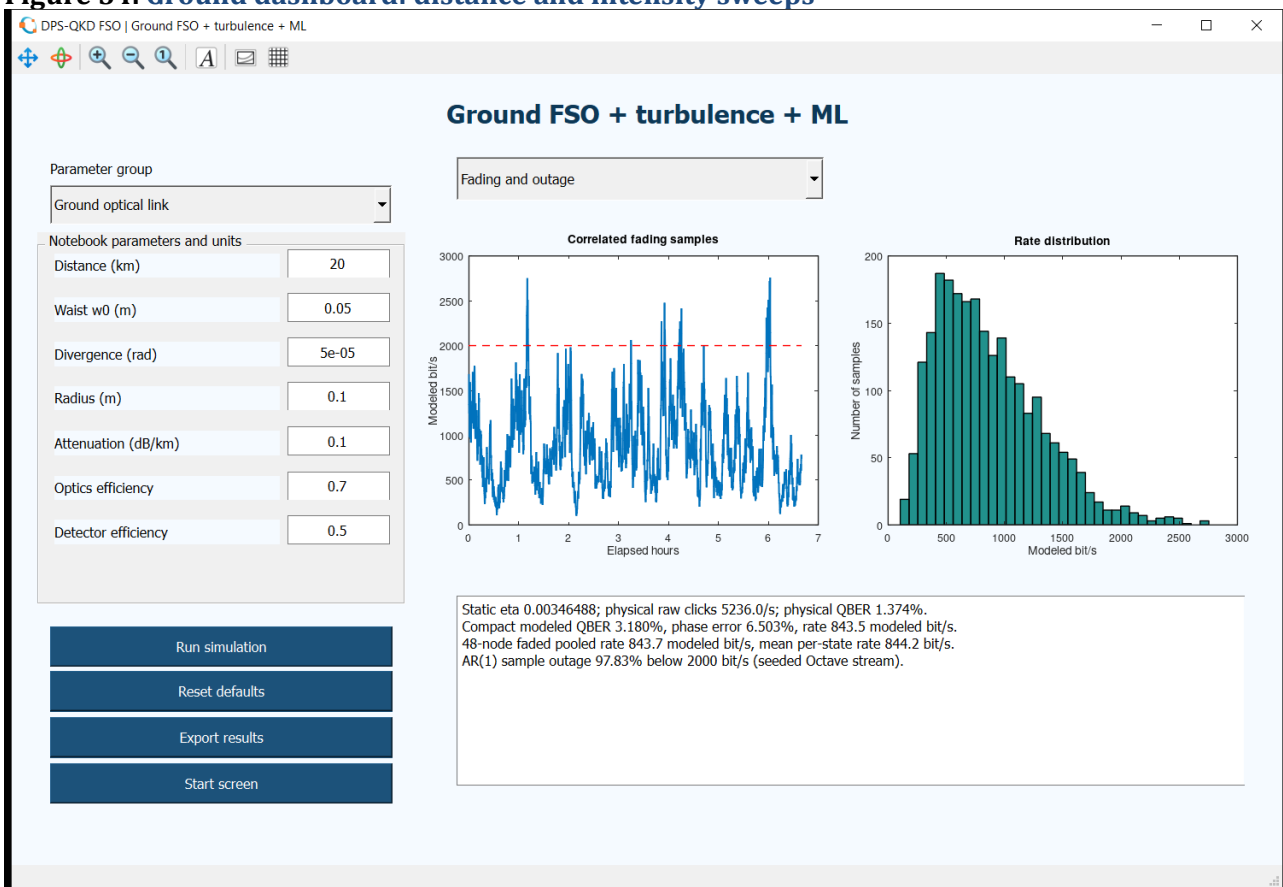


Figure 55. Ground dashboard: correlated modeled-rate outage

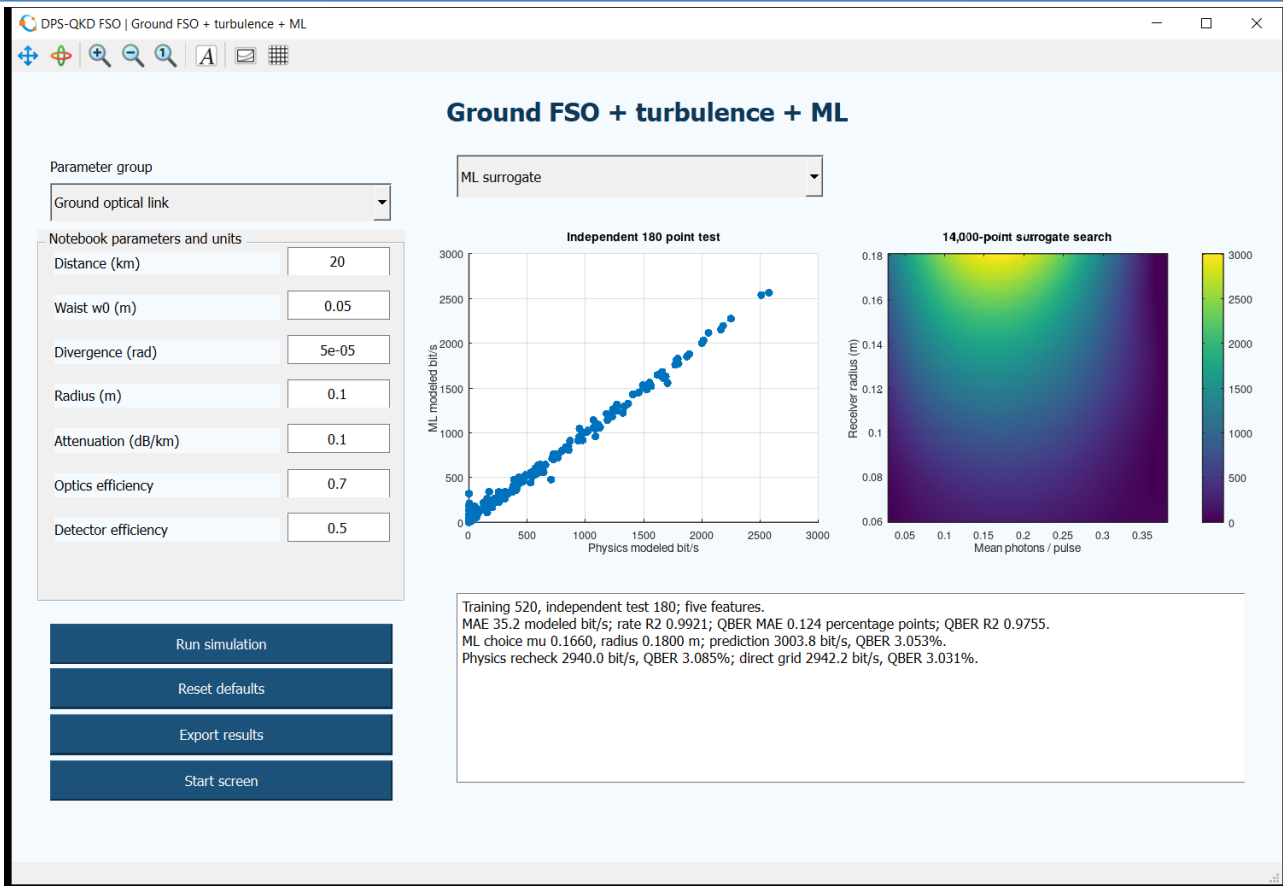


Figure 56. Ground dashboard: surrogate validation and design search

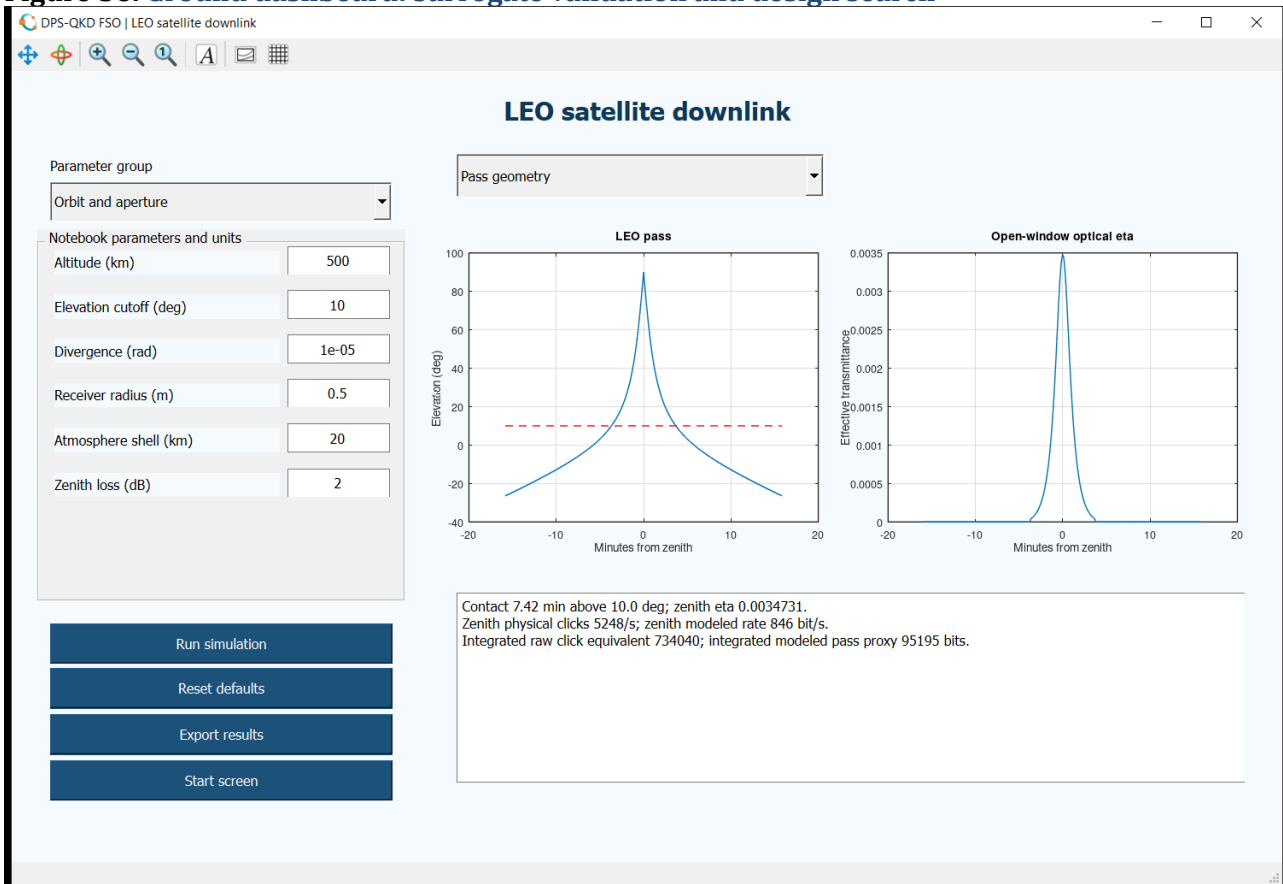


Figure 57. LEO dashboard: elevation and effective efficiency

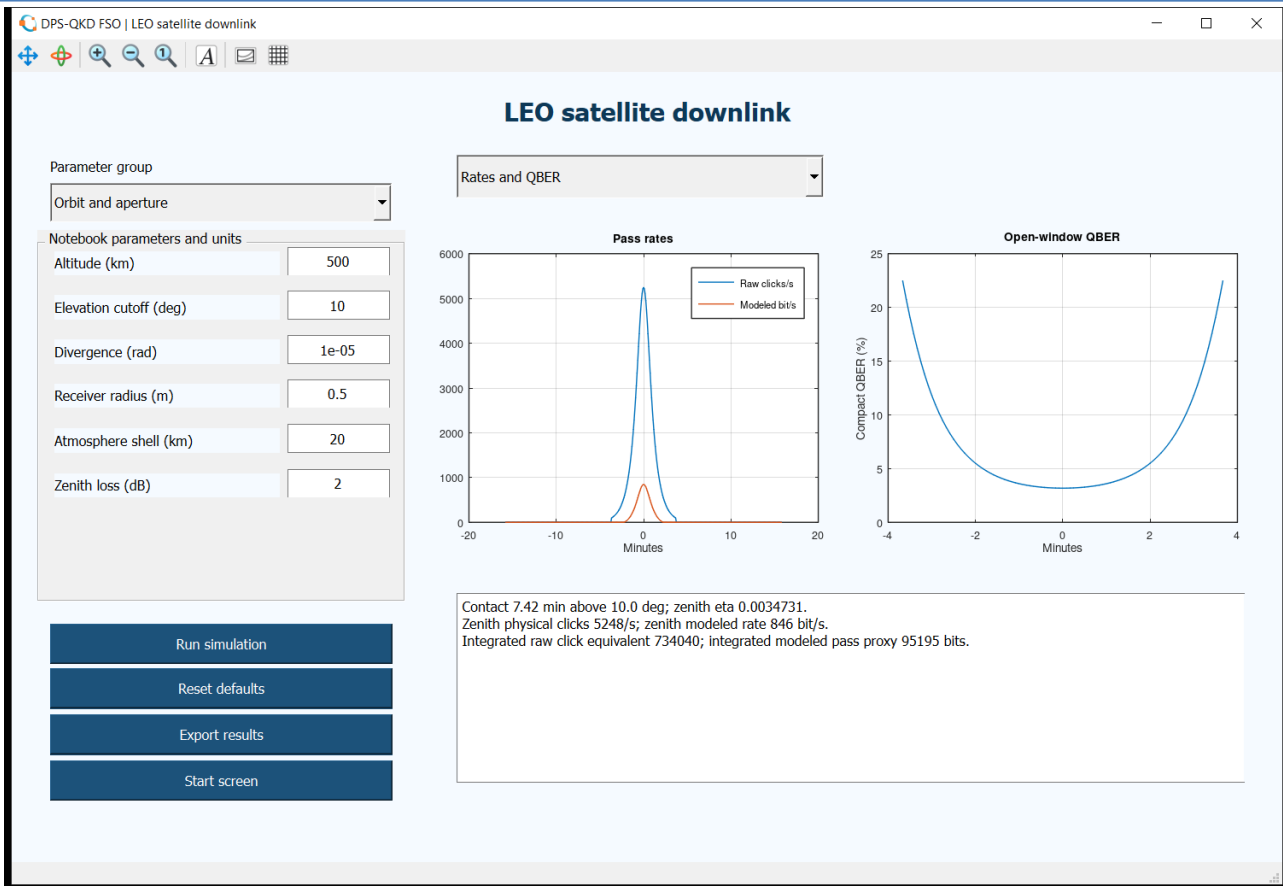


Figure 58. LEO dashboard: raw clicks, modeled rate and compact QBER

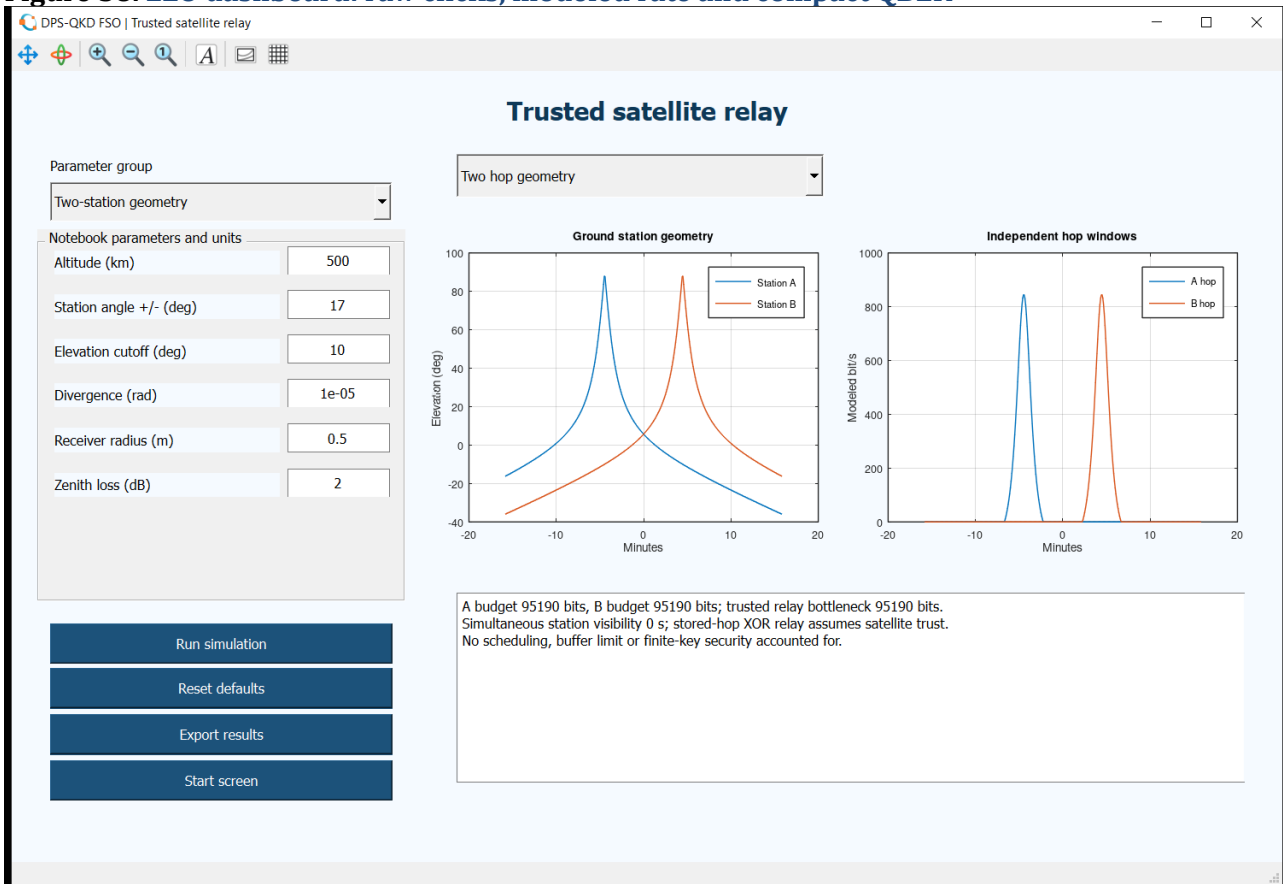


Figure 59. Trusted relay dashboard: geometry and separated windows

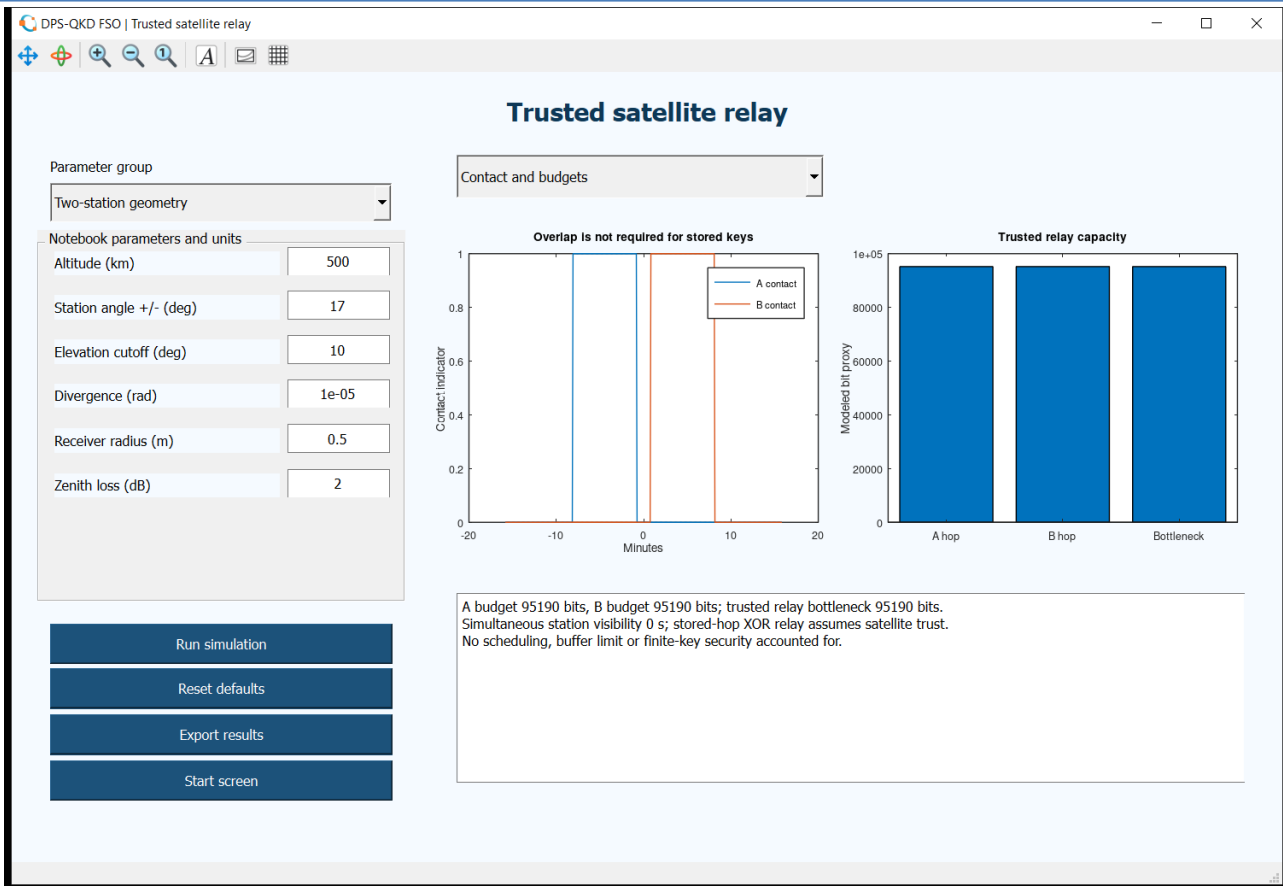


Figure 60. Trusted relay dashboard: integrated hop budgets

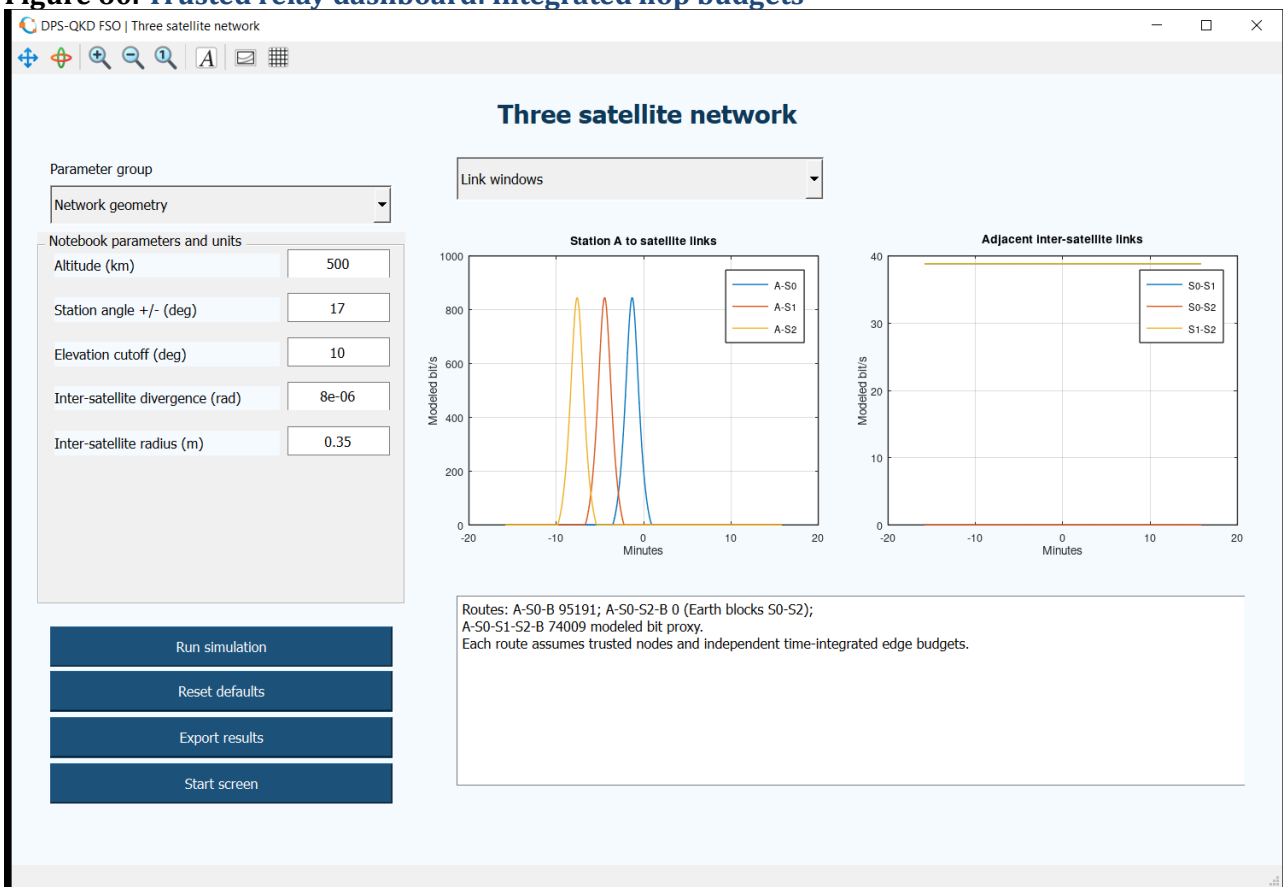


Figure 61. Network dashboard: ground-link windows and inter-satellite rates

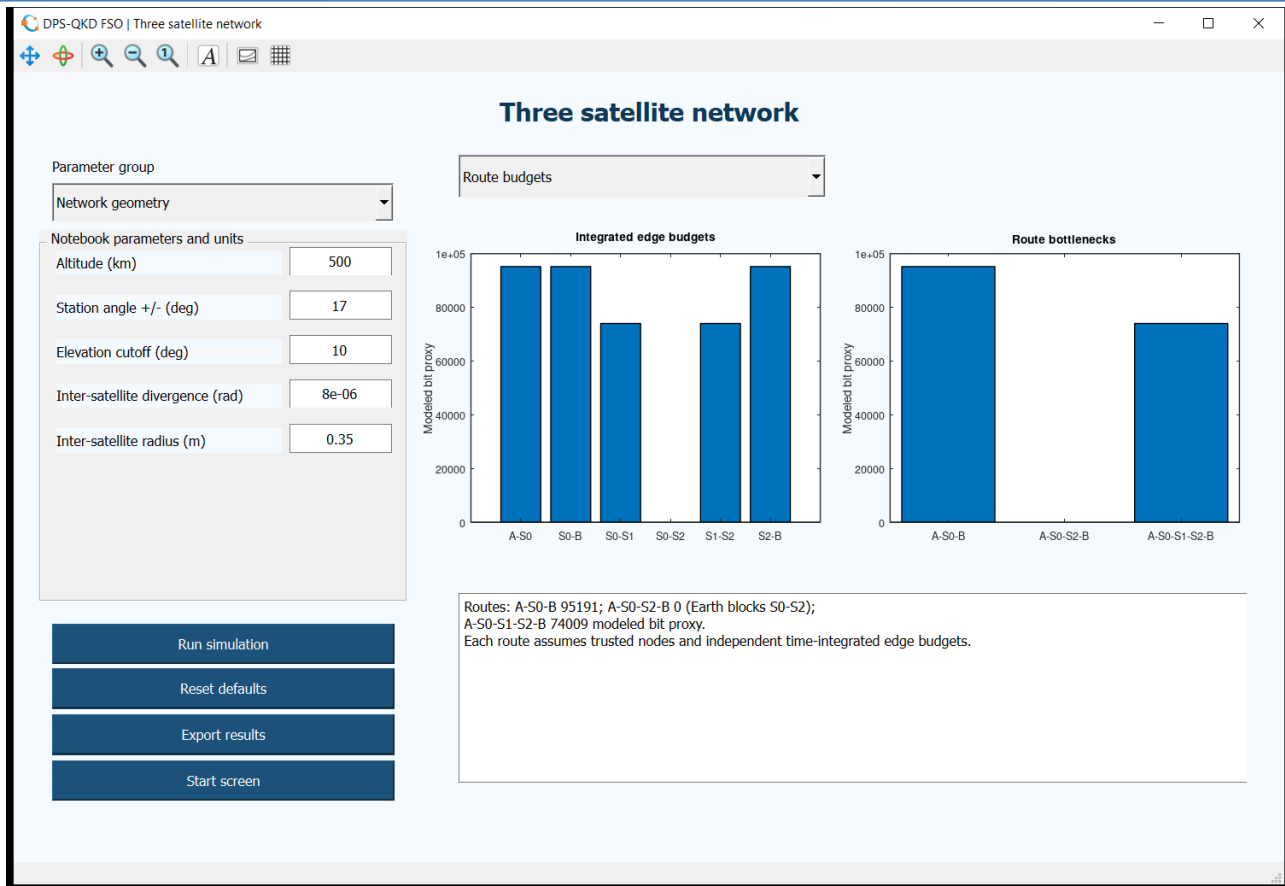


Figure 62. Network dashboard: integrated edge and route budgets

Consolidated Results and Discussion

Project-Scope Summary

The core project is the DPS-QKD QBER and modeled rate study for the terrestrial FSO link, extended to turbulence and ML design search. The satellite and trusted network chapters reuse the model for additional scenarios.

The BB84 and BBM92 examples introduce protocol ideas; the numerical DPS-FSO study supplies the project results.

Consolidated Results

Table 21. Consolidated numerical results and interpretation.

Study	Current numerical result	Interpretation
20 km physical receiver	5,235.96 detections/s; QBER 1.374%	Independent two-detector click model.
20 km compact model	QBER 3.180%; 843.5 modeled bit/s	Separate photon-number and phase-proxy approximation.
48-node fading model	843.7 pooled; 844.2 mean conditional modeled bit/s	Different orders of averaging and nonlinear rate evaluation.
ML proposal, physics checked	2,940.0 modeled bit/s; QBER 3.085%	$\mu = 0.1660$ and receiver radius 0.18 m.
Direct finite-grid optimum	2,942.2 modeled bit/s; QBER 3.031%	$\mu = 0.1609$ and receiver radius 0.18 m.
Single LEO pass	7.42 min sampled contact; 95,195 proxy bits	Deterministic pass and five-second sample sum.



One trusted relay	95,190 proxy bits; zero simultaneous contact	Minimum of independent integrated hop budgets.
Three-satellite chain	74,009 proxy bits	Adjacent inter-satellite edges limit the integrated route.

These results answer different engineering questions. Rates and accumulated budgets have different units, and the ML proposal changes receiver aperture relative to the baseline. The study therefore compares mechanisms and assumptions rather than ranking all values as equivalent measures of secure service. The underlying references provide protocol and post-processing context; numerical agreement verifies this implementation without supplying an implementation-matched security proof.

Conclusion and Future Work

This project developed a reproducible DPS-QKD simulation framework that connects protocol-level concepts to free-space optical propagation, detector behavior, atmospheric fading, machine-learning-assisted design search and trusted satellite extensions. The terrestrial 20 km study provides the baseline link budget; the turbulence model demonstrates state-by-state nonlinear evaluation and detection-weighted error pooling; the ML surrogate accelerates design exploration while retaining a physics-model recheck; and the satellite and banking sections show how the same framework can be extended to wider-area key-distribution architectures. All reported key-rate quantities remain engineering proxies under the stated assumptions rather than certified finite-key security bounds.

Answers to the Research Questions

1. Parameter sensitivity: within the model, increasing receiver aperture and interferometer visibility improves useful detection performance, while greater distance, background probability and deep fading generally reduce available signal and can increase QBER. Mean photon number has a trade-off: values that are too small reduce useful detections, whereas larger values increase the multiphoton fraction.
2. Order of averaging: a fluctuating channel should not in general be replaced by one average transmission value before nonlinear detector and entropy calculations. The report therefore evaluates each fading state first and pools detection/error statistics afterward. At the present weak-to-moderate baseline the pooled and mean-conditional rate proxies are close, but this numerical closeness is scenario-specific rather than a general equivalence.
3. ML-assisted search: the RBF kernel-ridge surrogate reproduces the simulated five-parameter performance surface with high held-out agreement inside the sampled domain and proposes an operating point close to the direct finite-grid optimum. The physics-model recheck remains authoritative, and deployment would additionally require measured channel and device data.

Future Work

Future work should extend the project from engineering proxies toward implementation-matched security analysis and experimental validation. Priorities include finite-key DPS security bounds, a full iterative LDPC reconciliation implementation with explicit leakage/verification accounting, measured Bahrain atmospheric and daylight-background data, detector afterpulsing/dead-time characterization, pointing-acquisition-tracking dynamics, real satellite ephemerides and weather availability, key-inventory scheduling across trusted relays, and a hardware proof of concept that tests vendor-supported KMS/HSM and QKD/PQC integration interfaces.

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I acknowledge the use of OpenAI's ChatGPT and Codex in the development and documentation of this project. ChatGPT assisted with diagrams, technical explanations and report refinement, while Codex supported the development, debugging and testing of Python and GNU Octave simulations. The project objectives, technical



requirements and design decisions were directed by me. I reviewed the AI-assisted outputs and take responsibility for the final content, results and interpretations.

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